Opening the Black Box*

A Theory of the Value of Data in Prediction

Giovanni Rizzi†

Latest Draft

Abstract

This paper develops a theory of the value of data for prediction. An agent observes a target variable and some characteristics (covariates) of a sample of individuals to predict the outcome for a target individual based on her characteristics. The main findings are: (i) covariates exhibit economies of scope, as the value of one covariate is higher when others are also observed; (ii) covariates and observations are complements when datasets are small but may become substitutes in large datasets. These results imply three policy conclusions. First, prediction markets feature barriers to entry even without fixed costs: data collection is profitable only beyond a minimum scale, and more advanced training technologies are viable only when prediction is sufficiently valuable. Second, competition between prediction sellers endogenously produces quality and technology asymmetries, especially when a large share of demand is contestable. Third, in data-acquisition markets, economies of scope create a force toward concentration: app concentration increases the value of users, so asymmetries propagate across market sides and may tip the market to a single ecosystem. Data can thus generate concentration through both entry barriers and amplification of small initial advantages.

JEL Classification: C11, D83, L12, L40.

Keywords: Value of Data, Prediction, Economies of Scope, Data Markets, Digital Platforms.

*I thank Patrick Rey for his guidance, and Jean-Pierre Florens and Doh-Shin Jeon for their suggestions. I also thank Jad Beyhum, Michele Bisceglia, Zhijun Chen, Alexandre de Corniere, Andrei Hagiu, Johannes Horner, Bruno Jullien, Hiroaki Kaido, Simon Loertscher, Friedrich Lucke, Leonardo Madio, Thierry Magnac, Nour Meddahi, Giovanni Morzenti, Juan Ortner, Christoph Reidl, Andrew Rhodes, Francois Salanie, Maximilian Schafer, Sara Shahanaghi, Tim Simcoe, Alex Smolin, Marshall Van Alstyne, Davide Viviano, as well as seminar participants at the European Association for Research in Industrial Economics Conference (Valencia, 2025), the Questrom Digital Platforms Seminar, and at TSE and Boston University.

†Toulouse School of Economics, University of Toulouse Capitole, France. E-mail: giovanni.rizzi@tse-fr.eu

1 Introduction

In digital markets, data drive competitive advantage because they allow firms to make better predictions: ChatGPT and Gemini predict the next response a user is likely to find helpful given a prompt and context; Amazon and Uber predict where demand will arise; Google and Meta predict which ad a user will click; and Netflix predicts what content a user will enjoy most. Over the last decade, many digital markets have become dominated by a few firms that control vast amounts of data, attracting the attention of competition authorities. For instance, according to the 2021 U.S. House Report, "data advantages [...] can reinforce dominance and serve as a barrier to entry,"[1] a warning echoed in the EU Data Act proposal[2] and in major policy reports.[3]

These concerns are partly fueled by earlier statements from industry leaders: as Google's CEO declared in 2009, "Scale is the key. We just have so much scale in terms of the data we can bring to bear."[4] However, more recently, technology firms have downplayed policymakers' claims, arguing that data exhibit diminishing returns to scale by the Law of Large Numbers, and that data concentration in the hands of few firms reflects their technological advantage on rivals rather than entry barriers.[5]

To evaluate these competing narratives, we must consider two distinct ways in which firms acquire data: either by observing more individuals (i.e., increasing sample size) or by observing more attributes of each individual (i.e., increasing the covariates). This paper asks a simple but unresolved question: how do predictions improve when a firm acquires (i) more observations and/or (ii) more covariates? Empirically and conceptually, these two channels—*economies of scale* and *economies of scope*—are often conflated. My goal is to disentangle them.

I develop a Bayesian model that cleanly disentangles these two dimensions. A firm predicts a target variable for a target individual—for example, the price of a house. The target variable depends on many potential covariates—such as size, number of rooms, and year of construction—each of which affects the outcome through an unknown weight. The firm begins with prior uncertainty about these weights and can collect two types of data: (1) *Training data*: covariates and target variables for a sample of individuals, used to learn the relationship between covariates and target variables; (2) *target covariates*: a subset of the covariates of the target individual, used to apply what the firm has learned from the training data.

Consequently, the firm designs a dataset by choosing three objects: (i) the number of observations in the training data N (the number of past house prices), (ii) the set of *training covariates* $\mathcal{T}$ (which house characteristics on past purchases), and (iii) the set of *target covariates* $\mathcal{P}$ to observe for the target individual (which characteristics to observe on the house whose price

[1] U.S. House of Representatives (2020), Investigation of Competition in Digital Markets, 117-40, pp. 36–38.
[2] European Commission (2023), Data Act proposal, COM(2023) 193 final.
[3] Digital Platforms (2019) and UK Competition and Markets Authority (2019).
[4] Schmidt (2009), "How Google Plans to Stay Ahead in Search," Bloomberg, 2 October.
[5] Varian (2018) and Bajari et al. (2019).

must be predicted).
Because collecting either type of data is costly, the dataset design choice reflects the trade-off between better predictions and higher data collection costs.
Given any dataset design, I derive the optimal Bayesian predictor and characterize the value of data as the expected reduction in out-of-sample mean squared error. This value depends on two objects: the information that training data provide about the weights and the target covariates to which these weights are applied.
I show that these components yield a closed-form expression for the value of a dataset design in the high-dimensional limit, for sequences of asymptotically non-degenerate designs. Dataset value increases when training covariates reduce residual noise, when additional observations improve the precision of weight estimates, and when target covariates allow the firm to apply these estimates to the target individual. Two results follow. First, training covariates exhibit *economies of scope*: the marginal value of an additional training covariate is higher when more training covariates are already observed, because broader training scope reduces residual noise and thereby improves learning about all trained weights. Second, training covariates and observations are *complements* when datasets are small, but may become *substitutes* when datasets are large. With small datasets, adding a covariate raises the marginal value of an observation. With large datasets, additional covariates mainly reduce residual noise and can lower the marginal value of further observations. This substitution result disappears when idiosyncratic noise is sufficiently large.

Optimal Dataset Design. I then study the firm's optimal data-collection problem when the value of a prediction is determined by the firm's ex ante uncertainty about the target variable. I show that this problem is isomorphic to a production problem in which the firm chooses a level of output sold at an exogenous price. This yields a microfoundation for the cost of prediction accuracy. The model implies barriers to entry even in the absence of fixed costs: because observations and covariates are complementary at low scale, a strictly positive level of uncertainty is required for data collection to be profitable. The model also distinguishes between two training regimes. Under *same-scope training*, the firm trains on the same covariates that it observes at prediction time. Under *expanded-scope training*, it trains on a strictly larger set of covariates. I show that moving from same-scope to expanded-scope training may itself require a minimum level of uncertainty, generating barriers to expansion.

Competition in Data-driven Markets. I then apply the model to two competitive environments. First, I study two horizontally differentiated firms that sell predictions to shoppers and captive buyers. Before competing in prices, each firm chooses its dataset by paying the microfounded cost of data collection. I show that firms endogenously choose different dataset sizes, so market competition generates endogenous asymmetries in prediction quality. I then characterize how the value of prediction accuracy affects entry and the firms' choice between

same-scope and expanded-scope training.

Second, I study competition for data acquisition when firms make predictions for internal use. Users generate observations, and firms compete for them in a Hotelling framework. Apps generate covariates, and a subset of them can choose which ecosystem to join. I show that the statistical properties of data shape competition on both sides of the market. Economies of scope in covariates create an incentive for contestable apps to concentrate in a single ecosystem. This concentration of apps increases the value of attracting users, because more covariates raise the marginal value of observations. As a result, asymmetries in app shares can spill over into the user market and be amplified there. When a sufficiently large share of apps is contestable and users are not too differentiated, the equilibrium tips to a single ecosystem. In this sense, data behave as a natural monopoly: competitive forces tend to concentrate both covariates and observations in one ecosystem.

Roadmap. Section 2 introduces the data-generating process and the firm's dataset-design problem. Section 3 derives the optimal predictor, characterizes the value of data, and studies returns to scale and scope together with complementarities across design margins. Section 4 analyzes the firm's optimal dataset design under data-collection costs. Section 5 studies competition between prediction sellers when the cost of data collection is microfounded by the model. Section 6 studies datasets as a competitive bottleneck when two firms independently acquire data to make predictions. Section 7 concludes. Proofs are in Appendix A.

Related Literature. There is a rich information-design literature on the value of data, starting with Bergemann, Bonatti, and Smolin (2018) and continuing with Jones and Tonetti (2020), Bergemann, Bonatti, and Gan (2022), Bergemann and Bonatti (2024), and Acemoğlu et al. (2022). These papers characterize how much an agent is willing to pay for more informative experiments by modeling the choice of an information structure. In contrast, my setting is more structured than a Blackwell experiment and allows me to distinguish observations, training covariates, and target covariates. This lets me map the value of data to transparent statistical objects (covariate variances, posterior variances, and a misspecification penalty) and separate the roles of sample size, scope, and covariate selection. Methodologically, the paper is related to Montiel Olea et al. (2022), Iyer and Ke (2024), and Dasaratha, Ortner, and Zhu (2025), who study competition between models with different covariates; here, covariates and observations are modeled jointly and training and prediction data are separated, which generates structural non-convexities and complementarities in the value of data. Finally, Strzalecki (2024) interpret deviations from Bayesian updating via a primitive misspecification parameter; in my framework, an analogous distortion arises endogenously as standard Bayesian behavior under incomplete data collection, providing a microfoundation for likelihood discounting.

I also contribute to the broader literature on economies of scope and scale in data. Most models fix the set of covariates and study returns to observations, as in Bajari et al. (2019) and

Goldfarb and Tucker (2011). By endogenizing which covariates are collected, my framework rationalizes empirical evidence on complementarities and economies of scope in Schaefer and Sapi (2023) and Carballa-Smichowski et al. (2025b). Schaefer (2025) develops a complementary frequentist approach and shows that the eigenspectrum of the covariate matrix shapes returns to scale, in line with my comparative statics. Allcott et al. (2025) estimate diminishing returns to additional observations and limited complementarities across search queries. Lee and Wright (2023) develop an empirical model of a recommender system using Reddit data on jokes and show that there are diminishing returns to the sample size but complementarities between sample size and number of covariates. While Radner and Stiglitz (1984) attribute increasing returns to information costs, here non-convexities and increasing returns arise purely from the statistical structure of prediction, even under separable reduced-form data-collection costs. These results provide microfoundations for work in IO and macroeconomics that takes increasing returns to data as primitive, either via feedback between data and demand or by assuming complementarities across datasets (Hagiu and Wright 2023; Prüfer and Schottmüller 2021; Farboodi and Veldkamp 2025; Aral, Brynjolfsson, and Wu 2008; Cong, He, and Yu 2021; Carballa-Smichowski et al. 2025a; De Corniere and Taylor 2025; Calzolari, Cheysson, and Rovatti 2025): in my framework, economies of scope and complementarities emerge from the way training data generate information about weights, without relying on demand-side feedback or exogenous complementarities.

2 Setup

An agent (she) seeks to predict a target variable for a target individual (he). To do so, she can collect (i) training data, consisting of target variables and covariates for a sample of individuals, to learn the relationship between target variables and covariates; and (ii) covariates of the target individual, to apply this relationship and form a prediction. The agent chooses which covariates to observe, trading off prediction accuracy and data-collection costs.

2.1 Data-Generating Process

An agent predicts a real-valued target variable y_0 for a target individual indexed by 0 and drawn from a countably infinite population $\mathbb{N}_0$. We consider a sequence of data-generating processes (DGPs) indexed by $K \in \mathbb{N}_+$, where the set of potential covariates is $\mathcal{K} \equiv \{1, ..., K\}$. For each individual $i \in \mathbb{N}_0$, the target variable satisfies

$$y_i = \sum_{k=1}^{K} \beta_k x_{ik} + \eta_i,$$

where $x_{ik} \in \mathbb{R}$ is covariate k of individual i, β_k is an unknown weight, and η_i is an idiosyncratic error term.

Weights. The weights are i.i.d. and independent of the covariates and errors:

$$\beta_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \sigma_\beta^2\right),$$

where $\sigma_\beta^2 \in [0, \infty)$ reflects prior uncertainty about the weights.

Covariates. Covariates are mutually independent and i.i.d. across individuals:

$$x_{ik} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \frac{\sigma_x^2}{K}\right).$$

The scaling by $1/K$ keeps the total covariate variance $\sum_{k=1}^{K} \mathrm{Var}(x_{ik})$ equal to σ_x^2, independently of K. The parameter $\sigma_x^2 > 0$ reflects the overall signal in covariates.

Idiosyncratic Errors. The idiosyncratic errors are i.i.d. across individuals and independent of weights and covariates:

$$\eta_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\eta^2 \sigma_\beta^2),$$

where $\sigma_\eta^2 \geq 0$ reflects the importance of the ***idiosyncratic noise*** (e.g., measurement error or non-linearity in the DGP).
Under these assumptions, for all $i \in \mathbb{N}_0$, it holds that $\mathrm{Var}[y_i] = \sigma_\beta^2(\sigma_x^2 + \sigma_\eta^2)$, which is finite and does not depend on K.[6]

Notation. For K fixed, let $\boldsymbol{x}_i \equiv (x_{ik})_{k=1}^{K} \in \mathbb{R}^K$ denote the covariate vector of individual i, and let $\boldsymbol{\beta} \equiv (\beta_k)_{k=1}^{K} \in \mathbb{R}^K$ denote the weight vector. For any vector $\boldsymbol{v} \in \mathbb{R}^K$ and index set $\mathcal{K}' \subseteq \mathcal{K}$, let $\boldsymbol{v}_{\mathcal{K}'} \equiv (v_k)_{k \in \mathcal{K}'}$ denote the corresponding subvector.

2.2 Data and Predictor

The agent chooses three objects which determine the characteristics of the observed dataset:

- a ***sample size***, $N \in \mathbb{N}_0$, of individuals on which to observe the target variable and resulting in a ***target vector*** $\boldsymbol{y}_N \in \mathbb{R}^N$;[7]
- a ***set of training covariates***, $\mathcal{T} \subseteq \mathcal{K}$, observed on the N individuals in the sample and resulting in a training matrix $\boldsymbol{X}_{N,\mathcal{T}} \in \mathbb{R}^{N \times T}$, with $T \equiv |\mathcal{T}|$; and

[6] The linear specification should be interpreted as a reduced-form representation of the predictive component of a possibly nonlinear data-generating process. More generally, any square-integrable target variable admits a linear projection onto the covariates, and in high dimension with many weak predictors, smooth nonlinear functions are well approximated by their first-order expansion.

[7] Because the variables in the DGP are i.i.d. across individuals, the joint distribution of the training sample depends only on its size. Hence there is no loss of generality in assuming the sample consists of the individuals $\{1, \ldots, N\} \subset \mathbb{N}_0$.

- a ***set of target covariates***, $\mathcal{P} \subseteq \mathcal{K}$, observed on the target individual 0 and resulting in a target vector $\boldsymbol{x}_{0,\mathcal{P}} \in \mathbb{R}^P$, with $P \equiv |\mathcal{P}|$.[8]

The agent's data-collection strategy is summarized by the following object.

Definition 1. A ***dataset design*** is a triple

$$\mathcal{D} \equiv (N, \mathcal{T}, \mathcal{P}),$$

where N is the sample size, $\mathcal{T}$ is the set of training covariates and $\mathcal{P}$ is the set of prediction covariates.

Given a design $\mathcal{D}$, the ***training data*** consist of the realization

$$(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}),$$

where the i-th row of $\boldsymbol{X}_{N,\mathcal{T}}$ collects the covariates $(x_{ik})_{k\in\mathcal{T}}$ for individual i, and the i-th entry of $\boldsymbol{y}_N$ equals y_i. The ***dataset*** is a realization

$$\boldsymbol{d} \equiv ((\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}), \boldsymbol{x}_{0,\mathcal{P}}),$$

which includes the training data and the prediction covariates of the target individual. Under design $\mathcal{D}$, the random dataset $\boldsymbol{d}$ is distributed according to the prior described in Section 2.1, which I denote by P_D, i.e. $\boldsymbol{d} \sim P_D$. We denote the *empty design* by $\varnothing$.

Predictor. Predictions depend on the realized dataset. Given a design $\mathcal{D} = (N, \mathcal{T}, \mathcal{P})$, a ***predictor*** is a measurable map

$$\hat{y}_D : \mathbb{R}^{N\times(1+T)} \times \mathbb{R}^P \to \mathbb{R}$$

that assigns a prediction to each realized dataset. In other words, under any fixed design $\mathcal{D}$, $\hat{y}_D$ is a random variable induced by the randomness in $\boldsymbol{d}$, and the predictor specifies how the agent uses the observed data to form a prediction.

2.3 Agent's Problem

The agent chooses a costly dataset design $\mathcal{D}$ and a predictor $\hat{y}_D$, which is costless, to minimize her ex-ante prediction loss.

If the agent makes a point prediction $\hat{y} \in \mathbb{R}$ of the target variable y_0, the realized prediction error is $y_0 - \hat{y}$, and she incurs quadratic loss

$$\ell(y_0, \hat{y}) = (y_0 - \hat{y})^2 .$$

[8] Allowing $\mathcal{T}$ and $\mathcal{P}$ to differ makes it possible to study the overlap between prediction and training covariates.

Therefore, the agent is penalized proportionally more for larger prediction errors. Because predictor choice is costless, the agent chooses the ***optimal predictor***[9]

$$\hat{y}_D^*(\boldsymbol{d}) \equiv \arg\min_{\hat{y}_D} \mathbb{E}\left[\ell(y_0, \hat{y}_D(\boldsymbol{d})) \mid \boldsymbol{d}\right].$$

For quadratic loss, $\hat{y}_D^*$ exists and is unique up to realizations of $\boldsymbol{d}$ with probability 0. Hence, I can define the ***value of a dataset d*** as

$$v(\boldsymbol{d}) \equiv \underbrace{\min_{\hat{y}} \mathbb{E}\left[\ell(y_0, \hat{y})\right]}_{\text{minimum prior risk}} - \underbrace{\mathbb{E}\left[\ell\left(y_0, \hat{y}_D^*(\boldsymbol{d})\right) \mid \boldsymbol{d}\right]}_{\text{minimum posterior risk}},$$

which is the reduction of risk that the agent achieves by observing $\boldsymbol{d}$ and using it to compute $\hat{y}_D^*(\boldsymbol{d})$, relative to the outside option of not collecting any data and making the ex-ante optimal prediction.
The value of a dataset design is the ex-ante expected value of the dataset it induces.

Definition 2. The ***value of a design*** $\mathcal{D}$ is

$$V(\mathcal{D}) \equiv \mathbb{E}_{\boldsymbol{d} \sim P_{\mathcal{D}}}\left[v(\boldsymbol{d})\right].$$

I assume that the cost of a dataset design $\mathcal{D}$ depends only on its dimensions:

$$C(\mathcal{D}) = C(N, T, P),$$

where C is weakly increasing in each argument, nonnegative, and $C(0, 0, 0) = 0$.
Thus, $V(\mathcal{D}) - C(\mathcal{D})$ is the agent's ex-ante net payoff from choosing design $\mathcal{D}$, relative to the outside option of collecting no data. The agent therefore solves

$$\max_{\mathcal{D}} \; V(\mathcal{D}) - C(\mathcal{D}).$$

3 The Value of Datasets

3.1 Optimal Predictor

In this section I characterize the optimal predictor in a DGP with a fixed, finite number of potential covariates $K \in \mathbb{N}_+$. The agent chooses how to use the target vector $\boldsymbol{x}_{\mathcal{P}}$ and the training data $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$ to form a prediction.
The optimal predictor separates (i) *learning across training individuals* from the training sample, which updates beliefs about $\boldsymbol{\beta}$, and (ii) *using within-target information* $\boldsymbol{x}_{\mathcal{P}}$, which maps

[9]Because of its tractability, quadratic loss is a standard assumption in econometrics and machine learning. Furthermore, it has desirable theoretical properties discussed in Brier (1950) and axiomatized in Selten (1998).

posterior beliefs on weights into a prediction.
Under quadratic loss, the optimal predictor is the posterior mean of the target variable, a standard result in Bayesian decision theory.

Proposition 1 (Optimal predictor)**.** *The optimal predictor is*

$$\hat{y}_D^*(\boldsymbol{d}) = \mathbb{E}\left[y_0 \mid \boldsymbol{d}\right] = \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^\top \mathbb{E}[\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}} \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})].$$

As is well known, under quadratic loss the optimal predictor is the posterior mean. Because target variables are conditionally independent across individuals given $\boldsymbol{\beta}$, the optimal predictor depends on the training data $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$ only through the posterior distribution of the weights.
Furthermore, covariates and weights are mutually independent. Hence, for any $k \in \mathcal{P} \setminus \mathcal{T}$, the likelihood of $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$ does not depend on β_k, so its posterior coincides with its prior, which is zero, implying

$$\mathbb{E}[\beta_k \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})] = 0, \qquad \forall k \in \mathcal{P} \setminus \mathcal{T}.$$

Therefore, covariates observed for the target individual but not included in training do not enter the optimal predictor. I refer to the vector $\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}}$ the ***target weights***.

Corollary 1 (Value of dataset)**.** *The value of a dataset $\boldsymbol{d}$ is*

$$v(\boldsymbol{d}) = Var[y_0] - Var[y_0 \mid \boldsymbol{d}].$$

3.2 Asymptotic Designs

3.2.1 Nested representation of dataset design

Proposition 1 has an immediate implication for data design. Target covariates excluded from training never affect the optimal predictor, and therefore do not affect dataset value. Since such covariates are weakly costly, any design is weakly dominated by one in which every target covariate is also included in training.

Corollary 2 (Nested representation)**.** *For any design $(N, \mathcal{T}, \mathcal{P})$, the design $(N, \mathcal{T}, \mathcal{P} \cap \mathcal{T})$ yields weakly higher net payoff.*

Therefore, I can restrict attention to nested designs with $\mathcal{P} \subseteq \mathcal{T}$. Under index-wise symmetry, covariate labels are immaterial and only cardinalities matter. Thus, after relabeling, any such design $\mathcal{D}$ can be represented without loss by

$$D = (N, T, P), \qquad P \leq T,$$

with

$$\mathcal{T} = \{1, \dots, T\}, \qquad \mathcal{P} = \{1, \dots, P\}.$$

Given K and conditional on the resulting dataset design $D(K)$, for any individual i in the training sample, we can decompose the target variable as

$$y_i^{(K)} = \sum_{k=1}^{T(K)} \beta_k^{(K)} x_{ik}^{(K)} + \varepsilon_i^{(K)},$$

where $\sum_{k=1}^{T(K)} \beta_k^{(K)} x_{ik}^{(K)}$ is the variation due to the training covariates and

$$\varepsilon_i^{(K)} \equiv \sum_{k=T(K)+1}^{K} \beta_k^{(K)} x_{ik}^{(K)} + \eta_i^{(K)},$$

is the *training residual* due to the omitted covariates and the idiosyncratic error $\eta_i^{(K)}$. Its unconditional distribution is a sum of products of independent gaussians. Using the law of large numbers, I show that, in the limit as $K \to \infty$, the distribution of the residual also becomes gaussian.

3.2.2 Asymptotic Gaussianity of Residual

To do so, I define a *sequence of dataset designs* as a collection $\{D(K)\}_{K \in \mathbb{N}_+}$ and make the following asymptotic assumption.

Assumption 1 (Sequence of Designs)**.** *We restrict attention to sequences* $\{D(K)\}_{K \in \mathbb{N}_+}$ *for which there exist constants* $(n, t, p) \in (0, \infty) \times (0, 1]^2$ *such that, as* $K \to \infty$*,*

$$\frac{N(K)}{K} \to n, \qquad \frac{T(K)}{K} \to t, \qquad \frac{P(K)}{K} \to p.$$

We call (n, t, p) *the* **design margins** *of the sequence, with* n *the* **training scale***,* t *the* **training scope***, and* p *the* **target scope***.*

Assumption 1 restricts attention to nondegenerate asymptotic environments in which neither sample size, nor training scope, nor target scope becomes asymptotically negligible relative to the number of potential covariates K.[10] Therefore, it reflects the abundance of data that characterizes modern machine learning environments, where both the number of observations and the number of covariates are typically large. Furthermore, working in high dimension delivers well-behaved closed form expressions for the value of a dataset.

The following result shows that the training residual converges in distribution to a Gaussian as $K \to \infty$.

[10] A similar assumption is standard in random matrix theory (see, e.g., Marčenko and Pastur (1967), Bai and Silverstein (2009), and Hastie et al. (2020)). I depart from the literature by introducing the variable K which allows me to control three distinct margins $N(K)$, $T(K)$ and $P(K)$.

Lemma 1. *For any sequence of training covariate sizes* $\{T(K)\}_{K\in\mathbb{N}_+}$ *satisfying Assumption 1 and any fixed* $m \in \mathbb{N}_+$,

$$(\varepsilon_1^{(K)}, \dots, \varepsilon_m^{(K)})^\top \underset{K\to\infty}{\overset{d}{\to}} \mathcal{N}(\mathbf{0}_m, \sigma_\beta^2 \sigma_\varepsilon^2(t) \cdot \boldsymbol{I}_m),$$

where the (relative) residual noise *is*

$$\sigma_\varepsilon^2(t) \equiv (1-t)\sigma_x^2 + \sigma_\eta^2.$$

Under Assumption 1, the residual contains a large number of omitted covariates which are mutually independent. As the covariate variance is σ_x^2/K, for $K \to \infty$ each omitted covariate has a negligible impact on the residual and a central-limit result then implies that the residual behaves asymptotically like a Gaussian noise term. The residual noise $\sigma_\varepsilon^2(t)$ is decreasing in t: a larger training scope means fewer omitted covariates enter the residual. As covariates are homoskedastic and independent and each has the same marginal impact on the residual noise, so $\sigma_\varepsilon^2(t)$ is linear in t. The parameter σ_x^2 governs the strength of this effect, capturing the variance of omitted covariates. Since the omitted covariates are independent across individuals, the residuals are also asymptotically independent.

By Lemma 1, estimating the target weights $\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}}$ reduces to a textbook Bayesian linear regression (e.g., DeGroot (1970) and Berger (1990)), and the optimal predictor is

$$\hat{y}_D^*(\boldsymbol{d}) = \boldsymbol{x}_{0,\mathcal{P}\cap\mathcal{T}}^\top \underbrace{\left(\left(\boldsymbol{X}_{N,\mathcal{T}}{}^\top \boldsymbol{X}_{N,\mathcal{T}} + \sigma_\varepsilon^2(t)\cdot \boldsymbol{I}_T\right)^{-1} \boldsymbol{X}_{N,\mathcal{T}}{}^\top \boldsymbol{y}_N\right)_{\mathcal{P}\cap\mathcal{T}}}_{=\mathbb{E}\left[\boldsymbol{\beta}_{\mathcal{P}\cap\mathcal{T}} \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})\right]}.$$

The posterior mean of the weights is a shrinkage of the maximum-likelihood estimator estimator

$$\left(\boldsymbol{X}_{N,\mathcal{T}}{}^\top \boldsymbol{X}_{N,\mathcal{T}}\right)^{-1} \boldsymbol{X}_{N,\mathcal{T}}{}^\top \boldsymbol{y}_N$$

towards the prior mean $\mathbf{0}_T$. The strength of this shrinkage is governed by the residual noise $\sigma_\varepsilon^2(t)$. Intuitively, if $\sigma_\varepsilon^2(t) \approx 0$, the weights are estimated almost perfectly, so the posterior mean coincides with the maximum-likelihood estimator.

As $\sigma_\varepsilon^2(t)$ decreases, the agent becomes more confident of using the training data to update its prior, and the posterior mean strays further away from the prior mean. This implies that by observing new training covariates, the agent reduces the shrinking for the posterior means of all the weights, meaning that, even if the covariates are independent, they help in estimating each other's weights.

Remark 1 (Training scope as endogenous likelihood precision). This provides a Bayesian interpretation of the likelihood-discounting logic studied by Strzalecki (2024). There, conservatism resulting from misspecification of the DGP is represented by an exogenous parameter that directly downweights the likelihood. Here, an analogous conservatism arises endogenously: when training scope is narrow, the likelihood is less precise because residual noise

is higher. Thus the agent behaves more conservatively while remaining fully Bayesian under the induced reduced-form model.

3.3 The Value of a Dataset

Under Assumption 1, define the ***(relative) information on trained weights*** as

$$\mathcal{I}(n,t) \equiv \lim_{K\to\infty} \frac{\operatorname{tr}\left[\operatorname{Var}[\boldsymbol{\beta}_{\mathcal{T}(K)}] - \operatorname{Var}\left[\boldsymbol{\beta}_{\mathcal{T}(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right]\right]}{\operatorname{tr}[\operatorname{Var}[\boldsymbol{\beta}_{\mathcal{T}(K)}]]}, \tag{1}$$

when the limit exists. The weight information measures what fraction of the variance in the weights of the target covariates is learned under Assumption 1, when training on $T(K)$ training covariates and $N(K)$ observations as $K \to \infty$. Thus $\mathcal{I}(n,t) = 0$ means training is uninformative, while $\mathcal{I}(n,t) = 1$ means training is perfectly informative (i.e., the posterior variance of weights is zero).

Theorem 1 (Asymptotic Value of a Dataset)**.** *Let $\{D(K)\}_{K\in\mathbb{N}_+}$ be a sequence of designs satisfying Assumption 1. Then*

$$v\left(\boldsymbol{d}_{D(K)}\right) \xrightarrow{p} \mathcal{V}(n,t,p),$$

where

$$\mathcal{V}(n,t,p) = \underbrace{\sigma_x^2 \min\{p,t\}}_{\textit{target signal}} \times \underbrace{\sigma_\beta^2 \mathcal{I}(n,t)}_{\textit{information about predictor weights}},$$

and

$$\mathcal{I}(n,t) = \frac{n+1+\frac{\sigma_\eta^2}{\sigma_x^2} - \sqrt{\left(n+1+\frac{\sigma_\eta^2}{\sigma_x^2}\right)^2 - 4nt}}{2t}.$$

The proof relies on concentration of the target vector and the training matrix in high-dimension: by a central-limit theorem type argument, the variance of infinite-dimensional objects converges to the mean variance. Specifically, the closed form of $\mathcal{I}(n,t)$ follows from the expectation of the eigenvalue distribution of a Gaussian matrix, which follows a Marchenko-Pastur distribution and has a closed form if (and only if) covariates are homoskedastic and independent. Several observations are in order.

Decomposition of Dataset Value. The value of a dataset is the product of two terms:

- The target signal $\sigma_x^2 \min\{p,t\}$ is the part of the variance of the target variable which is due to the overlapping covariates.
- The weight information $\sigma_\beta^2 \mathcal{I}(n,t)$ measures the reduction of variance of the predictor weights brought forth by training data of scale and scope (n,t).

If any design margin is zero, the dataset has no value:

$$\mathcal{V}(0, t, p) = \mathcal{V}(n, 0, p) = \mathcal{V}(n, t, 0) = 0,$$

for every feasible (n, t, p).[11] This is because if $n = 0$ or $t = 0$, the agent learns nothing about the weights linking the covariates to the target variable, so $\mathcal{I}(n, t) = 0$. Conversely, if $p = 0$, the agent observes no target covariates so estimating their weights is useless. Therefore, around $(0, 0, 0)$, the design margins are hard complements. By contrast, as the scale grows large, the predictor weights are learned almost perfectly, i.e. $\lim_{n\to\infty} \mathcal{I}(n, t) = 1$, and

$$\lim_{n\to\infty} \mathcal{V}(n, t, p) = \sigma_\beta^2 \sigma_x^2 \min\{p, t\}.$$

Therefore, for large scale, the training and target covariates become again hard complements.

Arrow's information paradox. A classic obstacle to pricing datasets is that their value is itself uncertain: they are experience goods. A buyer would like to inspect a dataset before purchasing it in order to assess its value, but once it is inspected, the buyer no longer needs to pay for access. This is Arrow's information paradox (Arrow 1959). The problem is driven by the fact that the seller observes the realization of the dataset, while the buyer does not.
Theorem 1 resolves this difficulty in large datasets. When the target variable depends on many covariates, none of which has an overwhelming effect, datasets behave like standardized goods: conditional on their design (n, t, p), their value is known ex ante, so the seller's informational advantage disappears.[12] It follows that, when contracts can be written on design margins, large datasets are contractible objects and can be priced and traded on the basis of observable characteristics. Consequently, Theorem 1 implies that markets for large datasets are less prone to failure and more liquid than markets for small datasets, because their value is more transparent ex ante.

Privileged Information. The model allows the agent to train on more covariates than she observes for the target individual, that is, $t > p$. This design can be attractive because training-only covariates reduce the residual noise $\sigma_\varepsilon^2(t)$, which improves estimation on the predictor weights.
This pattern closely matches a standard practice in modern data science: models are often trained on a richer set of covariates than those available for the target individual, a setting

[11] The expression for $\mathcal{V}$ admits a unique continuous extension to

$$\{(n, t, p) \in \mathbb{R}_+ \times [0, 1] \times [0, t]\},$$

and I use the same notation for this extension.

[12] This relies on the high-dimensional law of large numbers embedded in the model and, in particular, on assumptions ruling out dominant covariates or heavy-tailed realizations. Independence, homoskedasticity, and gaussianity are sufficient conditions.

known as *Learning Using Privileged Information (LUPI).*[13]

Weight information as a signal share. The closed-form expression for $\mathcal{I}(n,t)$ admits a useful signal-share representation. For each K, define

$$\mathcal{S}_K \equiv \frac{1}{T(K)} \operatorname{tr}\left(\boldsymbol{X}_{N(K),\mathcal{T}(K)}^{\top} \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right),$$

and, for the target individual 0,

$$u_0^{(K)} \equiv y_0^{(K)} - \boldsymbol{x}_{0,\mathcal{T}(K)}^{\top} \mathbb{E}\left[\boldsymbol{\beta}_{\mathcal{T}(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right].$$

Define also

$$\mathcal{E}_K \equiv \frac{1}{\sigma_\beta^2} \operatorname{Var}\left(u_0^{(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right).$$

Proposition 2 (Signal-share representation of weight information)**.** *As $K \to \infty$,*

$$\mathcal{S}_K \xrightarrow{p} \mathcal{S}(n), \qquad \mathcal{E}_K \xrightarrow{p} \mathcal{E}(\mathcal{I}(n,t);t).$$

where

$$\mathcal{S}(n) \equiv n\sigma_x^2 \quad \textit{and} \quad \mathcal{E}(\mathcal{I};t) \equiv \sigma_\varepsilon^2(t) + t\sigma_x^2(1-\mathcal{I}).$$

Moreover, $\mathcal{I}(n,t)$ is the unique element of $[0,1]$ such that

$$\mathcal{I}(n,t) = \frac{\mathcal{S}(n)}{\mathcal{S}(n) + \mathcal{E}(\mathcal{I}(n,t);t)}.$$

The proposition shows that weight information is the unique self-consistent training signal share in the training problem. The training signal is linear in the scale n. The training noise is the sum of two components: the misspecification noise $\sigma_\varepsilon^2(t)$, which is decreasing in t, and a noise deriving from estimation $t\sigma_x^2(1-\mathcal{I})$, reflecting the posterior uncertainty about the trained weights, which is increasing in the number of weights reflected in the training scope t. Proposition 2 shows that the former effect dominates the latter, so $\mathcal{E}(\mathcal{I};t)$ is decreasing in t. Consequently, both training scope t and scale n have a positive impact on $\mathcal{I}(n,t)$. Moreover, the weight information $\mathcal{I}(n,t)$ is the unique fixed point of $\Phi(\cdot;n,t)$ in $[0,1]$:

$$\mathcal{I}(n,t) = \Phi(\mathcal{I}(n,t);n,t), \qquad \Phi(\mathcal{I};n,t) \equiv \frac{\mathcal{S}(n)}{\mathcal{S}(n) + \mathcal{E}(\mathcal{I};t)},$$

where $\Phi(\mathcal{I};n,t)$ is the signal-share.

[13] On privileged information, see Crankshaw et al. (2017), Vapnik and Izmailov (2015), and Lopez-Paz et al. (2016).

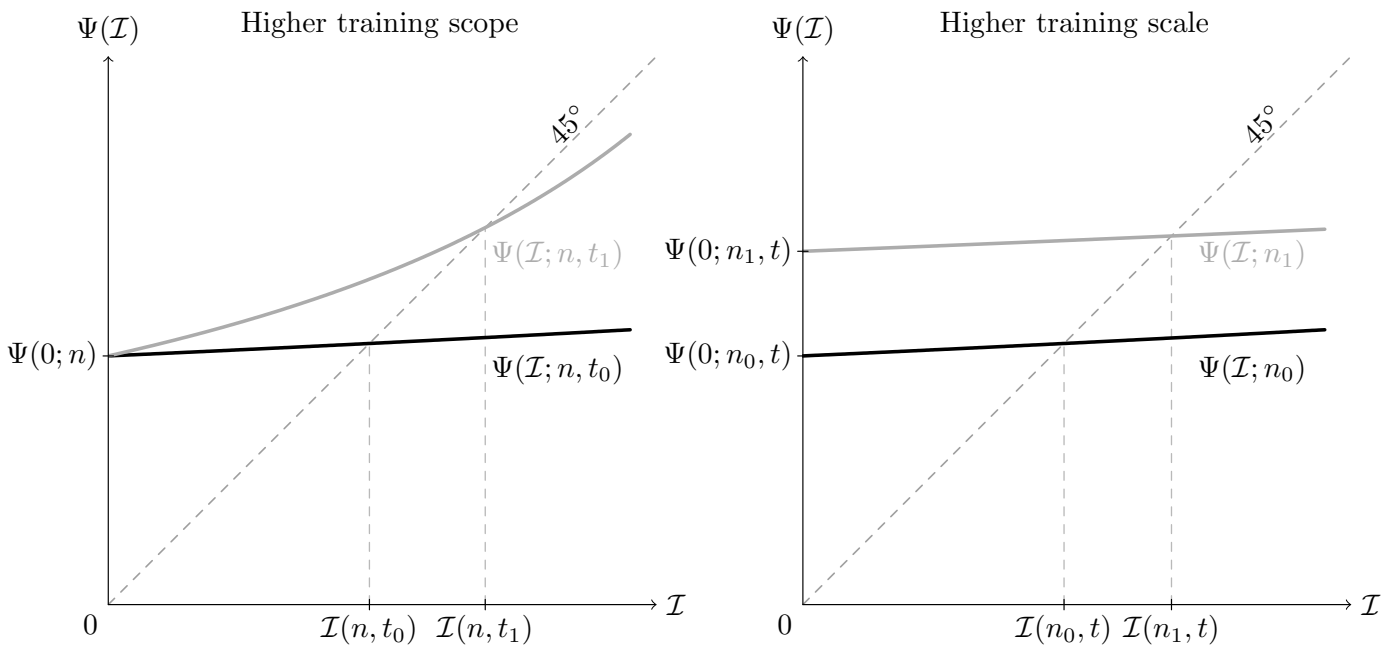

Figure 1: Weight information as training signal share. In each panel, $\mathcal{I}(n, t)$ is the intersection between $\Phi(\mathcal{I})$ and the 45° line. Left panel: holding training scale n fixed, a higher training scope t steepens $\Phi(\mathcal{I})$ without changing $\Phi(0)$. Right panel: holding training scope t fixed, a higher training scale n shifts $\Phi(\mathcal{I})$ upward.

Figure 1 shows that training scope t has an amplification effect on $\mathcal{I}(n, t)$: by increasing the training scope, the training noise goes down by an amount proportional to $\mathcal{I}$ itself. Broader training scope reduces misspecification noise, and the additional information thereby generated is greater if the weights are better learned to begin with. By contrast, training scale n works mainly through a level effect rather than by strengthening the feedback mechanism.

3.4 Returns to Design Dimensions

I study the returns to design margins by differentiating $\mathcal{V}(n, t, p)$. Let

$$\xi \equiv \frac{\sigma_\eta^2}{\sigma_x^2} \in [0, \infty)$$

denote the ratio of idiosyncratic noise to covariate signal. Thus, if $\xi > 1$, variation in the target variable is driven more by idiosyncratic noise than by the component explained by the potential covariates; if $\xi \leq 1$ the opposite is true. I then normalize $\sigma_x^2 = 1$. As the value function is linear in σ_β^2 we define the relative value as

$$V(n, t, p) \equiv \frac{\mathcal{V}(n, t, p)}{\sigma_\beta^2} \in [0, \min\{p, t\}]. \tag{2}$$

Using Theorem 1 and the novel parametrization,

$$V(n,t,p) = \min\{p,t\} \times \underbrace{\frac{n+\xi+1-\sqrt{(n+\xi+1)^2-4nt}}{2t}}_{=\mathcal{I}(n,t)}.$$

Conditional on $p \leq t$, V is linear in p: because target covariates are i.i.d. and contribute identically, the target scope has positive and constant returns up to $t = p$, then its returns fall to 0.

3.4.1 Diminishing Returns to Scale.

Observations have positive but diminishing marginal value.

Corollary 3 (Returns to Scale)**.** *The value is increasing in training scale n:*

$$\frac{\partial V}{\partial n}(n,t,p) = \min\{p,t\}\,\frac{\partial \mathcal{I}}{\partial n}(n,t) > 0.$$

Moreover, the value is concave in n:

$$\frac{\partial^2 V}{\partial n^2}(n,t,p) = \min\{p,t\}\,\frac{\partial^2 \mathcal{I}}{\partial n^2}(n,t) < 0.$$

This follows from Proposition 2. The training signal $\mathcal{S}(n)$ is linear in n, while the weight information $\mathcal{I}(n,t)$ is increasing and concave in $\mathcal{S}(n)$. It follows that $\mathcal{I}(n,t)$, and therefore $V(n,t,p)$, are increasing and concave in n.

The intuition is standard Bayesian learning. Additional observations improve the estimate of the coefficients because they provide more information about the training relationship. But as the sample grows, the posterior is already based on a larger body of evidence, so one more observation changes beliefs by less. Each additional observation therefore raises value, but by a smaller amount than the previous one. This yields diminishing returns to training scale. This pattern is consistent with the empirical evidence in Bajari et al. (2019), Goldfarb and Tucker (2011), Schaefer and Sapi (2023), and Lee and Wright (2023).

3.4.2 Economies of Scope in Training

Training covariates have positive and increasing marginal value.

Corollary 4 (Economies of Training Scope)**.** *The value is increasing in training scope t:*

$$\frac{\partial V}{\partial t}(n,t,p) = \underbrace{\mathbf{1}_{\{t<p\}}\,\mathcal{I}(n,t)}_{\textit{target-weight effect}} + \underbrace{\min\{p,t\}\,\frac{\partial \mathcal{I}}{\partial t}(n,t)}_{\textit{information effect}} > 0.$$

Moreover, the value is convex in t:

$$\frac{\partial^2 V}{\partial t^2}(n,t,p) = 2\,\mathbf{1}_{\{t<p\}}\,\frac{\partial \mathcal{I}}{\partial t}(n,t) + \min\{p,t\}\,\frac{\partial^2 \mathcal{I}}{\partial t^2}(n,t) > 0.$$

An increase in training scope raises value through two channels. First, when $t < p$, increasing t enlarges the set of coefficients that are both learned and used in prediction. This is the target-weight effect. Second, for all t, increasing training scope improves the estimation of the coefficients already used in prediction. This is the information effect. Thus, even when $t \geq p$, so that broader training scope no longer expands the set of target covariates, it still increases value by improving the precision with which the relevant weights are learned.

These two channels also generate convexity. When $t < p$, they reinforce each other: increasing training scope both adds more predictive weights and raises the value of learning them more precisely. When $t \geq p$, only the information effect remains, but value is still convex because $\mathcal{I}(n,t)$ is itself convex in t.

The intuition for the convexity of $\mathcal{I}(n,t)$follows from Proposition 2. Holding n fixed, increasing training scope lowers effective training noise. The misspecification component $\sigma_\varepsilon^2(t)$ decreases with t, while the signal-share representation implies that broader scope strengthens the feedback from current weight information into future learning. As a result, $\mathcal{I}(n,t)$ is increasing and convex in t.

These economies of scope are consistent with the empirical findings in Carballa-Smichowski et al. (2025b).

3.4.3 Are observations and covariates complements or substitutes?

We say that two design margins are *complements* (*substitutes*) if an increase in one raises (lowers) the marginal value of the other. The next result shows that scale and target scope are always complements, and so are training scope and target scope. By contrast, training scale and training scope are generically complements, but can become substitutes when the learning problem is sufficiently informative.

Corollary 5 (Complements and substitutes)**.** *For every $(n,t,p) \in \mathbb{R}_+ \times [0,1]^2$,*

$$\frac{\partial^2 V}{\partial n\,\partial p}(n,t,p) = \mathbf{1}_{\{p<t\}}\frac{\partial \mathcal{I}}{\partial n}(n,t) \geq 0, \qquad \frac{\partial^2 V}{\partial t\,\partial p}(n,t,p) = \mathbf{1}_{\{p<t\}}\frac{\partial \mathcal{I}}{\partial t}(n,t) \geq 0.$$

Moreover,

$$\frac{\partial^2 V}{\partial t\,\partial n}(n,t,p) = \mathbf{1}_{\{t<p\}}\frac{\partial \mathcal{I}}{\partial n}(n,t) + \min\{p,t\}\frac{\partial^2 \mathcal{I}}{\partial n\,\partial t}(n,t) \gtreqless 0 \iff t \lesseqgtr \hat{t}(n;\xi),$$

where $\hat{t}(n;\xi)$ is increasing in n and decreasing in ξ. Finally, if $\xi \geq 1$, then $\hat{t}(n;\xi) \geq 1$ and training scope and scale are complements.

Scale and target scope are complements. A larger training scale improves information about the coefficients, which raises the value of observing the target covariates to which those coefficients are applied. Training scope and target scope are also complements: when $p < t$, a broader training scope improves the precision with which the relevant target coefficients are learned, thereby raising the marginal value of target scope.

The interaction between training scale and training scope is more subtle. The signal-share representation in Proposition 2 clarifies why the sign of the cross-partial between n and t is ambiguous.

On the one hand, increasing training scope lowers effective training noise. This reduces the direct marginal value of an additional observation: when the learning problem is already less noisy, one more observation contributes less to the signal share. This force pushes toward substitutability between scale and scope.

On the other hand, increasing training scope strengthens the feedback from current weight information into future learning. Since the effective noise term depends on $\mathcal{I}$ through $-t\mathcal{I}$, a larger t makes improvements in weight information translate into larger reductions in effective training noise. As a result, an additional observation has a stronger indirect effect on learning, which pushes toward complementarity.

The threshold $\hat{t}(n; \xi)$ is the point at which these two forces exactly offset each other. For relatively narrow training scope, the amplification effect dominates, so training scale and training scope are complements. For sufficiently broad training scope, the direct noise-reduction effect dominates, so they become substitutes.

The threshold also depends on the noise parameter ξ. A higher ξ makes the learning problem less informative and enlarges the region in which scale and scope are complements. In particular, if $\xi > 1$, training scale and training scope are complements everywhere. The interaction of training scope and scale can be summarized in the following figure.

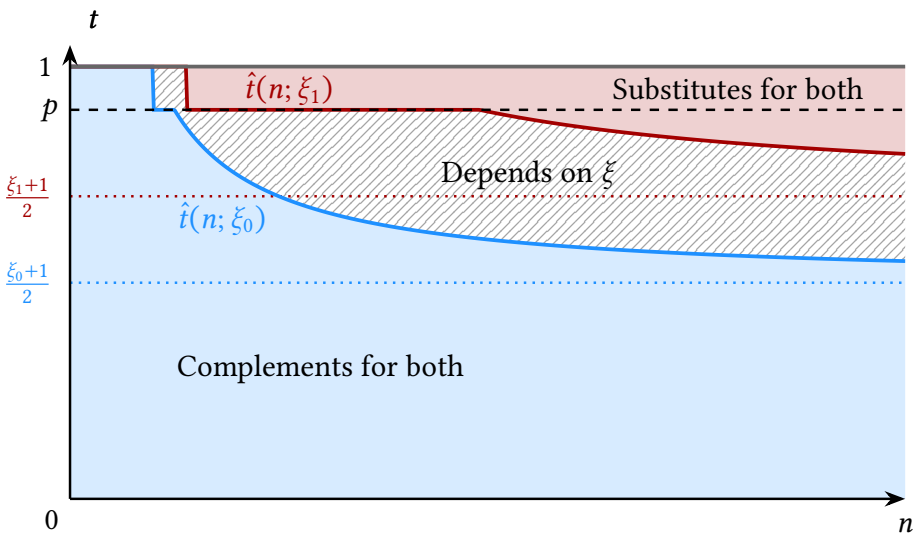

Figure 2: Threshold curves $\hat{t}(n;\xi_0)$ and $\hat{t}(n;\xi_1)$ for $\xi_0 < \xi_1 < 1$. Below both curves, observations and covariates are complements for both values of ξ; above both curves, they are substitutes for both values of ξ; and in the intermediate region between the two curves, the classification depends on ξ.

3.5 Marginal Rate of Technical Substitution in Training

The marginal rate of technical substitution measures the increase in training scale n needed to compensate for a marginal reduction in scope t while keeping value constant:

$$MRTS_{t,n}(n,t,p;\xi) \equiv -\frac{dn}{dt}\bigg|_{V=\bar{V}} = \frac{\partial_t V(n,t,p;\xi)}{\partial_n V(n,t,p;\xi)},$$

whenever the value function is differentiable. A high $MRTS_{t,n}$ means that a training covariate requires a large increase in observations, so scope is the local bottleneck. A low value means few additional observations compensate for the loss of a training covariate, so scale is the local bottleneck.

Proposition 3 (Marginal rate of technical substitution)**.** *For every $(n,t,p) \in \mathbb{R}_+ \times [0,1]^2$ with $t \neq p$, the marginal rate of technical substitution between training scope and scale is*

$$MRTS_{t,n}(n,t,p) = \begin{cases} \dfrac{n/t}{1-\mathcal{I}(n,t;\xi)}, & t < p, \\[2ex] \dfrac{\mathcal{I}^2(n,t;\xi)}{1-\mathcal{I}(n,t;\xi)}, & t > p. \end{cases}$$

At $t = p$, V is not differentiable, so the MRTS is not uniquely defined.

The common denominator $1 - \mathcal{I}(n,t;\xi)$ is the posterior variance on the predictor weights.

When this variance is large, additional observations are less effective at compensating for a loss of scope because the trained weights are still poorly learned.
The two regimes differ in the channel through which scope creates value.

- ***Equal Information:*** $t < p$**.** In this region, the numerator n/t is the number of observations per trained covariate. When this ratio is high, each trained covariate is supported by many observations, so a marginal loss of scope requires a relatively large increase in scale to be offset.

- ***Privileged information:*** $t > p$**.** In this region, extra training covariates are not deployed, so they affect value only through better estimates of the deployed weights. Therefore, $MRTS_{t,n}(n, t, p; \xi)$ only depends on $\mathcal{I}(n, t; \xi)$.

Thus, below $t = p$, the loss of training scope is costly partly because it shrinks the set of usable deployed covariates. Above $t = p$, the loss of training scope matters only through worse estimation, so replacing scope with scale becomes easier when the design is still weakly informative and harder once the deployed weights are already fairly well learned.

Corollary 6. *The $MRTS_{t,n}(n, t, p; \xi)$ is increasing in n and decreasing in ξ. Furthermore,*

$$\lim_{n\to\infty} MRTS_{t,n}(n, t, p; \xi) = +\infty.$$

4 Optimal Dataset Design, Costs and Entry

The firm chooses the dataset design margins (n, t, p) to solve the profit-maximization problem in Section 2.3. Expressing the problem in terms of the relative value function in Eq. (2), the firm's objective can then be written as

$$\max_{n,t,p} \ \sigma_\beta^2 V(n, t, p) - C(n, t, p).$$

This formulation is analogous to the problem of a competitive firm that produces output $V(n, t, p)$, sells it at the exogenous price σ_β^2, and faces input costs $C(n, t, p)$.

Cost function. I assume that the cost of dataset design is

$$C(n, t, p) \equiv \begin{cases} \Gamma(n, t), & \text{if } p \le \hat{p}, \\ +\infty, & \text{if } p > \hat{p}, \end{cases}$$

where $\Gamma : [0, \infty)\times[0, 1) \to \mathbb{R}_+$ is the training cost function and $\hat{p} \in [0, 1]$ is an exogenous upper bound on target scope. Thus, target covariates are costless up to $\hat{p}$, but unavailable beyond

that threshold.[14] This upper bound may capture technological limits on which covariates can be observed at the prediction stage, or regulatory restrictions on the use of target-side information.

I normalize $\Gamma(0,0) = 0$ and assume that Γ is twice continuously differentiable, strictly increasing, convex and weakly supermodular. In addition, I impose Inada-type conditions on marginal costs:

$$\lim_{n\to 0^+} \Gamma_n(n,t) = 0, \qquad \lim_{n\to\infty} \Gamma_n(n,t) = +\infty \quad \text{for every } t \in [0,1),$$

and

$$\lim_{t\to 0^+} \Gamma_t(n,t) = 0, \qquad \lim_{t\to 1^-} \Gamma_t(n,t) = +\infty \quad \text{for every } n \geq 0.$$

These conditions imply that both scale and training scope are costly margins, with marginal costs that are small near the origin and arbitrarily large near the boundary. They rule out degenerate solutions with unbounded scale or maximal training scope and guarantee well-behaved interior trade-offs along the relevant margins. I furthermore assume

$$\sup_{t\in(0,1)} \Gamma_n(n,t) < \infty \qquad \text{for every fixed } n \geq 0.$$

Finally, define the relative marginal cost of training scope as

$$MCR(n,t) \equiv \frac{\Gamma_t(n,t)}{\Gamma_n(n,t)}.$$

I assume that $MCR(n,t)$ is strictly increasing in t and strictly decreasing in n.[15] Intuitively, expanding training scope becomes relatively more expensive at broader scope, but relatively less expensive at larger scale.

Solution. I solve the firm's problem in two steps. First, I study the associated cost-minimization problem of producing a given level of relative value v at minimum cost. This yields the expenditure function

$$\tilde{C}(v) \equiv \min_{n,t,p}\{C(n,t,p) \,:\, V(n,t,p) \geq v\}.$$

Analyzing this problem clarifies the structure of the efficient dataset and the trade-off between scale and training scope. I then return to the firm's profit-maximization problem and characterize the optimal relative value

$$v^* \in [0, \hat{p})$$

[14] Because $V(n,t,p)$ is increasing in p only up to t, and target covariates are costless for $p \leq \hat{p}$, any optimum satisfies $p = \min\{t, \hat{p}\}$.

[15] Equivalently, I assume that for every (n,t), $\frac{\Gamma_{nt}(n,t)}{\Gamma_n(n,t)} < \frac{\Gamma_{tt}(n,t)}{\Gamma_t(n,t)}$ and $\frac{\Gamma_{nt}(n,t)}{\Gamma_t(n,t)} < \frac{\Gamma_{nn}(n,t)}{\Gamma_n(n,t)}$.

as the solution to

$$\max_{v \in [0,\hat{p})} \ \sigma_\beta^2 v - \tilde{C}(v),$$

where σ_β^2 is interpreted as the exogenous price of predictive value.

4.1 Cost-Minimizing Design

Because $V(n, t, p)$ is increasing in p only up to t, and target covariates are costless for $p \leq \hat{p}$ but infeasible for $p > \hat{p}$, the optimal target scope is

$$\tilde{p}(t) = \min\{t, \hat{p}\}.$$

Define therefore

$$\tilde{V}(n, t) \equiv V\big(n, t, \tilde{p}(t)\big),$$

and consider the expenditure function

$$\tilde{C}(v) \equiv \min_{n \geq 0,\ t \in [0,1)} \left\{ \Gamma(n, t) \ : \ \tilde{V}(n, t) \geq v \right\}, \qquad v \in [0, \hat{p}).$$

For $v = 0$, the optimum is the shutdown design $(n, t, p) = (0, 0, 0)$, so $\tilde{C}(0) = 0$.
Now fix $v \in (0, \hat{p})$ and $t \in (v, 1)$. Let $\tilde{n}(v, t)$ denote the minimum scale required to attain value v at scope t, i.e. the unique solution to

$$\tilde{V}(n, t) = v.$$

Inverting $\tilde{V}$ yields

$$\tilde{n}(v, t) = \begin{cases} \dfrac{v(\xi + 1 - v)}{t - v}, & \text{if } v < t \leq \hat{p}, \\[2ex] \dfrac{v\left(\xi + 1 - \dfrac{tv}{\hat{p}}\right)}{\hat{p} - v}, & \text{if } \hat{p} < t < 1. \end{cases}$$

The minimum scale $\tilde{n}(v, t)$ is increasing in v and decreasing in t. A positive value is attainable with finite scale only if $v < \min\{t, \hat{p}\}$, and the cost-minimization problem can be written as

$$\tilde{C}(v) = \min_{t \in (v,1)} \Gamma\big(\tilde{n}(v, t), t\big).$$

4.1.1 Isovalue Geometry

Define the marginal rate of technical substitution between training scope and scale along the isovalue curve as

$$\mu(v, t) \equiv MRTS_{t,n}(\tilde{n}(v, t), t, \tilde{p}(t)).$$

As in standard producer theory, $\mu(v, t)$ is the absolute value of the slope of the isovalue curve in the (n, t) plane:

$$\mu(v, t) = -\frac{d\tilde{n}(v, t)}{dt}.$$

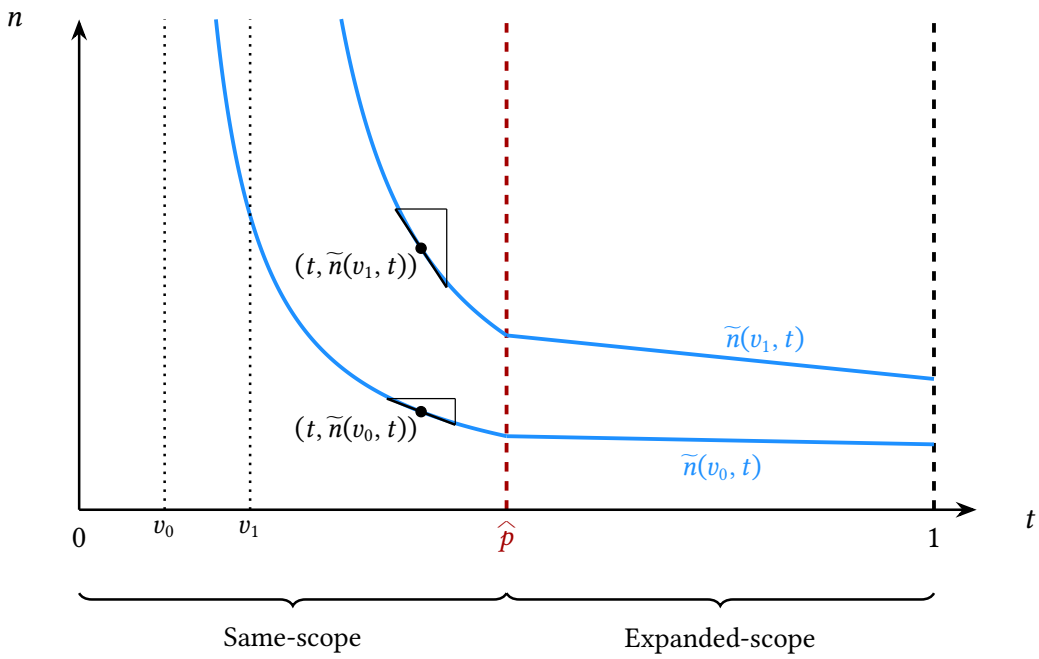

Figure 3: Isovalue curves $\tilde{n}(v, t)$ for two target values $v_0 < v_1$. The black tangent segments indicate the local slope of the isovalue curves at a common training scope t, corresponding to $\mu(v, t)$. The kink at $t = \hat{p}$ marks the transition from same-scope training to expanded-scope training.

Figure 3 plots the isovalue curves, showing how the geometry of the isovalue curves changes at the kink $t = \hat{p}$, where the target constraint starts to bind. The next proposition characterizes this geometry by describing how the trade-off between training scope and scale varies along an isovalue curve.

Proposition 4 (Marginal rate of technical substitution). *Fix a feasible target value $v \in (0, \hat{p})$ and a training scope $t \in (v, 1]$. Then*

$$\mu(v, t) = \begin{cases} \dfrac{v(\xi + 1 - v)}{(t - v)^2}, & \text{if } v < t < \hat{p}, \\[2ex] \dfrac{v^2}{\hat{p}(\hat{p} - v)}, & \text{if } \hat{p} < t \leq 1. \end{cases}$$

Therefore:

1. *For fixed v, $\mu(v,t)$ is strictly decreasing in t on $(v,\hat{p})$ and constant on $(\hat{p},1]$. Moreover,*

$$\lim_{t\to\hat{p}^-}\mu(v,t)=\frac{v(\xi+1-v)}{(\hat{p}-v)^2}>\lim_{t\to\hat{p}^+}\mu(v,t)=\frac{v^2}{\hat{p}(\hat{p}-v)}.$$

2. *For fixed $t>v$, $\mu(v,t)$ is strictly increasing in v.*

Proposition 4 shows that training scope lies in two distinct regions, corresponding to two different training strategies:

- ***Target-scope training ($t<\hat{p}$):*** the firm trains and targets on the same set of covariates. An increase in t then raises $\tilde{V}(n,t)$ through two channels: it expands target scope, since $\tilde{p}(t)=t$, and it increases weight information $\mathcal{I}(n,t)$. By contrast, an increase in n raises value only through weight information. Training scope therefore has an additional margin of value relative to scale, so the two inputs are imperfect substitutes and $\mu(v,t)$ declines with t.

- ***Expanded-scope training ($t>\hat{p}$):*** the firm collects additional covariates that are used in training but not at the target stage. In this region, an increase in t raises $\tilde{V}(n,t)$ only through weight information, just as an increase in n does. Training scope and scale therefore affect value through the same channel, and the isovalue curve becomes locally linear: the trade-off between the two margins is constant, and $\mu(v)$ is independent of t.

4.1.2 Marginal Cost along Isovalue Curve

Define the marginal cost ratio of training scope and scale along the isovalue curve as

$$\gamma(v,t)\equiv MCR(\tilde{n}(v,t),t)=\frac{\Gamma_t\left(\tilde{n}(v,t),t\right)}{\Gamma_n\left(\tilde{n}(v,t),t\right)},$$

which is the magnitude of the local slope of the isocost curve needed to deliver value v at scope t.

Lemma 2. *The function $\gamma(v,t)$ is continuous, strictly decreasing in v and strictly increasing in t.*
Furthermore, for every fixed $v\in(0,\hat{p})$,

$$\lim_{t\to v^+}\gamma(v,t)\in[0,\infty),\qquad \lim_{t\to 1^-}\gamma(v,t)=+\infty,$$

and for every fixed $t\in(0,1)$,

$$\lim_{v\to 0^+}\gamma(v,t)=+\infty,\qquad \lim_{v\to\min\{t,\hat{p}\}^-}\gamma(v,t)\in[0,\infty).$$

Lemma 2 implies that the relative cost of training scope increases as the firm moves to broader training scope, and decreases as it targets a higher value level.

4.1.3 Net scope savings

Define the ***net (training) scope savings***

$$\phi(v,t) \equiv \mu(v,t) - \gamma(v,t),$$

that is, the difference between the technological gain from expanding training scope and its relative marginal cost. The net scope savings determines the sign of the derivative of reduced cost: using the chain rule and the definition of $\phi(v,t)$,

$$\frac{d\Gamma}{dt}(\tilde{n}(v,t),t) = -\Gamma_n\big(\tilde{n}(v,t),t\big)\phi(v,t) \lesseqgtr 0 \iff \phi(v,t) \gtreqless 0.$$

Thus, $\phi(v,t)$ determines whether increasing training scope lowers or raises the minimum cost of delivering value v. Proposition 4 and Lemma 2 have sdtraighforward implications on the net scope savings $\phi(v,t)$.

Corollary 7. *The net scope savings $\phi(v,t)$ is strictly increasing and continuous in v and strictly decreasing in t, continuous for $t \neq \hat{p}$. Furthermore,*

$$\lim_{t\to v^+} \phi(v,t) = +\infty, \qquad \lim_{t\to 1^-} \phi(v,t) = -\infty, \tag{3}$$

$$\lim_{v\to 0^+} \phi(v,t) = -\infty, \qquad \lim_{v\to \min\{t,\hat{p}\}} \phi(v,t) = +\infty. \tag{4}$$

For every fixed $v \in (0,\hat{p})$, the function $\phi(v,t)$ has a downward jump at $t = \hat{p}$. Let

$$\phi_-(v) \equiv \lim_{t\to\hat{p}^-} \phi(v,t), \qquad \phi_+(v) \equiv \lim_{t\to\hat{p}^+} \phi(v,t).$$

Both $\phi_-(v)$ and $\phi_+(v)$ are strictly increasing in v, and each crosses zero exactly once. This defines two unique thresholds

$$0 < v_- < v_+ < \hat{p}$$

such that

$$\phi_-(v_-) = 0 \qquad \text{and} \qquad \phi_+(v_+) = 0.$$

The *constraint-binding threshold* v_- is the smallest value for which the minimized cost is decreasing in t to the left of $\hat{p}$. The *expanded-scope threshold* v_+ is the smallest value for which the minimized cost is decreasing in t to the right of $\hat{p}$.

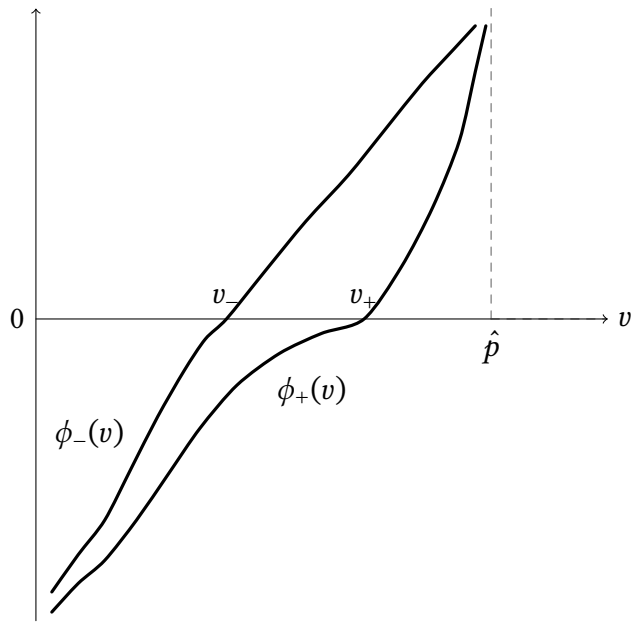

Figure 4: The left- and right-limit net scope savings at the target bound, $\phi_-(v)$ and $\phi_+(v)$. Both functions are strictly increasing, with $\phi_-(v) > \phi_+(v)$ for all $v \in (0, \hat{p})$. Their zero crossings define the thresholds v_- and v_+.

I can now solve for the cost-minimizing design.

Proposition 5 (Cost-minimizing design)**.** *For every target value $v \in (0, \hat{p})$, the reduced cost-minimization problem admits a unique minimizer. The optimal design is given by*

$$p^*(v) = \min\{t^*(v), \hat{p}\}, \qquad n^*(v) = \tilde{n}(v, t^*(v)),$$

where the optimal training scope satisfies

$$t^*(v) \in \begin{cases} \{t \in (v, \hat{p}) \,:\, \phi(v,t) = 0\}, & \text{if } 0 < v < v_-, \\ \{\hat{p}\}, & \text{if } v_- \le v \le v_+, \\ \{t \in (\hat{p}, 1) \,:\, \phi(v,t) = 0\}, & \text{if } v_+ < v < \hat{p}. \end{cases}$$

Figure 5 illustrates the sign pattern of $\phi(v, t)$ and the resulting optimizer $t^*(v)$. The two curved branches coincide with the zero set of $\phi(v, t)$, while the flat segment at $t = \hat{p}$ corresponds to the intermediate region in which the optimum is attained at the kink created by the downward jump of ϕ.

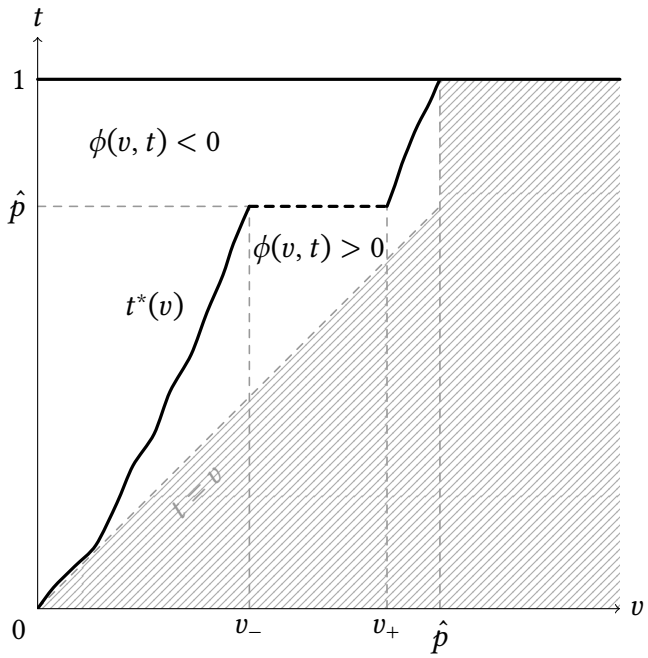

Figure 5: Phase diagram in the (v, t)-plane. The solid locus is the set of points at which $\phi(v, t) = 0$, while the dashed horizontal segment at $t = \hat{p}$ is the kink region in which the cost-minimizing training scope satisfies $t^*(v) = \hat{p}$. The hatched regions are infeasible.

Naturally, the conditional training scope $t^*(v)$ is strictly increasing for $v < v_-$ and $v > v_+$ and constant for $v \in [v_-, v_+]$. The conditional target scope $p^*(v)$ is strictly increasing for $v < v_-$ and constant $v > v_+$ and constant otherwise.

The comparative statics of $n^*(v)$ combine a positive direct effect and a negative substitution effect.

$$\frac{dn^*(v)}{dv} = \underbrace{\frac{\partial \tilde{n}(v, t^*(v))}{\partial v}}_{>0:\,\text{direct effect}} + \underbrace{\frac{\partial \tilde{n}(v, t^*(v))}{\partial t}\frac{dt^*(v)}{dv}}_{\leq 0:\,\text{substitution effect}}.$$

Holding training scope fixed, a higher value target requires more observations because $\tilde{V}(n, t)$ is increasing in n. But outside the kink region, a higher v also induces the firm to expand training scope, and this relaxes the need for additional scale. Algebraic manipulation shows that

$$\frac{dn^*(v)}{dv} < 0 \iff \frac{dt^*(v)}{dv} > \frac{1}{\tilde{V}_t(n^*(v), t^*(v))},$$

so $n^*(v)$ decreases if and only if the increase in training scope generates more than the entire extra unit of value required by the rise in v. Intuitively, as v increases, the firm may expand training covariates so aggressively that it can economize on observations.

To summarize, there are three regions:

1. **Unconstrained Region ($v < v_-$).** Training and target scope are equivalent and strictly increasing in v. The scale $n^*(v)$ increases at first but may subsequently decrease.

2. **(Constrained) Same-scope Region ($v_- \leq v \leq v_+$).** Training and target scope are constant and equivalent to $\hat{p}$. The scale $n^*(v)$ is strictly increasing and strictly convex in v.

3. **Expanded-scope Region ($v > v_+$).** The target constraint binds and training scope is larger than target scope. The scale $n^*(v)$ may decrease but will eventually increase.

4.2 Cost of Prediction and Minimum Efficient Scale

The previous subsection characterized the least-cost dataset design associated with each target value v. I now use that design to derive the cost schedule of predictive value itself. This makes it possible to study marginal cost, average cost, and the minimum efficient scale of prediction.

Corollary 8 (Expenditure function and marginal cost)**.** *For every $v \in (0, \hat{p})$, the expenditure function is*

$$\tilde{C}(v) = \Gamma(n^*(v), t^*(v)).$$

Moreover, for every $v \in (0, \hat{p}) \setminus \{v_-, v_+\}$,

$$MC(v) \equiv \tilde{C}'(v) = \frac{\Gamma_n}{\tilde{V}_n}(n^*(v), t^*(v)).$$

Its derivative satisfies

$$\frac{MC'(v)}{MC(v)} = \left(\frac{\Gamma_{nn}}{\Gamma_n} - \frac{\tilde{V}_{nn}}{\tilde{V}_n}\right)\frac{dn^*(v)}{dv} + \left(\frac{\Gamma_{nt}}{\Gamma_n} - \frac{\tilde{V}_{nt}}{\tilde{V}_n}\right)\frac{dt^*(v)}{dv}.$$

Hence, for $v \in (0, v_-) \cup (v_+, \hat{p})$,

$$MC'(v) \leq 0 \iff \frac{\partial \tilde{n}(v, t^*(v))}{\partial t}\frac{dt^*(v)}{dv} \leq \frac{dt^*(v)}{dv}\frac{\frac{\tilde{V}_{nt}}{\tilde{V}_n} - \frac{\Gamma_{nt}}{\Gamma_n}}{\frac{\Gamma_{nn}}{\Gamma_n} - \frac{\tilde{V}_{nn}}{\tilde{V}_n}} - \frac{\partial \tilde{n}(v, t^*(v))}{\partial v}.$$

Finally, for $v \in (v_-, v_+)$,

$$MC'(v) > 0.$$

Marginal cost equals the marginal cost of scale, Γ_n, divided by the marginal contribution of scale to predictive value, $\tilde{V}_n$. It therefore decreases only if, along the optimal design path, the marginal productivity of additional observations rises faster than their marginal cost. In the kink region, training scope is fixed at $t^*(v) = \hat{p}$, so this scope-saving channel is absent; only diminishing returns in scale remain, implying $MC'(v) > 0$.

The derivative of marginal cost reflects two opposing forces. On the one hand, higher target value makes the isovalue curve steeper in the scale dimension and may also move the firm

into regions where observations are more expensive at the margin, both of which increase $MC'(v)$. On the other hand, a higher target value induces the firm to expand training scope, which reduces the scale required to generate an additional unit of value and therefore dampens the rise in marginal cost. The overall sign of $MC'(v)$ is therefore generally ambiguous under a generic cost function.

4.3 Illustration

To illustrate the previous results in a transparent benchmark, I now specialize the training cost function to the linear form

$$\Gamma(n, t) = n + \gamma t,$$

where one unit of training scale costs one, and $\gamma > 0$ is the constant marginal cost of training scope relative to scale. Under this specification, isocost curves are linear and the relative marginal cost of training scope is constant:

$$\gamma(v, t) = \gamma.$$

This removes any variation in the cost-side trade-off across designs, so the structure of the optimal dataset is driven entirely by the geometry of the isovalue curves. As a result, the thresholds v_- and v_+, and the associated regions of same-scope and expanded-scope training, can be characterized in closed form.

Proposition 6 (Cost-minimizing design)**.** *The cost-minimization problem admits a solution. The cost-minimizing design is* $p^*(v) = \min\{t^*(v), \hat{p}\}$, $n^*(v) = \tilde{n}(v, t^*(v))$ *and*

$$t^*(v) \in \begin{cases} \left\{ v + \sqrt{\dfrac{v(\xi + 1 - v)}{\gamma}} \right\}, & \text{if } 0 \le v \le v_-(\hat{p}, \gamma, \xi), \\ \left\{ \hat{p} \right\}, & \text{if } v_-(\hat{p}, \gamma, \xi) \le v < v_+(\hat{p}, \gamma), \\ [\hat{p}, 1], & \text{if } v = v_+(\hat{p}, \gamma), \\ \{1\}, & \text{if } v_+(\hat{p}, \gamma) < v < \hat{p}. \end{cases}$$

The result gives rise to three regions, illustrated in Figure 6.

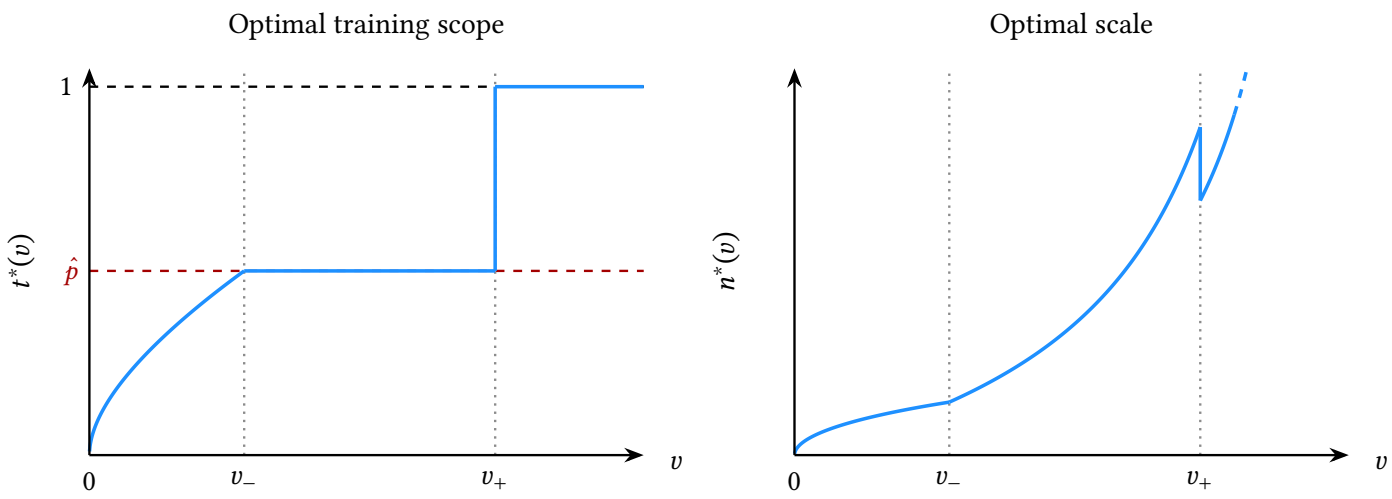

Figure 6: Cost-minimizing training scope $t^*(v)$ and scale $n^*(v)$ as functions of the value v, for fixed $(\hat{p}, \gamma, \xi) = (0.5, 2, 2)$.

At $v = v_+(\hat{p}, \gamma)$, the optimal design switches from same-scope training, $t^* = p^* = \hat{p}$, to expanded-scope training, $t^* = 1 > p^* = \hat{p}$. For $v < v_+$, the firm trains only on covariates available for the target individual when prediction occurs. For $v > v_+$, it becomes optimal to invest in a richer offline training pipeline using covariates that are unavailable, too costly, or too delayed to be deployed at the time of prediction. Since $MRTS_{tn}(v, t)$ is constant for $t > \hat{p}$, the adoption of this training technology is bang-bang, so the firm jumps directly to $t^* = 1$. The resulting increase in training efficiency allows it to reduce scale discontinuously, generating a downward jump in $n^*(v)$.

4.4 Cost of Prediction and Minimum Efficient Scale

The previous subsection characterized the least-cost dataset design associated with each target value v. I now use that design to derive the cost schedule of predictive value itself. This makes it possible to study marginal cost, average cost, and the minimum efficient scale of prediction.

Corollary 9 (Cost of predictive value)**.** *For every target $v \in (0, \hat{p})$,*

$$\tilde{C}(v) = \begin{cases} \gamma v + 2\sqrt{\gamma v(\xi + 1 - v)}, & \text{if } v \in (0, v_-(\hat{p}, \gamma, \xi)), \\ \gamma \hat{p} + \dfrac{v(\xi + 1 - v)}{\hat{p} - v}, & \text{if } v \in [v_-(\hat{p}, \gamma, \xi), v_+(\hat{p}, \gamma)], \\ \gamma + v\left(\dfrac{\xi}{\hat{p} - v} + \dfrac{1}{\hat{p}}\right), & \text{if } v \in (v_+(\hat{p}, \gamma), \hat{p}). \end{cases}$$

The cost function $\tilde{C}(v)$ has two kinks: the first, at $v_-(\hat{p}, \gamma, \xi)$, occurs when the target constraint

$\hat{p}$ starts to bind; the second, at $v_+(\hat{p}, \gamma)$, when the firm switches to expanded-scope training. Define marginal cost as

$$MC(v) \equiv \tilde{C}'(v),$$

whenever $\tilde{C}$ is differentiable.

Corollary 10 (Marginal Cost)**.** *$MC(v)$ is strictly decreasing on $v \in (0, v_-(\hat{p}, \gamma, \xi))$, strictly increasing on $(v_-(\hat{p}, \gamma, \xi), v_+(\hat{p}, \gamma))$ and $(v_+(\hat{p}, \gamma), \hat{p})$, and falls discontinuously at $v_+(\hat{p}, \gamma)$.*

These properties reflect the three data-design regimes studied above.

- In the unconstrained region $(0, v_-(\hat{p}, \gamma, \xi))$ the firm expands training and target scope jointly. Because of the economies of scope in t of Corollary 4 and the complementarities between t, p and n of Corollary 5, producing higher value comes at a lower marginal cost.
- In the constrained same-scope region $[v_-(\hat{p}, \gamma, \xi), v_+(\hat{p}, \gamma))$ the firm stops expanding the scope, and further expansions are achieved by increasing scale: since there are diminishing returns to scale by Corollary 3, the marginal cost is increasing.
- At $v_+(\hat{p}, \gamma)$ the firm increases its training scope to $t^* = 1$ and cuts back its scale, reducing marginal cost discontinuously, since there are diminishing returns to scale.
- In the expanded-scope region $(v_+(\hat{p}, \gamma), \hat{p})$, further expansions are achieved by increasing scale, which again results in increasing marginal cost.

Define the average cost per unit of relative value as

$$AC(v) \equiv \frac{\tilde{C}(v)}{v}.$$

A minimum efficient scale (MES), when it exists, is any

$$v_{MES} \in \arg\min_{v \in (0, \hat{p})} AC(v; \hat{p}).$$

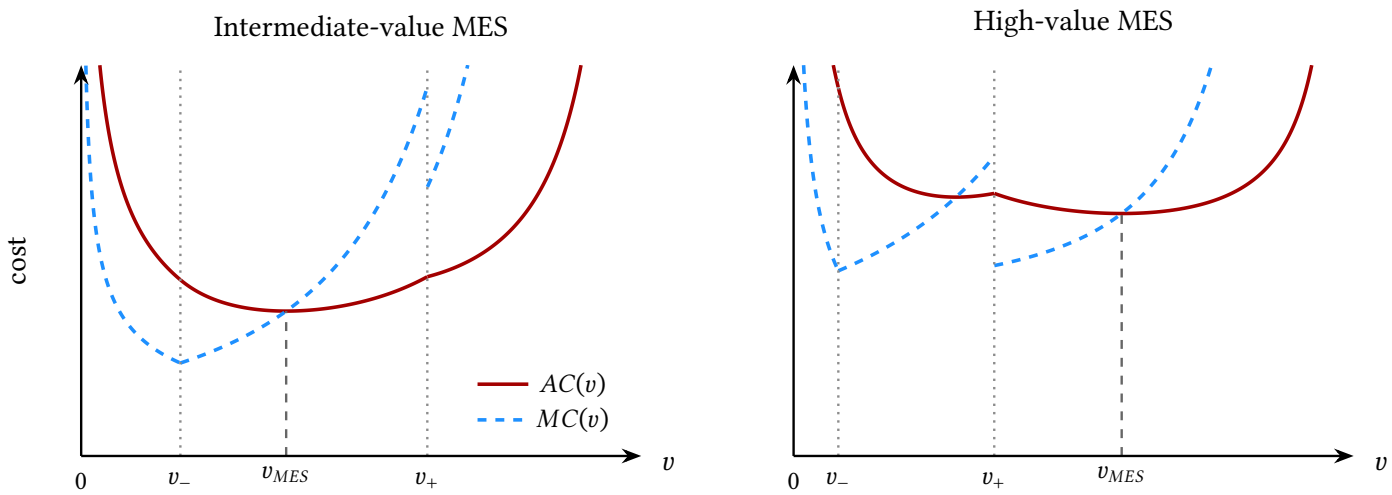

Figure 7: Average and marginal cost as functions of target value v. The left panel illustrates a case in which the minimum efficient scale v_{MES} lies in the same-scope region $[v_-, v_+)$, while the right panel illustrates a case in which v_{MES} lies in the expanded-scope region $[v_+, \hat{p})$.

Proposition 7 (Minimum efficient scale). *A minimum efficient scale exists and cannot lie in the unconstrained-scope region* $(0, v_-)$. *Moreover*

- *If* $\xi \geq 1$ *or* $\gamma \geq (1-\xi)^2/\xi$, *the minimum efficient scale is in the same-scope region:*

$$v_{MES}(\hat{p}, \gamma, \xi) \in [v_-(\hat{p}, \gamma, \xi), v_+(\hat{p}, \gamma)).$$

- *Otherwise, let*

$$p^\dagger(\gamma, \xi) = \frac{\gamma + 4\sqrt{\xi}\left(\sqrt{\gamma} + \sqrt{\xi}\right)}{\gamma + 4} \in [0, 1).$$

The minimum efficient scale is

$$v_{MES}(\hat{p}, \gamma, \xi) \in \begin{cases} [v_-(\hat{p}, \gamma, \xi), v_+(\hat{p}, \gamma)), & \text{if } \hat{p} \leq p^\dagger(\gamma, \xi), \\ (v_+(\hat{p}, \gamma), \hat{p}), & \text{if } \hat{p} > p^\dagger(\gamma, \xi). \end{cases}$$

Proposition 7 implies that an entrant must therefore operate at a scale at which the target constraint binds, so that $p^* = \hat{p}$.

The proposition also identifies two distinct entry technologies. If the target constraint is tight, i.e. $\hat{p} \leq p^\dagger(\gamma, \xi)$, then the MES lies in the same-scope region, and efficient entry occurs with scope -constrained training: $t^* = p^* = \hat{p}$. By contrast, if the target constraint is lax, i.e. $\hat{p} > p^\dagger(\gamma, \xi)$, then the MES lies in the expanded-scope region, and efficient entry requires

expanded-scope training: $t^* = 1 > p^* = \hat{p}$. Since $p^\dagger(\gamma, \xi)$ is increasing in both arguments on the relevant domain, higher noise ξ and a higher relative price of training covariates γ enlarge the set of target constraints $\hat{p}$ for which efficient entry occurs in the same-scope region.

4.5 Profit Maximization and Entry

Having characterized the expenditure function $\tilde{C}(v)$, I now return to the firm's original problem. Since σ_β^2 is the exogenous price of predictive value, the firm chooses the target value v to maximize revenue net of the expenditure required to attain it:

$$\Pi(\sigma_\beta^2) \equiv \max_{v \in [0, \hat{p})} \left\{ \sigma_\beta^2 v - \tilde{C}(v) \right\}.$$

Prediction is profitable if and only if the price of predictive value exceeds minimum average cost:

$$\sigma_\beta^2 \geq \min_{v \in (0, \hat{p})} AC(v) = AC(v_{MES}).$$

By Proposition 7, the low-value region is never cost-minimizing. It is therefore sufficient to compare the same-scope and expanded-scope branches of the expenditure function.
Let

$$\underline{\sigma}_\beta^2(\hat{p}) \equiv \min_{v \in [v_-(\hat{p}, \gamma, \xi), v_+(\hat{p}, \gamma))} AC(v) = 1 + \frac{\left(\sqrt{\gamma \hat{p}} + \sqrt{\xi + 1 - \hat{p}} \right)^2}{\hat{p}}$$

denote the lowest price at which prediction is profitable on the same-scope branch, and let

$$\bar{\sigma}_\beta^2(\hat{p}) \equiv \min_{v \in [v_+(\hat{p}, \gamma), \hat{p})} AC(v) = \frac{1 + \left(\sqrt{\gamma} + \sqrt{\xi} \right)^2}{\hat{p}}$$

denote the lowest price at which prediction is profitable on the expanded-scope branch.
When both branches are profitable, define $\tilde{\sigma}_\beta^2$ implicitly by

$$\Pi_M(\tilde{\sigma}_\beta^2) = \Pi_H(\tilde{\sigma}_\beta^2),$$

where Π_M and Π_H are the indirect profit functions associated with the same-scope and expanded-scope branches. Thus, $\tilde{\sigma}_\beta^2$ is the price at which the firm is indifferent between the two technologies.

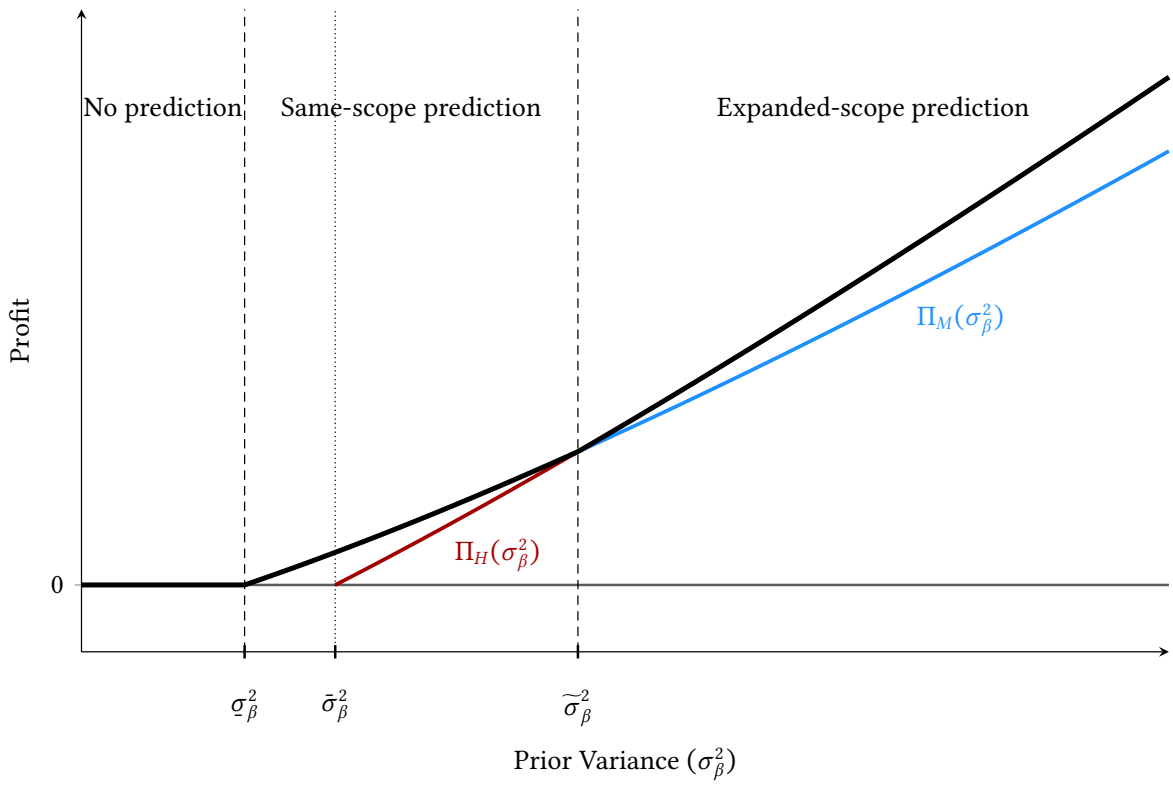

Figure 8: Profit as a function of the price of predictive value, σ_β^2, for $\hat{p} \leq p^\dagger(\gamma, \xi)$. The blue and red curves give profit on the same-scope and expanded-scope branches, and the thick black curve is the upper envelope. At $\underline{\sigma}_\beta^2$, prediction becomes profitable; at $\bar{\sigma}_\beta^2$, the expanded-scope branch becomes profitable; and at $\tilde{\sigma}_\beta^2$, the firm switches from the same-scope to the expanded-scope technology.

Proposition 8 (Profit-maximizing relative value). *Fix* $(\gamma, \hat{p}, \xi)$.
If $\hat{p} > p^\dagger(\gamma, \xi)$, *then the unique global maximizer is*

$$v^*(\sigma_\beta^2) = \begin{cases} 0, & \text{if } \sigma_\beta^2 < \bar{\sigma}_\beta^2, \\ \hat{p} - \sqrt{\dfrac{\xi \hat{p}}{\sigma_\beta^2 - 1/\hat{p}}}, & \text{if } \sigma_\beta^2 \geq \bar{\sigma}_\beta^2. \end{cases}$$

If $\hat{p} \leq p^\dagger(\gamma, \xi)$, *then the unique global maximizer is*

$$v^*(\sigma_\beta^2) = \begin{cases} 0, & \text{if } \sigma_\beta^2 < \underline{\sigma}_\beta^2, \\ \hat{p} - \sqrt{\dfrac{\hat{p}(\xi + 1 - \hat{p})}{\sigma_\beta^2 - 1}}, & \text{if } \underline{\sigma}_\beta^2 \leq \sigma_\beta^2 < \tilde{\sigma}_\beta^2, \\ \hat{p} - \sqrt{\dfrac{\xi \hat{p}}{\sigma_\beta^2 - 1/\hat{p}}}, & \text{if } \sigma_\beta^2 > \tilde{\sigma}_\beta^2. \end{cases}$$

At $\sigma_\beta^2 = \tilde{\sigma}_\beta^2$, both branch maximizers are optimal.

The following figure illustrates the three regions.

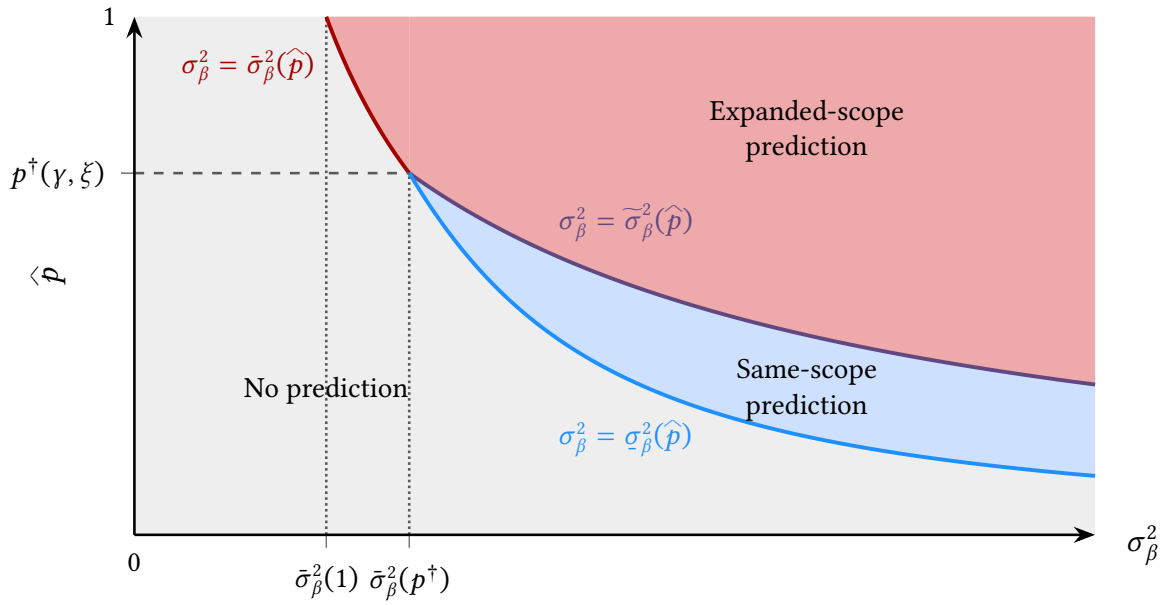

Figure 9: Prediction regions in the $(\sigma_\beta^2, \hat{p})$ plane. Prediction is not profitable in the gray region, follows the same-scope design in the blue region, and follows the expanded-scope design in the red region.

Since the prediction region expands in σ_β^2, prediction is profitable only for target variables with sufficiently high ex ante uncertainty. Furthermore, as it expands also in $\hat{p}$ laxer target constraints allow prediction to be profitable on target variables with lower prior uncertainty. If target constraints are lax, noise is low and training covariates are relatively cheap, i.e. $\hat{p} > p^\dagger(\gamma, \xi)$, whenever entry is profitable, expanded-scope is always preferable to same-scope prediction.

By contrast, if target constraints are tight, noise is low or training covariates are relatively expensive, i.e. $\hat{p} \leq p^\dagger(\gamma, \xi)$, the same-scope prediction is profitable for intermediate values of $\hat{p}$: laxer target constraints will incentivize firms to shift from same-scope to expanded-scope production, i.e. to shift from same- to expanded-scope training. Furthermore, expanded-scope prediction is more prevalent when the target variable has high prior uncertainty σ_β^2.

5 Shoppers, Price Discrimination, and Data Collection

Suppose now that two firms, E_1 and E_2, compete to sell predictions to users. Relative to the Hotelling formulation, this alternative replaces horizontal differentiation with a distinction between captive and contestable demand.

5.1 Setup

Firms. Each firm $i \in \{1, 2\}$ chooses a dataset design in period 1. By Proposition 6, this is equivalent to choosing a relative predictive value

$$v_i \in [0, \hat{p})$$

at cost $\tilde{C}(v_i)$, where $\tilde{C}$ is the expenditure function characterized in Corollary 9. Conditional on v_i, the firm implements the cost-minimizing margins (n_i^*, t_i^*, p_i^*) described in Proposition 6.

Users. There is a unit mass of users. A fraction $\mu \in [0, 1]$ are shoppers, and a fraction $1 - \mu$ are captive. Captive users are split evenly across firms, so each firm has a mass $(1 - \mu)/2$ of captive users. A captive user attached to firm i buys only from firm i and obtains utility

$$U_i^C = \sigma_\beta^2 v_i - P_i^C.$$

A shopper compares the two firms and obtains utility

$$U_i^S = \sigma_\beta^2 v_i - P_i^S$$

from buying from firm i.
It is useful to define the quality gap

$$\Delta(v_1, v_2) \equiv \sigma_\beta^2 (v_1 - v_2).$$

Timing and solution concept. The game occurs in two periods:

- Period 1 (dataset investment): firm i chooses v_i paying $\tilde{C}(v_i)$.
- Period 2 (price competition): firm i observes v_j and chooses P_i^S and P_i^C; payoffs are realized.

I characterize subgame perfect Nash equilibria in pure strategies with trembling-hand refinement when necessary.

5.2 Price competition

Fix $(v_1, v_2) \in [0, \hat{p})^2$. Price competition is standard and separates across the two demand segments.
On the captive segment, each firm is a monopolist over its attached users, so the equilibrium is unique and each firm extracts the users' entire willingness to pay:

$$P_i^{C*} = \sigma_\beta^2 v_i, \qquad R_i^C(v_i) = \frac{1 - \mu}{2} \sigma_\beta^2 v_i.$$

On the shopper segment, the relevant outcome is the trembling-hand-perfect Bertrand equilibrium. If one firm offers strictly higher predictive value, say $v_i > v_j$, that firm serves the entire shopper mass and charges the quality advantage over its rival:

$$P_i^{S*} = \sigma_\beta^2(v_i - v_j), \qquad P_j^{S*} = 0,$$

so that

$$R_i^S(v_i, v_j) = \mu\sigma_\beta^2(v_i - v_j), \qquad R_j^S(v_j, v_i) = 0.$$

If $v_1 = v_2$, price competition on the shopper segment yields zero profits for both firms:

$$(R_1^S, R_2^S) = (0, 0).$$

5.3 Dataset Investment

Using the results in the previous section, firm i's period-1 payoff is

$$\Pi_i(v_i, v_j) = \frac{1-\mu}{2}\sigma_\beta^2 v_i + \mu\sigma_\beta^2[v_i - v_j]_+ - \tilde{C}(v_i), \tag{5}$$

where $[x]_+ \equiv \max\{x, 0\}$.
It is convenient to define the follower and leader shadow values of predictive quality:

$$\rho_F \equiv \frac{1-\mu}{2}\sigma_\beta^2, \qquad \rho_L \equiv \frac{1+\mu}{2}\sigma_\beta^2.$$

5.3.1 Asymmetric Equilibria

Proposition 9 (No symmetric pure equilibrium)**.** *Suppose $\mu > 0$. Let $\tilde{C}$ be continuous on $[0, \hat{p})$, differentiable on $(0, \hat{p}) \setminus \{v_+\}$, and admit one-sided derivatives at v_+. If*

$$\tilde{C}'_-(v_+) > \tilde{C}'_+(v_+),$$

then there is no symmetric pure-strategy equilibrium with

$$v_1 = v_2 \in (0, \hat{p}).$$

Intuitively, symmetric equilibria give zero-payoffs so both firms have an incentive to deviate to lower v to save marginally on prediction cost.

Proposition 10 (Asymmetric pure equilibria)**.** *Define*

$$v_F \in \arg\max_{v \in [0, \hat{p})} \left\{ \rho_F v - \tilde{C}(v) \right\},$$

and

$$v_L \in \arg\max_{v \in [0,\hat{p})} \left\{ \rho_L v - \tilde{C}(v) \right\},$$

which are unique by Proposition 8 and such that

$$v_L > v_F.$$

Then the period-1 game has exactly two pure-strategy Nash equilibria:

$$(v_1^*, v_2^*) = (v_L, v_F) \qquad \textit{and} \qquad (v_1^*, v_2^*) = (v_F, v_L).$$

The two equilibria involve one firm capturing all the shoppers and the other capturing only its captive consumers.

5.3.2 Prediction Technology

From the cost characterization in the previous section, an interior candidate must lie on one of the two relevant technology branches.

The first is the *same-scope* branch, on which the firm does not use privileged training covariates beyond those used at prediction. On that branch, the candidate value for a firm of type $r \in \{F, L\}$ is

$$v_M^r = \hat{p} - \sqrt{\frac{\hat{p}(\xi + 1 - \hat{p})}{\rho_r - 1}},$$

provided the branch is active, that is, provided the relevant marginal-return condition holds and the solution lies in the admissible interval $[v_-, v_+)$.

The second is the *expanded-scope* branch, on which the firm trains on a broader set of covariates than it observes at prediction. On that branch, the candidate value is

$$v_H^r = \hat{p} - \sqrt{\frac{\xi \hat{p}}{\rho_r - 1/\hat{p}}},$$

again provided the corresponding branch condition holds and the solution lies in the admissible interval $(v_+, \hat{p})$.

Since the relevant marginal returns are $\rho_L > \rho_F$, the leader has a stronger incentive to invest in predictive quality than the follower.

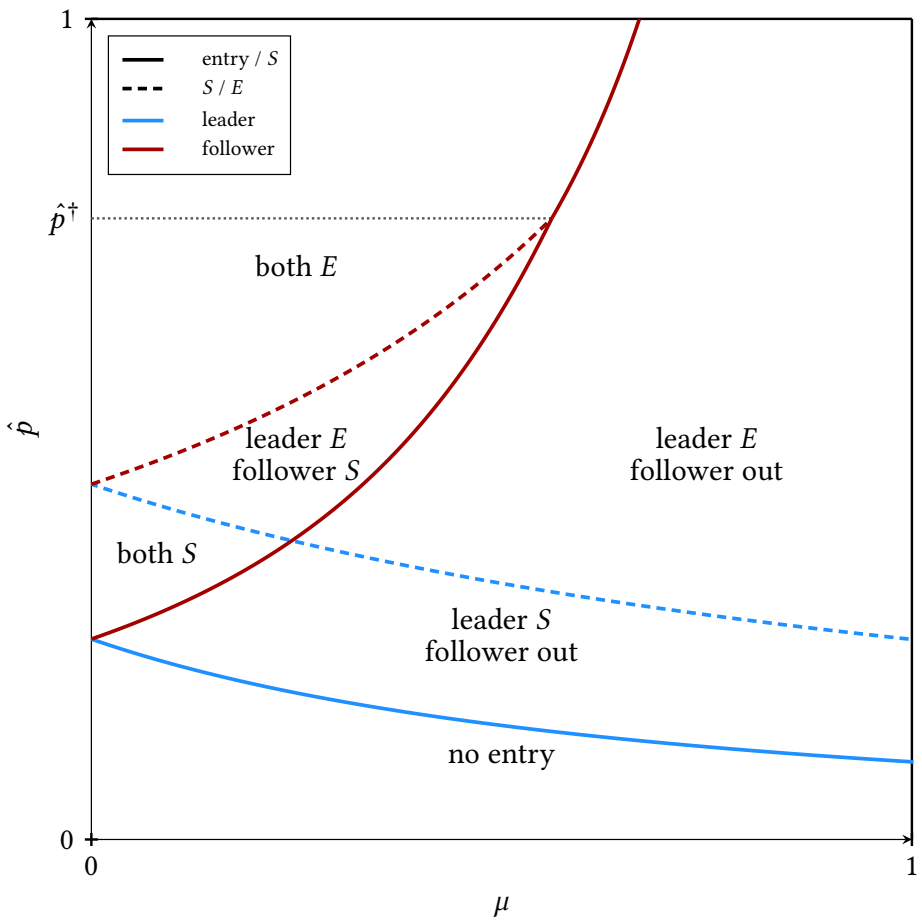

Figure 10: Equilibrium technology regimes in the $(\mu, \hat{p})$ plane. The blue curves correspond to the leader and the red curves to the follower. Solid curves mark the entry thresholds, while dashed curves mark the switch from same-scope training (M) to expanded-scope training (H). The horizontal line at $\hat{p}^\dagger$ separates the region in which both technologies may arise from the region in which only expanded-scope training is viable.

Figure 10 summarizes the equilibrium regime map in the $(\mu, \hat{p})$ plane. The solid curves are the entry thresholds and the dashed curves are the technology-upgrade thresholds from same-scope training (S) to expanded-scope training (E); blue refers to the leader and red to the follower. As the shopper share μ rises, the return to quality increases for the leader and decreases for the follower. Accordingly, the leader's thresholds move downward while the follower's move upward. This generates a monotone transition from similar to distinct technologies: for low $\hat{p}$, the market moves from no entry to leader-only entry and then to both firms using S; for intermediate $\hat{p}$, the leader switches to E before the follower, creating a region in which the leader uses expanded-scope training while the follower remains on same-scope training; and for larger μ, the follower eventually exits while the leader remains active. The horizontal line at $\hat{p}^\dagger$ marks the boundary above which same-scope training ceases to be sustainable, so that the only active technology is E. The figure therefore makes clear that a larger shopper seg-

ment amplifies technology asymmetries: it advances both entry and upgrading for the leader, and delays both for the follower.

6 Data Duopsony

Whereas the previous section studied competition to sell prediction, the current section will focuse on competition for data inputs. We study the case of two ecosystems that compete to acquire users, which produce observations, and apps, which produce covariates. For simplicity we assume ecosystem do not compete in the sale of predictions. Furthermore, for tractability we assume that firms adopt the same-covariate technology (Section 4 has shown this can be rationalized with intermediate σ_β^2). The equilibrium allocation of inputs determines dataset designs (n_i, k_i) and thus prediction quality $\mathcal{V}(n_i, k_i)$.
The model reflects a three-sided platform which must attract users and third-party apps to generate data with which to make predictions which is monetized in a reduced form. It can represent Apple and Android app-stores which generate revenue by selling predictions on which advertisement a user will click on, and compete to attract third-party apps and users which generate data that makes their predictions more accurate.

6.1 Setup

Two ecosystems, E_1 and E_2, compete for users and apps whose data they use to make predictions. Users generate observations, while apps generate covariates. The equilibrium allocation of users and apps determines, for each ecosystem $i \in \{1, 2\}$, a dataset design (n_i, k_i).

Apps. There is a unit mass of apps, which can join either ecosystem. Each ecosystem E_i offers a transfer $a_i \geq 0$ to third-party apps, whose outside option is normalized to zero. Third-party apps single-home. Let $s_i(a_i, a_j) \in [0, 1]$ denote the share of third-party apps joining ecosystem E_i. The total app mass attached to E_i is therefore

$$m_i(a_i, a_j) \equiv \frac{1-\lambda}{2} + \lambda s_i(a_i, a_j),$$

and the corresponding covariate mass is

$$k_i(a_i, a_j) \equiv \kappa\, m_i(a_i, a_j),$$

where each app generates a mass $\kappa \in [0, 1]$ of covariates.

Users. There is a unit mass of users. Each user is characterized by a type $\theta \sim \mathcal{U}([0, 1])$, which captures her relative preference for E_1 over E_2. Ecosystem E_i offers utility $u_i \geq 0$ in the

form of services. A user of type θ obtains payoff

$$U_{1\theta}(u_1, u_2) = u_1 - \tau\theta, \qquad U_{2\theta}(u_1, u_2) = u_2 - \tau(1-\theta),$$

where $\tau > 0$ measures horizontal differentiation between ecosystems.
Users single-home and there is full coverage. Let $q_i(u_i, u_j) \in [0, 1]$ denote the share of users joining ecosystem E_i, so that

$$q_1(u_1, u_2) + q_2(u_1, u_2) = 1.$$

Each user generates a mass $\nu \geq 0$ of observations, so the observation mass collected by E_i is

$$n_i(u_i, u_j) \equiv \nu\ q_i(u_i, u_j).$$

Ecosystems. Given a dataset design (n_i, k_i), ecosystem E_i obtains value $\mathcal{V}(n_i, k_i)$, where

$$\mathcal{V}(n, k) = \frac{n + 1 - \sqrt{(n+1)^2 - 4nk}}{2}.$$

I assume that the same covariates are used for training and prediction, so that a design is summarized by (n_i, k_i).
Ecosystem E_i's profit is

$$\Pi_i(a_i, a_j, u_i, u_j) = \mathcal{V}\big(n_i(u_i, u_j), k_i(a_i, a_j)\big) - u_i q_i(u_i, u_j) - \lambda a_i s_i(a_i, a_j).$$

Timing. In Stage 1, ecosystems simultaneously choose transfers (a_1, a_2) to third-party apps. This determines (s_1, s_2) and hence (m_1, m_2) and (k_1, k_2). In Stage 2, taking (k_1, k_2) as given, ecosystems simultaneously choose utilities (u_1, u_2). Users then choose which ecosystem to join, determining (q_1, q_2) and hence (n_1, n_2).

6.2 Competition for Users

Fix an app allocation inherited from Stage 1. Let $m \in [0, 1]$ denote the mass of apps attached to E_1, so that $1 - m$ is the mass of apps attached to E_2. Hence the covariate masses available to the two ecosystems are

$$k_1 = \kappa m, \qquad k_2 = \kappa(1 - m).$$

In Stage 2, ecosystems simultaneously choose utilities (u_1, u_2). Users single-home and there

is full coverage. Therefore, given (u_i, u_j), user demand for ecosystem E_i is

$$q_i(u_i, u_j) = \begin{cases} 1 & \text{if } u_i > u_j + \tau, \\ \dfrac{1}{2} + \dfrac{u_i - u_j}{2\tau} & \text{if } |u_i - u_j| < \tau, \\ 0 & \text{if } u_i < u_j - \tau. \end{cases}$$

Given m, ecosystem E_i solves

$$\max_{u_i \geq 0} \left\{ \mathcal{V}\big(n_i(u_i, u_j), k_i\big) - u_i q_i(u_i, u_j) \right\},$$

where $n_i(u_i, u_j) \equiv \nu\, q_i(u_i, u_j)$.

Assumption 2. *Let us assume that*

$$\nu\, \mathcal{V}_n(\nu, \kappa) > 3\tau.$$

Intuitively, Assumption 2 states that the user differentiation is small enough that if one ecosystem has access to the entire app stock, so that its covariate mass is κ while the rival has none, then the app-rich ecosystem finds it profitable to attract all users. Equivalently, users are not so differentiated as to allow the zero-app rival cannot profitably sustain a positive user share in the most unequal app allocation possible.[16]

Proposition 11 (User subgame equilibrium)**.** *Under Assumption 2, there exists a unique threshold, $\tilde{m}(\nu, \kappa, \tau) \in [0, 1/2]$, such that*

$$\nu\, (\mathcal{V}_n(\nu, \kappa(1 - \tilde{m})) - \mathcal{V}_n(0, \kappa\tilde{m})) = 3\tau.$$

For every app allocation $m \in [0, 1]$, the user subgame admits a unique trembling-hand perfect equilibrium. The equilibrium user shares are

$$\hat{q}_1(m) = \begin{cases} 0, & \text{if } m \in [0, \tilde{m}], \\ \hat{\theta}(m), & \text{if } m \in (\tilde{m}, 1 - \tilde{m}), \\ 1, & \text{if } m \in [1 - \tilde{m}, 1], \end{cases} \qquad \hat{q}_2(m) = 1 - \hat{q}_1(m),$$

where, for $m \in (\tilde{m}, 1 - \tilde{m})$, the scalar $\hat{\theta}(m) \in (0, 1)$ is the unique solution to

$$\nu \left(\mathcal{V}_n(\nu\, \hat{\theta}(m), \kappa m) - \mathcal{V}_n(\nu(1 - \hat{\theta}(m)), \kappa(1 - m)) \right) = 6\tau \left(\hat{\theta}(m) - \frac{1}{2} \right).$$

[16]Because Stage 2 is solved for a general fixed app allocation, this assumption is independent of how the aggregate app stock is decomposed between proprietary and third-party apps, and therefore imposes no restriction on λ.

Equivalently, the equilibrium observation masses are

$$(\hat{n}_1(m), \hat{n}_2(m)) = (\nu\,\hat{q}_1(m), \nu\,\hat{q}_2(m)),$$

and the equilibrium utilities are

$$(\hat{u}_1(m), \hat{u}_2(m)) = \begin{cases} \big(\nu\,\kappa m, \nu\,\kappa m + \tau\big), & \text{if } m \in [0, \tilde{m}], \\ (\nu\,\mathcal{V}_n(\nu\,\hat{q}_1(m), \kappa m) - 2\tau\,\hat{q}_1(m), \nu\,\mathcal{V}_n(\nu\,\hat{q}_2(m), \kappa(1-m)) - 2\tau\,\hat{q}_2(m)), & \text{if } m \in (\tilde{m}, 1-\tilde{m}), \\ \big(\nu\,\kappa(1-m) + \tau, \nu\,\kappa(1-m)\big), & \text{if } m \in [1-\tilde{m}, 1]. \end{cases}$$

The equilibrium configuration then depends on the app allocation inherited from Stage 1. If $m < \tilde{m}$, E_1 holds relatively few covariates, so the value of attracting its first user is below the value for E_2 of retaining its last user. Hence, E_1 cannot profitably compete for users, and the market tips to E_2. The equilibrium utility is then the lowest utility that still deters E_1 from entering the user market, namely the marginal value of a first user for E_1, equal to κm. Symmetrically, if $m > 1 - \tilde{m}$, the market tips to E_1. For intermediate app allocations, by contrast, both ecosystems hold enough covariates to compete for users. The equilibrium is then interior. Because both ecosystems have diminishing marginal returns to observations, and observations are proportional to the mass of users, users split across ecosystems until the difference in marginal value of an additional observation for the two platforms is equal to the marginal transportation cost the marginal user faces to change ecosystem. The equilibrium utility $\hat{u}_1(m)$ offered by E_i is the marginal value minus a markdown

$$2\tau\,\hat{q}_i.$$

Intuitively, the larger τ the larger the oligopsony power, and the larger the amount of users $\hat{q}_i$ the more E_i has an incentive to distort downward the compensation of the marginal user to reduce the transportation rent of inframarginal users.

6.3 Competition for Apps

For any total app mass $m \in \left[\frac{1-\lambda}{2}, \frac{1+\lambda}{2}\right]$ attached to E_1, let

$$\mathcal{W}_1(m) \equiv \mathcal{V}\big(\nu\,\hat{q}_1(m), \kappa m\big) - \hat{u}_1(m)\hat{q}_1(m), \tag{6}$$

$$\mathcal{W}_2(m) \equiv \mathcal{V}\big(\nu\,\hat{q}_2(m), \kappa(1-m)\big) - \hat{u}_2(m)\hat{q}_2(m) \tag{7}$$

denote the continuation payoffs induced by the Stage 2 user subgame.
If the mass of third-party apps joining E_1 in Stage 1 is $s(a_1, a_2)$, ecosystem E_i's payoff is

$$\hat{\Pi}_1(a_1, a_2) = \mathcal{W}_1(m(s(a_1, a_2))) - \lambda a_1 s(a_1, a_2), \qquad \hat{\Pi}_2(a_1, a_2) = \mathcal{W}_2(m(s(a_1, a_2))) - \lambda a_2 (1 - s(a_1, a_2)).$$

Define the incremental continuation value of winning rather than losing the entire third-party block by

$$\Delta \equiv \mathcal{W}_1\left(\frac{1+\lambda}{2}\right) - \mathcal{W}_1\left(\frac{1-\lambda}{2}\right) = \mathcal{W}_2\left(\frac{1-\lambda}{2}\right) - \mathcal{W}_2\left(\frac{1+\lambda}{2}\right). \tag{8}$$

where the equality holds by symmetry.

Proposition 12 (App-subgame equilibrium)**.** *In any pure-strategy equilibrium of the Stage 1 app subgame,*

$$a_1^* = a_2^* = \frac{\Delta}{\lambda}.$$

Moreover, the only pure-strategy equilibrium allocations of third-party apps are the two corner outcomes

$$\left(s_1^*, s_2^*\right) \in \{(0,1),(1,0)\}.$$

In either equilibrium, the winner is bid down to its losing payoff.

Each ecosystem is willing to bid up to its incremental continuation value per third-party app, namely Δ/λ. Hence equilibrium transfers are pinned down at that level. At equal bids, however, a split allocation of third-party apps is inefficient from each ecosystem's point of view because continuation payoffs are strictly convex in the total app mass m. Therefore any interior split is unstable: one ecosystem strictly prefers a corner outcome and can secure it by an arbitrarily small deviation. As a result, only the two tipping allocations survive in pure strategies.

Wlog, focus on the corner equilibrium $(s_1^*, s_2^*) = (0,1)$, in which all third-party apps join E_2. Then the mass of apps of E_1 is

$$m^* = m(0) = \frac{1-\lambda}{2}.$$

By Proposition 11, the user market tips in favor of E_2 if and only if

$$m^* < \tilde{m}(\nu, \kappa, \tau).$$

Size of the Tipping Region. The threshold $\tilde{m}(\nu, \kappa, \tau)$ measures how much asymmetry in app adoption can be sustained before the user market tips. Equivalently, it measures the size of the tipping region on one side of the symmetric allocation.

Naturally, $\tilde{m}(\nu, \kappa, \tau)$ is decreasing in τ: more user differentiation reduces tipping in the user market.

For every $\tau > 0$, the user market is a smooth Hotelling game with a unique equilibrium allocation, and the equilibrium shares converge as $\tau \downarrow 0$ to the allocation that equalizes the marginal value of an additional user across ecosystems.

We analyze the limit game $\tau \downarrow 0$.

When ν rises, each user contributes more observations. As there are diminishing returns to observations, this lowers competition for users on both sides. As a result, a larger asymmetry in app adoption is needed for one ecosystem to be unable to compete, so the tipping region shrinks. This threshold is decreasing in ν, the mass of observations generated by each user:

$$\frac{\partial \tilde{m}(\nu, \kappa)}{\partial \nu} < 0.$$

By contrast, its dependence on κ, the mass of covariates generated by each app, is generally non-monotone. In particular for $\tau \downarrow 0$,

$$\frac{\partial \tilde{m}(\nu, \kappa)}{\partial \kappa} > 0 \iff \nu \leq \sqrt{5} - 1 \quad \text{or} \quad \left(\sqrt{5} - 1 < \nu < 2 \text{ and } 0 < \kappa < \frac{4 - \nu}{2} - \frac{\sqrt{\nu^2 - \nu}}{\sqrt{2}} \right). \tag{9}$$

The effect of κ is ambiguous. On the one hand, when each app generates more covariates, the disadvantaged ecosystem values its first user more because it is in the region in which observations and covariates are complements. This force acts against tipping. On the other hand, κ has an ambiguous effect on the value for the advantaged ecosystem of retaining its last user: if ν is small, it is also in the complementarity region, so κ tends to reinforce tipping; if ν is large, the advantaged ecosystem is in the substitution region, so κ hinders tipping.

Corollary 11 (Sensitivity of Users to Apps)**.** *If $\hat{\theta} \in (0, 1)$, then is continuously differentiable in m, and*

$$\hat{\theta}'(m) = -\frac{\kappa}{\nu} \frac{\mathcal{V}_{nk}(\nu\,\hat{\theta}(m), \kappa m) + \mathcal{V}_{nk}(\nu(1 - \hat{\theta}(m)), \kappa(1 - m))}{\mathcal{V}_{nn}(\nu\,\hat{\theta}(m), \kappa m) + \mathcal{V}_{nn}(\nu(1 - \hat{\theta}(m)), \kappa(1 - m)) - \frac{6\tau}{\nu^2}}. \tag{10}$$

Furthermore, under Assumption 2 and for $\tau \downarrow 0$,

$$\hat{\theta}'(m) > 0.$$

The derivative $\hat{\theta}'(m)$ measures the marginal change in ecosystem E_1's equilibrium user share induced by a marginal reallocation of apps from E_2 to E_1. It therefore captures the strength of the app-to-user spillover.

Two forces determine this spillover. The numerator captures how an increase in the app base affects the marginal value of attracting users in the two ecosystems, through the cross-partial $\mathcal{V}_{nk}$. The denominator captures diminishing returns to users, through $\mathcal{V}_{nn} < 0$, which dampens the reallocation of users across ecosystems. The second part of the corollary shows that in any user-subgame equilibrium the numerator must be positive meaning that a marginal shift of apps toward E_i increases its equilibrium user share.

Equilibrium Users. The graph of $\hat{\theta}$ reflects all the possible user subgame equilibria as a function of the app subgame outcome m:

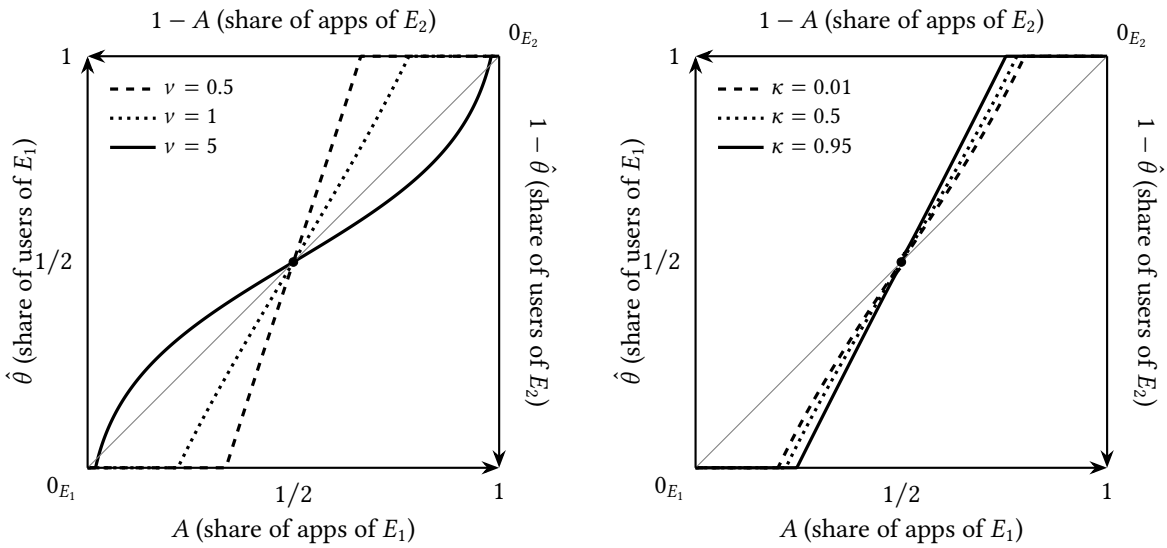

Figure 11: In each panel, the horizontal axis measures the share of apps attracted by ecosystem E_1, while the vertical axis is the resulting equilibrium share of users attracted by E_1; the top and right axes report the corresponding shares for ecosystem E_2. The left panel plots $\hat{\theta}(\hat{\omega})$ for fixed $\kappa = 1/2$ and the right panel plots $\hat{\theta}(\hat{\omega})$ for fixed $\nu = 1/2$. The 45° line is shown for reference.

Figure 11 summarizes how asymmetries in app adoption propagate into asymmetries in user adoption. The kinks mark the boundaries of the tipping regions, i.e. the threshold values $\tilde{m}(\nu, \kappa)$ and $1 - \tilde{m}(\nu, \kappa)$. Thus, the figure makes transparent that the user market acts as an amplification device: sufficiently large asymmetries in app adoption are translated into corner outcomes on the user side, while smaller asymmetries generate an interior reallocation of users across ecosystems. The 45-degree line provides a natural benchmark. When $\hat{\theta}(m)$ is steeper than the 45-degree line, a marginal asymmetry in apps translates into a larger asymmetry in users; when it is flatter, the induced user asymmetry is attenuated.

The left panel shows that increasing ν compresses the tipping region and flattens the schedule around the symmetric allocation. When each user contributes more observations, an additional user has lower marginal value because of diminishing returns on the observation side. As a result, ecosystems compete less aggressively for users, and a given asymmetry in apps is less likely to tip the user market. This is especially visible when comparing $\nu = 0.5$ and $\nu = 5$: for low ν, the graph is steep around $m = 1/2$ and reaches the corner regions quickly, whereas for high ν the mapping is more gradual and the interior region is substantially wider.

The right panel shows that increasing κ has the opposite effect at $\nu = 1$: the schedule becomes steeper and the tipping region expands. Intuitively, when each app brings more covariates, app asymmetries become more consequential for the value of attracting users, so a given

difference in app shares translates into a larger difference in user shares.

In summary, app-side advantages matter most for user-market concentration when there are few observations per users and many covariates per app, and matter less when each user already generates a large amount of observations.

7 Conclusion

This paper develops a general framework for understanding the value of data in prediction by explicitly modeling covariates. The analysis shows how economies of scope across covariates, interactions between covariates and observations, and complementarities between training and prediction can generate increasing returns, offering a microfoundation for the rich-get-richer effects often observed in data-driven markets.

These forces have direct implications for policy and strategy. Prediction technologies may display natural monopsony characteristics, as concentrating covariates within one firm can raise efficiency. Privacy regulation that fragments data supply may inadvertently reinforce monopsony power, creating a trilemma between privacy, competition, and efficiency. The framework also highlights that not all data pooling agreements are alike: pooling lists of users with the same covariates can be anticompetitive, whereas pooling different covariates on similar users raise welfare by eliminating double marginalization. Exclusivity deals, such as those signed between AI labs and data providers, may profitably foreclose entry by depriving rivals of essential complements. For firms, the results imply that prediction entails substantial sunk costs: early on, investment should balance user acquisition and attribute enrichment, while specialization and integration become optimal at a larger scale.

More broadly, the analysis cautions against treating data as homogeneous. Policies promoting open data without regard to dataset composition may miss crucial efficiency margins, whereas access remedies such as FRAND-priced APIs or federated learning preserve economies of scope.

My work opens two natural avenues for future research. The first is empirical. I aim to develop a methodology to test my results on real datasets. While the existing empirical literature[17] provides partial support to my findings, it suffers from two limitations: (i) most studies focus on a single dataset, whereas uncovering general properties requires comparing multiple datasets along common dimensions; and (ii) no existing work systematically tests all the properties identified in my model. Once these empirical properties are validated, my framework could serve as the foundation for a practical formula for data valuation, in the spirit of the Black–Scholes–Merton formula for derivatives.[18] The second avenue is theoretical. Embedding my static model into a dynamic Wald sampling framework would allow me

[17] See Bajari et al. (2019), Schaefer and Sapi (2023), Lee and Wright (2023), Yoganarasimhan (2020), and Carballa-Smichowski et al. (2025b)

[18] See Black and Scholes (1973), Merton (1973)

to microfound data-enabled learning and analyze when feedback loops generate convergent data-collection strategies versus when they diverge.

References

Acemoğlu, Daron et al. (2022). "Too much data: Prices and inefficiencies in data markets." In: *American Economic Journal: Microeconomics* 14.4, pp. 218–256.

Aghion, Philippe and Patrick Bolton (1987). "Contracts as a Barrier to Entry." In: *The American Economic Review* 77.3, pp. 388–401.

Allcott, Hunt et al. (2025). *Sources of market power in web search: Evidence from a field experiment.* Tech. rep. National Bureau of Economic Research.

Aral, Sinan, Erik Brynjolfsson, and DJ Wu (2008). "Which came first, IT or productivity? The virtuous cycle of investment and use in enterprise systems." In.

Arrow, Kenneth (1959). *Economic Welfare and the Allocation of Resources for Invention.* Santa Monica, CA: RAND Corporation.

Bai, Z. and J.W. Silverstein (2009). *Spectral Analysis of Large Dimensional Random Matrices.* Springer Series in Statistics. Springer New York.

Bajari, Patrick et al. (2019). "The impact of big data on firm performance: An empirical investigation." In: *AEA papers and proceedings* 109, pp. 33–37.

Bergemann, Dirk and Alessandro Bonatti (2024). "Data, Competition, and Digital Platforms." In: *American Economic Review* 114.8, pp. 2553–2595.

Bergemann, Dirk, Alessandro Bonatti, and Tan Gan (2022). "The economics of social data." In: *The RAND Journal of Economics* 53.2, pp. 263–296.

Bergemann, Dirk, Alessandro Bonatti, and Alex Smolin (2018). "The Design and Price of Information." In: *American Economic Review* 108.1, pp. 1–48.

Berger, James O. (1990). *Statistical decision theory.* Springer, pp. 277–284.

Black, Fischer and Myron Scholes (1973). "The Pricing of Options and Corporate Liabilities." In: *Journal of Political Economy* 81.3, pp. 637–654.

Brier, Glenn W. (1950). "Verification of Forecasts Expressed in Terms of Probability." In: *Monthly Weather Review* 78.1, pp. 1–3.

Calzolari, Giacomo, Anatole Cheysson, and Riccardo Rovatti (2025). "Machine data: market and analytics." In: *Management Science.*

Carballa-Smichowski, Bruno et al. (2025a). *Data Sharing or Analytics Sharing?* TSE Working Paper 25-1615. Toulouse School of Economics.

Carballa-Smichowski, Bruno et al. (2025b). "Economies of scope in data aggregation: Evidence from health data." In: *Information Economics and Policy* 71, p. 101146.

Cong, Lin William, Zhiguo He, and Changhua Yu (2021). "Data as Capital." In: *Review of Financial Studies* 34.6, pp. 2895–2936.

Cornière, Alexandre de and Greg Taylor (2024). "Data-Driven Mergers." In: *Management Science* 70.9, pp. 6473–6482.

Crankshaw, Daniel et al. (2017). "Clipper: A Low-Latency Online Prediction Serving System." In: *14th USENIX Symposium on Networked Systems Design and Implementation (NSDI 17)*. USENIX Association, pp. 613–627.

Dasaratha, Krishna, Juan Ortner, and Chengyang Zhu (2025). "Markets for Models." In: *arXiv preprint arXiv:2503.02946*.

De Corniere, Alexandre and Greg Taylor (2025). "Data and Competition: A Simple Framework." In: *Forthcoming, RAND Journal of Economics*.

DeGroot, Morris H. (1970). *Optimal Statistical Decisions*. en. New York: McGraw-Hill.

Digital Platforms, Stigler Committee on (2019). *Final Report*. Stigler Center for the Study of the Economy and the State, University of Chicago Booth School of Business.

Farboodi, Maryam and Laura Veldkamp (2025). *A model of the Data Economy*. Tech. rep. R&R, Review of Economic Studies.

Goldfarb, Avi and Catherine Tucker (2011). "Privacy Regulation and Online Advertising." In: *Management Science* 57.1, pp. 57–71.

Gu, Yiquan, Leonardo Madio, and Carlo Reggiani (Sept. 2021). "Data brokers co-opetition." In: *Oxford Economic Papers* 74.3, pp. 820–839.

Hagiu, Andrei and Julian Wright (2023). "Data-enabled learning, network effects, and competitive advantage." In: *The RAND Journal of Economics* 54.4, pp. 638–667.

Hastie, Trevor et al. (2020). "Surprises in High-Dimensional Ridgeless Least Squares Interpolation." In.

Iyer, Ganesh and Tianshu Ke (2024). "Competition and Algorithmic Complexity in Predictive Analytics." In: *Marketing Science* 43.2, pp. 215–233.

Jones, Charles I. and Christopher Tonetti (2020). "Nonrivalry and the Economics of Data." In: *American Economic Review* 110.9, 2819–58.

Katz, Michael L. and Carl Shapiro (Aug. 1986). "How to License Intangible Property*." In: *The Quarterly Journal of Economics* 101.3, pp. 567–589.

Lee, Gunhaeng and Julian Wright (2023). "Recommender systems and the Value of User Data." In: *National University of Singapore Working Paper*.

Lerner, Josh and Jean Tirole (2004). "Efficient Patent Pools." In: *American Economic Review* 94.3, 691–711.

Lopez-Paz, David et al. (2016). "Unifying Distillation and Privileged Information." In: *International Conference on Learning Representations (ICLR)*.

Marčenko, Vladimir A. and Leonid A. Pastur (1967). "Distribution of eigenvalues for some sets of random matrices." In: *Mathematics of the USSR-Sbornik* 1.4, pp. 457–483.

Merton, Robert C. (1973). "Theory of Rational Option Pricing." In: *The Bell Journal of Economics and Management Science* 4.1, pp. 141–183.

Montiel Olea, José Luis et al. (Apr. 2022). "Competing Models." In: *The Quarterly Journal of Economics* 137.4, 2419–2457.

Nocke, Volker, Martin Peitz, and Konrad Stahl (Dec. 2007). "Platform Ownership." In: *Journal of the European Economic Association* 5.6, pp. 1130–1160.

Prüfer, Jens and Christoph Schottmüller (2021). "Competing with big data." In: *The Journal of Industrial Economics* 69.4, pp. 967–1008.

Radner, Roy and Joseph Stiglitz (1984). "A Nonconcavity in the Value of Information." In: *Bayesian models in economic theory* 5, pp. 33–52.

Schaefer, Maximilian (2025). *When Should we Expect Non-Decreasing Returns from Data in Prediction Tasks?*

Schaefer, Maximilian and Geza Sapi (2023). "Complementarities in learning from data: Insights from general search." In: *Information Economics and Policy* 65, p. 101063.

Schmidt, Eric (Oct. 2, 2009). *How Google Plans to Stay Ahead in Search*. Accessed: 2025-11-12. Bloomberg. URL: https://www.bloomberg.com/news/articles/2009-10-02/how-google-plans-to-stay-ahead-in-search.

Selten, Reinhard (1998). "Axiomatic Characterization of the Quadratic Scoring Rule." In: *Experimental Economics* 1.1, pp. 43–62.

Strzalecki, Tomasz (2024). *Variational Bayes and non-Bayesian Updating*.

UK Competition and Markets Authority (2019). *Unlocking Digital Competition: Report of the Digital Competition Expert Panel*. London: Competition and Markets Authority.

Vapnik, Vladimir and Rauf Izmailov (2015). "Learning Using Privileged Information: Similarity Control and Knowledge Transfer." In: *Journal of Machine Learning Research* 16.61, pp. 2023–2049.

Varian, Hal (2018). "Artificial Intelligence, Economics, and Industrial Organization." In: *The Economics of Artificial Intelligence: An Agenda*. NBER Chapters. National Bureau of Economic Research, Inc, pp. 399–419.

Xu, Chen et al. (2020). "Privileged Features Distillation at Taobao Recommendations." In: *Proceedings of the 26th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pp. 2590–2598.

Yang, Shuo et al. (2022). "Toward Understanding Privileged Features Distillation in Learning-to-Rank." In: *Advances in Neural Information Processing Systems*. Vol. 35.

Yoganarasimhan, Hema (2020). "Search personalization using machine learning." In: *Management Science* 66.3, pp. 1045–1070.

A Proofs

A.1 Joint Law

Lemma 3 (Exact finite-K law of the observed dataset)**.** *Fix $K \in \mathbb{N}_+$ and a dataset design*

$$\mathcal{D} = (N, \mathcal{T}, \mathcal{P}),$$

with $T \equiv |\mathcal{T}|$ and $P \equiv |\mathcal{P}|$. Let

$$\boldsymbol{d} = ((\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}), \boldsymbol{x}_{0,\mathcal{P}})$$

denote the observed dataset induced by $\mathcal{D}$. Then:

1. *Writing $\mathcal{T}^c \equiv \mathcal{K} \setminus \mathcal{T}$, the training targets admit the exact decomposition*

$$\boldsymbol{y}_N = \boldsymbol{X}_{N,\mathcal{T}}\boldsymbol{\beta}_{\mathcal{T}} + \boldsymbol{r}_{N,\mathcal{T}} + \boldsymbol{\eta}_N, \qquad \boldsymbol{r}_{N,\mathcal{T}} \equiv \boldsymbol{X}_{N,\mathcal{T}^c}\boldsymbol{\beta}_{\mathcal{T}^c}.$$

   Moreover, conditional on $(\boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}})$,

$$\boldsymbol{X}_{N,\mathcal{T}}\boldsymbol{\beta}_{\mathcal{T}} \sim \mathcal{N}\left(0,\ \sigma_\beta^2 \boldsymbol{X}_{N,\mathcal{T}}\boldsymbol{X}_{N,\mathcal{T}}^\top\right),$$

   and

$$\boldsymbol{\eta}_N \sim \mathcal{N}\left(0,\ \sigma_\eta^2\sigma_\beta^2 I_N\right),$$

   with $\boldsymbol{X}_{N,\mathcal{T}}\boldsymbol{\beta}_{\mathcal{T}}$, $\boldsymbol{r}_{N,\mathcal{T}}$, and $\boldsymbol{\eta}_N$ mutually independent conditional on $(\boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}})$.

2. *Conditional on the omitted coefficients $\boldsymbol{\beta}_{\mathcal{T}^c}$, the omitted-covariate component is exactly Gaussian:*

$$\boldsymbol{r}_{N,\mathcal{T}} \mid \boldsymbol{\beta}_{\mathcal{T}^c}, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}} \sim \mathcal{N}\left(0,\ \frac{\sigma_x^2}{K}\sum_{k\notin\mathcal{T}}\beta_k^2\, I_N\right).$$

   In particular,

$$\mathbb{E}[\boldsymbol{r}_{N,\mathcal{T}} \mid \boldsymbol{\beta}_{\mathcal{T}^c}, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}] = 0.$$

3. *After integrating out $\boldsymbol{\beta}_{\mathcal{T}^c}$, the vector $\boldsymbol{r}_{N,\mathcal{T}}$ remains conditionally centered and conditionally homoskedastic:*

$$\mathbb{E}[\boldsymbol{r}_{N,\mathcal{T}} \mid \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}] = 0,$$

   and

$$\mathrm{Var}(\boldsymbol{r}_{N,\mathcal{T}} \mid \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}) = \sigma_\beta^2\sigma_x^2\left(1 - \frac{T}{K}\right) I_N.$$

Equivalently, for every $i, j \in \{1, \dots, N\}$,

$$\mathrm{Cov}\big(r_i, r_j \mid \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}\big) = \begin{cases} \sigma_\beta^2 \sigma_x^2 \left(1 - \frac{T}{K}\right), & i = j, \\ 0, & i \neq j. \end{cases}$$

4. *In general, for finite K, the conditional law of $\boldsymbol{r}_{N,\mathcal{T}}$ given $(\boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}})$ is not Gaussian. Rather, it is a scale mixture of centered Gaussian vectors, where the random scale is*

$$\frac{\sigma_x^2}{K} \sum_{k \notin \mathcal{T}} \beta_k^2.$$

Consequently, the coordinates of $\boldsymbol{r}_{N,\mathcal{T}}$ are conditionally uncorrelated but generally not conditionally independent, and

$$\boldsymbol{y}_N \mid \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}$$

is in general not Gaussian either.

Proof. Fix $K \in \mathbb{N}_+$ and $\mathcal{D} = (N, \mathcal{T}, \mathcal{P})$.
First consider the observed covariates. Since the scalar covariates $\{x_{ik}\}_{i \in \mathbb{N}_0,\, k \in \mathcal{K}}$ are i.i.d.

$$x_{ik} \sim \mathcal{N}\left(0, \frac{\sigma_x^2}{K}\right),$$

for each $i \in \{1, \dots, N\}$ the row vector

$$\boldsymbol{x}_{i,\mathcal{T}} \equiv (x_{ik})_{k \in \mathcal{T}} \in \mathbb{R}^T$$

satisfies

$$\boldsymbol{x}_{i,\mathcal{T}} \sim \mathcal{N}\left(0, \frac{\sigma_x^2}{K} I_T\right),$$

and these rows are independent across i. Hence

$$\boldsymbol{X}_{N,\mathcal{T}} \sim \mathcal{MN}_{N \times T}\left(0,\, I_N,\, \frac{\sigma_x^2}{K} I_T\right).$$

Likewise,

$$\boldsymbol{x}_{0,\mathcal{P}} \equiv (x_{0k})_{k \in \mathcal{P}} \in \mathbb{R}^P$$

satisfies

$$\boldsymbol{x}_{0,\mathcal{P}} \sim \mathcal{N}\left(0, \frac{\sigma_x^2}{K} I_P\right).$$

Since $\boldsymbol{X}_{N,\mathcal{T}}$ depends on the coordinates $\{x_{ik} : i \in \{1, \dots, N\},\, k \in \mathcal{T}\}$ and $\boldsymbol{x}_{0,\mathcal{P}}$ depends on $\{x_{0k} : k \in \mathcal{P}\}$, and the primitive covariates are independent across individuals and coordinates, these two objects are independent. This proves part 1.

Next, for each $i \in \{1, \dots, N\}$,

$$y_i = \sum_{k \in \mathcal{T}} \beta_k x_{ik} + \sum_{k \notin \mathcal{T}} \beta_k x_{ik} + \eta_i.$$

Stacking across $i = 1, \dots, N$ gives

$$\boldsymbol{y}_N = \boldsymbol{X}_{N,\mathcal{T}} \boldsymbol{\beta}_{\mathcal{T}} + \boldsymbol{X}_{N,\mathcal{T}^c} \boldsymbol{\beta}_{\mathcal{T}^c} + \boldsymbol{\eta}_N.$$

Defining

$$\boldsymbol{r}_{N,\mathcal{T}} \equiv \boldsymbol{X}_{N,\mathcal{T}^c} \boldsymbol{\beta}_{\mathcal{T}^c},$$

we obtain

$$\boldsymbol{y}_N = \boldsymbol{X}_{N,\mathcal{T}} \boldsymbol{\beta}_{\mathcal{T}} + \boldsymbol{r}_{N,\mathcal{T}} + \boldsymbol{\eta}_N.$$

Conditional on $(\boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}})$, the vector $\boldsymbol{\beta}_{\mathcal{T}}$ remains distributed as

$$\boldsymbol{\beta}_{\mathcal{T}} \sim \mathcal{N}(0, \sigma_\beta^2 I_T),$$

because the weights are independent of all covariates. Therefore,

$$\boldsymbol{X}_{N,\mathcal{T}} \boldsymbol{\beta}_{\mathcal{T}} \mid \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}} \sim \mathcal{N}\left(0,\ \sigma_\beta^2 \boldsymbol{X}_{N,\mathcal{T}} \boldsymbol{X}_{N,\mathcal{T}}^\top\right).$$

Also,

$$\boldsymbol{\eta}_N \sim \mathcal{N}\left(0, \sigma_\eta^2 \sigma_\beta^2 I_N\right),$$

independently of all weights and covariates. Finally, $\boldsymbol{r}_{N,\mathcal{T}}$ depends only on $(\boldsymbol{X}_{N,\mathcal{T}^c}, \boldsymbol{\beta}_{\mathcal{T}^c})$, while $\boldsymbol{X}_{N,\mathcal{T}} \boldsymbol{\beta}_{\mathcal{T}}$ depends only on $(\boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{\beta}_{\mathcal{T}})$. Since the primitive collections

$$(\boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{\beta}_{\mathcal{T}}), \qquad (\boldsymbol{X}_{N,\mathcal{T}^c}, \boldsymbol{\beta}_{\mathcal{T}^c}), \qquad \boldsymbol{\eta}_N$$

are mutually independent, it follows that $\boldsymbol{X}_{N,\mathcal{T}} \boldsymbol{\beta}_{\mathcal{T}}$, $\boldsymbol{r}_{N,\mathcal{T}}$, and $\boldsymbol{\eta}_N$ are mutually independent conditional on $(\boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}})$. This proves part 2.

We now prove the conditional Gaussian shortcut. Conditional on $\boldsymbol{\beta}_{\mathcal{T}^c}$, each coordinate of $\boldsymbol{r}_{N,\mathcal{T}}$ can be written as

$$r_i = \sum_{k \notin \mathcal{T}} \beta_k x_{ik}, \qquad i = 1, \dots, N.$$

Given $\boldsymbol{\beta}_{\mathcal{T}^c}$, this is a linear combination of the omitted covariates $\{x_{ik}\}_{k \notin \mathcal{T}}$, which are jointly Gaussian and independent across i. Therefore, conditional on $(\boldsymbol{\beta}_{\mathcal{T}^c}, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}})$, the vector $\boldsymbol{r}_{N,\mathcal{T}}$ is Gaussian with mean zero. Its conditional covariance matrix is diagonal because the

omitted covariates are independent across individuals. For each i,

$$\mathrm{Var}(r_i \mid \boldsymbol{\beta}_{\mathcal{T}^c}, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}) = \sum_{k \notin \mathcal{T}} \beta_k^2 \,\mathrm{Var}(x_{ik}) = \frac{\sigma_x^2}{K} \sum_{k \notin \mathcal{T}} \beta_k^2,$$

and for $i \neq j$,

$$\mathrm{Cov}\big(r_i, r_j \mid \boldsymbol{\beta}_{\mathcal{T}^c}, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}\big) = 0.$$

Hence

$$\boldsymbol{r}_{N,\mathcal{T}} \mid \boldsymbol{\beta}_{\mathcal{T}^c}, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}} \sim \mathcal{N}\left(0,\ \frac{\sigma_x^2}{K} \sum_{k \notin \mathcal{T}} \beta_k^2 \, I_N\right).$$

This proves part 3.

We next integrate out $\boldsymbol{\beta}_{\mathcal{T}^c}$. By iterated expectations,

$$\mathbb{E}[\boldsymbol{r}_{N,\mathcal{T}} \mid \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}] = \mathbb{E}[\mathbb{E}[\boldsymbol{r}_{N,\mathcal{T}} \mid \boldsymbol{\beta}_{\mathcal{T}^c}, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}] | \, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}] = 0.$$

Similarly, using the conditional covariance formula,

$$\begin{aligned}\mathrm{Var}(\boldsymbol{r}_{N,\mathcal{T}} \mid \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}) &= \mathbb{E}[\mathrm{Var}(\boldsymbol{r}_{N,\mathcal{T}} \mid \boldsymbol{\beta}_{\mathcal{T}^c}, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}) | \, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}] \\ &\quad + \mathrm{Var}(\mathbb{E}[\boldsymbol{r}_{N,\mathcal{T}} \mid \boldsymbol{\beta}_{\mathcal{T}^c}, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}] | \, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}).\end{aligned}$$

The second term is zero, since the inner conditional mean is zero. Hence

$$\begin{aligned}\mathrm{Var}(\boldsymbol{r}_{N,\mathcal{T}} \mid \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}) &= \mathbb{E}\left[\frac{\sigma_x^2}{K} \sum_{k \notin \mathcal{T}} \beta_k^2 \, I_N\right] \\ &= \frac{\sigma_x^2}{K} \sum_{k \notin \mathcal{T}} \mathbb{E}[\beta_k^2] \, I_N \\ &= \frac{\sigma_x^2}{K}(K - T)\sigma_\beta^2 I_N \\ &= \sigma_\beta^2 \sigma_x^2 \left(1 - \frac{T}{K}\right) I_N.\end{aligned}$$

This proves part 4.

Finally, part 3 shows that conditional on $(\boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}})$, the vector $\boldsymbol{r}_{N,\mathcal{T}}$ is a mixture of centered Gaussian vectors with random covariance

$$\frac{\sigma_x^2}{K} \sum_{k \notin \mathcal{T}} \beta_k^2 \, I_N.$$

Unless this random scale is degenerate, such a mixture is not Gaussian in general. Thus, for finite K, the conditional law of $\boldsymbol{r}_{N,\mathcal{T}}$ given $(\boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}})$ is generally not Gaussian.

This representation also clarifies dependence across individuals. Conditional on $\boldsymbol{\beta}_{\mathcal{T}^c}$, the co-

ordinates $r_1, \dots, r_N$ are independent Gaussian with common variance

$$\frac{\sigma_x^2}{K} \sum_{k \notin \mathcal{T}} \beta_k^2.$$

After integrating out $\boldsymbol{\beta}_{\mathcal{T}^c}$, they remain uncorrelated because the covariance matrix is proportional to I_N, but they are generally not independent because they share the same random variance factor. Since

$$\boldsymbol{y}_N = \boldsymbol{X}_{N,\mathcal{T}} \boldsymbol{\beta}_{\mathcal{T}} + \boldsymbol{r}_{N,\mathcal{T}} + \boldsymbol{\eta}_N,$$

with the first and third terms conditionally Gaussian and the middle term generally non-Gaussian, it follows that

$$\boldsymbol{y}_N \mid \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{x}_{0,\mathcal{P}}$$

is in general not Gaussian either. This proves part 5. □

A.2 Proof of Proposition 1

Fix a design $\mathcal{D} = (N, \mathcal{T}, \mathcal{P})$ and a realized dataset

$$\boldsymbol{d} = ((\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}), \boldsymbol{x}_{0,\mathcal{P}}).$$

Under quadratic loss, the posterior risk of any predictor $\hat{y} \in \mathbb{R}$ is

$$\mathbb{E}\left[(y_0 - \hat{y})^2 \mid \boldsymbol{d}\right].$$

Completing the square around the posterior mean yields

$$\mathbb{E}\left[(y_0 - \hat{y})^2 \mid \boldsymbol{d}\right] = \mathbb{E}\left[(y_0 - \mathbb{E}[y_0 \mid \boldsymbol{d}])^2 \mid \boldsymbol{d}\right] + (\hat{y} - \mathbb{E}[y_0 \mid \boldsymbol{d}])^2.$$

Therefore the unique minimizer is

$$\hat{y}_{\mathcal{D}}^*(\boldsymbol{d}) = \mathbb{E}[y_0 \mid \boldsymbol{d}].$$

I now compute this conditional expectation. By the DGP,

$$y_0 = \sum_{k=1}^{K} \beta_k x_{0k} + \eta_0.$$

Since η_0 is independent of $\boldsymbol{d}$ and centered,

$$\mathbb{E}[\eta_0 \mid \boldsymbol{d}] = 0.$$

Next consider a generic coordinate k.

If $k \notin \mathcal{P}$, then x_{0k} is unobserved, independent of $(\boldsymbol{d}, \beta_k)$, and centered. Hence

$$\mathbb{E}[\beta_k x_{0k} \mid \boldsymbol{d}] = \mathbb{E}[\beta_k \, \mathbb{E}[x_{0k} \mid \boldsymbol{d}, \beta_k] \mid \boldsymbol{d}] = 0.$$

If $k \in \mathcal{P}$, then x_{0k} is observed and independent of $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{\beta})$. Therefore

$$\mathbb{E}[\beta_k x_{0k} \mid \boldsymbol{d}] = x_{0k} \, \mathbb{E}[\beta_k \mid \boldsymbol{d}] = x_{0k} \, \mathbb{E}[\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}],$$

where the second equality uses the independence of $\boldsymbol{x}_{0,\mathcal{P}}$ from $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}, \boldsymbol{\beta})$.
Thus

$$\mathbb{E}[y_0 \mid \boldsymbol{d}] = \sum_{k \in \mathcal{P}} x_{0k} \, \mathbb{E}[\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}].$$

It remains to show that

$$\mathbb{E}[\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}] = 0 \qquad \text{for all } k \notin \mathcal{T}.$$

Fix such a coordinate $k \notin \mathcal{T}$. Conditional on the observed training covariates $\boldsymbol{X}_{N,\mathcal{T}}$, the training outcomes satisfy

$$y_i = \boldsymbol{x}_{i,\mathcal{T}}^\top \boldsymbol{\beta}_{\mathcal{T}} + \beta_k x_{ik} + \sum_{j \notin \mathcal{T},\, j \neq k} \beta_j x_{ij} + \eta_i, \qquad i = 1, \dots, N.$$

Since $x_{ik} \sim \mathcal{N}(0, \sigma_x^2/K)$ is unobserved, centered, and independent of all other primitives, the conditional distribution of $(y_i)_{i=1}^N$ given $(\boldsymbol{X}_{N,\mathcal{T}}, \beta_k)$ is the same under $\beta_k = b$ and $\beta_k = -b$. Equivalently, the conditional likelihood of $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$ is an even function of β_k:

$$L(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}} \mid \beta_k = b) = L(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}} \mid \beta_k = -b).$$

Because the prior of β_k is centered Gaussian, it is also symmetric:

$$\pi_k(b) = \pi_k(-b).$$

Hence Bayes' rule implies that the posterior density of β_k given $(\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})$ is symmetric around zero. Therefore,

$$\mathbb{E}[\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}] = 0.$$

Substituting back, only coordinates in $\mathcal{P} \cap \mathcal{T}$ remain:

$$\hat{y}_D^*(\boldsymbol{d}) = \mathbb{E}[y_0 \mid \boldsymbol{d}] = \sum_{k \in \mathcal{P} \cap \mathcal{T}} x_{0k} \, \mathbb{E}[\beta_k \mid \boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}}].$$

Equivalently,

$$\hat{y}_D^*(\boldsymbol{d}) = \boldsymbol{x}_{0,\mathcal{P} \cap \mathcal{T}}^\top \mathbb{E}[\boldsymbol{\beta}_{\mathcal{P} \cap \mathcal{T}} \mid (\boldsymbol{y}_N, \boldsymbol{X}_{N,\mathcal{T}})].$$

This proves the claim.

A.3 Proof of Lemma 1

Fix K and write $T(K) \equiv |\mathcal{T}(K)|$. For each $i \in \{1, \dots, N(K)\}$ define

$$\varepsilon_i^{(K)} = \sum_{k=T(K)+1}^{K} \beta_k^{(K)} x_{ik}^{(K)} + \eta_i^{(K)}.$$

Fix any integer $m \geq 1$ and consider the first m components $\boldsymbol{\varepsilon}_{[m]}^{(K)} = (\varepsilon_1^{(K)}, \dots, \varepsilon_m^{(K)})^\top$.

Step 1 (conditional Gaussianity). Conditional on the coefficients $\boldsymbol{\beta}^{(K)} = (\beta_1^{(K)}, \dots, \beta_K^{(K)})$, the vectors $\boldsymbol{x}_k^{(K)} \equiv (x_{1k}^{(K)}, \dots, x_{mk}^{(K)})^\top$ are i.i.d. across k with

$$\boldsymbol{x}_k^{(K)} \mid \boldsymbol{\beta}^{(K)} \sim \mathcal{N}\left(\mathbf{0}_m, \frac{\sigma_x^2}{K} \boldsymbol{I}_m\right),$$

and are independent of $\boldsymbol{\eta}_{[m]}^{(K)} = (\eta_1^{(K)}, \dots, \eta_m^{(K)})^\top$. Hence, conditional on $\boldsymbol{\beta}^{(K)}$,

$$\sum_{k=T(K)+1}^{K} \beta_k^{(K)} \boldsymbol{x}_k^{(K)} \sim \mathcal{N}\left(\mathbf{0}_m, \frac{\sigma_x^2}{K} \sum_{k=T(K)+1}^{K} (\beta_k^{(K)})^2 \cdot \boldsymbol{I}_m\right),$$

and since $\boldsymbol{\eta}_{[m]}^{(K)} \sim \mathcal{N}(\mathbf{0}_m, \sigma_\eta^2 \sigma_\beta^2 \boldsymbol{I}_m)$ is independent, we obtain the conditional law

$$\boldsymbol{\varepsilon}_{[m]}^{(K)} \mid \boldsymbol{\beta}^{(K)} \sim \mathcal{N}\left(\mathbf{0}_m,\ s_K^2\, \boldsymbol{I}_m\right), \qquad s_K^2 \equiv \sigma_\beta^2 \sigma_\eta^2 + \frac{\sigma_x^2}{K} \sum_{k=T(K)+1}^{K} (\beta_k^{(K)})^2.$$

Step 2 (limit of the random variance). By Assumption 1, $T(K)/K \to t$, so $(K - T(K))/K \to 1 - t$. By the law of large numbers,

$$\frac{1}{K - T(K)} \sum_{k=T(K)+1}^{K} (\beta_k^{(K)})^2 \xrightarrow{p} \mathbb{E}[(\beta_1^{(K)})^2] = \sigma_\beta^2.$$

Therefore,

$$\frac{1}{K} \sum_{k=T(K)+1}^{K} (\beta_k^{(K)})^2 = \frac{K - T(K)}{K} \cdot \frac{1}{K - T(K)} \sum_{k=T(K)+1}^{K} (\beta_k^{(K)})^2 \xrightarrow{p} (1 - t)\sigma_\beta^2,$$

and thus

$$s_K^2 \xrightarrow{p} \sigma_\beta^2((1 - t)\sigma_x^2 + \sigma_\eta^2) \equiv \sigma_\beta^2 \sigma_\varepsilon^2(t).$$

Step 3 (convergence in distribution). For any fixed $\boldsymbol{w} \in \mathbb{R}^m$, the conditional characteristic function equals

$$\mathbb{E}\left[\exp\left(i\boldsymbol{w}^\top \boldsymbol{\varepsilon}_{[m]}^{(K)}\right) \mid \boldsymbol{\beta}^{(K)}\right] = \exp\left(-\tfrac{1}{2}s_K^2\|\boldsymbol{w}\|^2\right).$$

Taking expectations and using $s_K^2 \to \sigma_\beta^2\sigma_\varepsilon^2(t)$ in probability,

$$\mathbb{E}\left[\exp\left(i\boldsymbol{w}^\top \boldsymbol{\varepsilon}_{[m]}^{(K)}\right)\right] = \mathbb{E}\left[\exp\left(-\tfrac{1}{2}s_K^2\|\boldsymbol{w}\|^2\right)\right] \to \exp\left(-\tfrac{1}{2}\sigma_\beta^2\sigma_\varepsilon^2(t)\|\boldsymbol{w}\|^2\right),$$

which is the characteristic function of $\mathcal{N}(\boldsymbol{0}_m, \sigma_\beta^2\sigma_\varepsilon^2(t)\boldsymbol{I}_m)$. Hence,

$$(\varepsilon_1^{(K)}, \dots, \varepsilon_m^{(K)})^\top \xrightarrow{d} \mathcal{N}\left(\boldsymbol{0}_m, \sigma_\beta^2\sigma_\varepsilon^2(t)\boldsymbol{I}_m\right).$$

Since m is arbitrary, this establishes that the residuals are asymptotically Gaussian with asymptotically independent components and variance $\sigma_\beta^2\sigma_\varepsilon^2(t)$, where $\sigma_\varepsilon^2(t) = (1-t)\sigma_x^2 + \sigma_\eta^2$.

A.4 Closed for Weight information

Proposition 13 (Closed form for weight information)**.** *Under Assumption 1,*

$$\mathcal{I}(n,t) = \frac{n+1+\frac{\sigma_\eta^2}{\sigma_x^2} - \sqrt{\left(n+1+\frac{\sigma_\eta^2}{\sigma_x^2}\right)^2 - 4nt}}{2t} = \frac{2n}{n+1+\frac{\sigma_\eta^2}{\sigma_x^2} + \sqrt{\left(n+1+\frac{\sigma_\eta^2}{\sigma_x^2}\right)^2 - 4nt}}.$$

Proof. Fix a sequence of nested designs

$$D(K) = (N(K), T(K), P(K)), \qquad P(K) \le T(K),$$

such that

$$\frac{N(K)}{K} \to n, \qquad \frac{T(K)}{K} \to t, \qquad \frac{P(K)}{K} \to p.$$

Under the asymptotic Gaussian regression representation, conditional on $\boldsymbol{X}_{N(K),\mathcal{T}(K)}$, the posterior covariance of the trained weights is

$$\mathrm{Var}\left(\boldsymbol{\beta}_{\mathcal{T}(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right) = \sigma_\beta^2\left(I_{T(K)} + \frac{1}{\sigma_\varepsilon^2(t)}\boldsymbol{X}_{N(K),\mathcal{T}(K)}^\top \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right)^{-1},$$

where

$$\sigma_\varepsilon^2(t) = (1-t)\sigma_x^2 + \sigma_\eta^2.$$

By exchangeability of trained coordinates, the average posterior variance reduction is the

same on every coordinate in $\mathcal{T}(K)$. Hence the definition of $\mathcal{I}(n,t)$ can be written as

$$\mathcal{I}(n,t) = \lim_{K\to\infty} \frac{1}{T(K)} \operatorname{tr}\left[I_{T(K)} - \frac{1}{\sigma_\beta^2} \operatorname{Var}\left(\boldsymbol{\beta}_{\mathcal{T}(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right)\right].$$

Substituting the posterior covariance gives

$$\mathcal{I}(n,t) = \lim_{K\to\infty} \frac{1}{T(K)} \operatorname{tr}\left[I_{T(K)} - \left(I_{T(K)} + \frac{1}{\sigma_\varepsilon^2(t)} \boldsymbol{X}_{N(K),\mathcal{T}(K)}^\top \boldsymbol{X}_{N(K),\mathcal{T}(K)} \right)^{-1}\right].$$

Using

$$I - (I+A)^{-1} = A(I+A)^{-1},$$

this becomes

$$\mathcal{I}(n,t) = \lim_{K\to\infty} \frac{1}{T(K)} \operatorname{tr}\left[\boldsymbol{X}_{N(K),\mathcal{T}(K)}^\top \boldsymbol{X}_{N(K),\mathcal{T}(K)} \left(\boldsymbol{X}_{N(K),\mathcal{T}(K)}^\top \boldsymbol{X}_{N(K),\mathcal{T}(K)} + \sigma_\varepsilon^2(t) I_{T(K)} \right)^{-1}\right]. \tag{1}$$

Now write

$$\boldsymbol{X}_{N(K),\mathcal{T}(K)} = \sqrt{\frac{\sigma_x^2}{K}} \boldsymbol{Z}_K,$$

where $\boldsymbol{Z}_K$ is an $N(K) \times T(K)$ matrix with i.i.d. standard normal entries. Then

$$\boldsymbol{X}_{N(K),\mathcal{T}(K)}^\top \boldsymbol{X}_{N(K),\mathcal{T}(K)} = \frac{\sigma_x^2 N(K)}{K} \left(\frac{1}{N(K)} \boldsymbol{Z}_K^\top \boldsymbol{Z}_K \right).$$

Define

$$S_K \equiv \frac{1}{N(K)} \boldsymbol{Z}_K^\top \boldsymbol{Z}_K, \qquad \gamma_K \equiv \frac{T(K)}{N(K)}, \qquad \alpha_K \equiv \frac{\sigma_\varepsilon^2(t)}{\sigma_x^2} \frac{K}{N(K)}.$$

Since

$$\frac{T(K)}{N(K)} \to \frac{t}{n}, \qquad \frac{K}{N(K)} \to \frac{1}{n},$$

we have

$$\gamma_K \to \gamma \equiv \frac{t}{n}, \qquad \alpha_K \to \alpha \equiv \frac{(1-t)\sigma_x^2 + \sigma_\eta^2}{n\sigma_x^2}.$$

Therefore (1) can be rewritten as

$$\mathcal{I}(n,t) = \lim_{K\to\infty} \frac{1}{T(K)} \operatorname{tr}\left[S_K (S_K + \alpha_K I_{T(K)})^{-1} \right].$$

Let μ_{S_K} denote the empirical spectral distribution of S_K. By the Marchenko-Pastur theorem,

$$\mu_{S_K} \Rightarrow \mu_\gamma \qquad \text{a.s.},$$

where μ_γ is the Marchenko-Pastur law with aspect ratio

$$\gamma = \frac{t}{n}.$$

Hence

$$\mathcal{I}(n,t) = \int \frac{x}{x+\alpha}\, d\mu_\gamma(x). \tag{2}$$

Now

$$\frac{x}{x+\alpha} = 1 - \alpha \frac{1}{x+\alpha},$$

so

$$\mathcal{I}(n,t) = 1 - \alpha\, m_\gamma(-\alpha),$$

where m_γ is the Stieltjes transform of the Marchenko-Pastur law:

$$m_\gamma(z) = \int \frac{1}{x-z}\, d\mu_\gamma(x) = \frac{1-\gamma-z-\sqrt{(z-a_\gamma)(z-b_\gamma)}}{2\gamma z},$$

with

$$a_\gamma = (1-\sqrt{\gamma})^2, \qquad b_\gamma = (1+\sqrt{\gamma})^2.$$

Evaluating at $z = -\alpha$,

$$m_\gamma(-\alpha) = \frac{\sqrt{(\alpha+a_\gamma)(\alpha+b_\gamma)} - (1-\gamma+\alpha)}{2\gamma\alpha}.$$

Thus

$$\mathcal{I}(n,t) = \frac{1+\gamma+\alpha-\sqrt{(\alpha+a_\gamma)(\alpha+b_\gamma)}}{2\gamma}. \tag{3}$$

It remains to simplify the right-hand side. Since

$$\gamma = \frac{t}{n}, \qquad \alpha = \frac{(1-t)\sigma_x^2 + \sigma_\eta^2}{n\sigma_x^2} = \frac{1-t+\frac{\sigma_\eta^2}{\sigma_x^2}}{n},$$

we get

$$1+\gamma+\alpha = 1 + \frac{t}{n} + \frac{1-t+\frac{\sigma_\eta^2}{\sigma_x^2}}{n} = \frac{n+1+\frac{\sigma_\eta^2}{\sigma_x^2}}{n}.$$

Also,

$$\alpha + a_\gamma = \frac{1-t+\frac{\sigma_\eta^2}{\sigma_x^2}}{n} + \left(1-\sqrt{\frac{t}{n}}\right)^2 = \frac{n+1+\frac{\sigma_\eta^2}{\sigma_x^2} - 2\sqrt{nt}}{n},$$

and similarly

$$\alpha + b_\gamma = \frac{n+1+\frac{\sigma_\eta^2}{\sigma_x^2} + 2\sqrt{nt}}{n}.$$

Therefore,

$$\sqrt{(\alpha + a_\gamma)(\alpha + b_\gamma)} = \frac{\sqrt{\left(n + 1 + \frac{\sigma_\eta^2}{\sigma_x^2}\right)^2 - 4nt}}{n}.$$

Substituting into (3),

$$\mathcal{I}(n, t) = \frac{\frac{n+1+\frac{\sigma_\eta^2}{\sigma_x^2}}{n} - \frac{\sqrt{\left(n+1+\frac{\sigma_\eta^2}{\sigma_x^2}\right)^2 - 4nt}}{n}}{2t/n},$$

which simplifies to

$$\mathcal{I}(n, t) = \frac{n + 1 + \frac{\sigma_\eta^2}{\sigma_x^2} - \sqrt{\left(n + 1 + \frac{\sigma_\eta^2}{\sigma_x^2}\right)^2 - 4nt}}{2t}.$$

This proves the claim. □

A.5 Proof of Theorem 1

Fix a sequence of nested designs $\{D(K)\}_{K \in \mathbb{N}_+}$ satisfying Assumption 1, and write

$$D(K) = (N(K), T(K), P(K)), \qquad P(K) \leq T(K),$$

with

$$\frac{N(K)}{K} \to n, \qquad \frac{T(K)}{K} \to t, \qquad \frac{P(K)}{K} \to p.$$

Let

$$\Sigma_{K,P}^{post} \equiv \mathrm{Var}\left(\boldsymbol{\beta}_{\mathcal{P}(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right).$$

Because the design is nested, the target variable can be written as

$$y_0^{(K)} = \boldsymbol{x}_{0,\mathcal{P}(K)}^\top \boldsymbol{\beta}_{\mathcal{P}(K)} + \sum_{k=P(K)+1}^{K} \beta_k x_{0k} + \eta_0.$$

Conditional on the realized dataset $\boldsymbol{d}_{D(K)}$, the second term and η_0 remain unpredictable, while uncertainty about $\boldsymbol{\beta}_{\mathcal{P}(K)}$ is summarized by $\Sigma_{K,P}^{post}$. Hence

$$\mathrm{Var}\left(y_0^{(K)} \mid \boldsymbol{d}_{D(K)}\right) = \boldsymbol{x}_{0,\mathcal{P}(K)}^\top \Sigma_{K,P}^{post} \boldsymbol{x}_{0,\mathcal{P}(K)} + \sigma_\beta^2 \left[\left(1 - \frac{P(K)}{K}\right)\sigma_x^2 + \sigma_\eta^2\right].$$

Since

$$\mathrm{Var}(y_0^{(K)}) = \sigma_\beta^2(\sigma_x^2 + \sigma_\eta^2),$$

Corollary [Value of a dataset] yields

$$v(\boldsymbol{d}_{D(K)}) = \sigma_\beta^2 \frac{P(K)}{K}\sigma_x^2 - \boldsymbol{x}_{0,\mathcal{P}(K)}^\top \Sigma_{K,P}^{post} \boldsymbol{x}_{0,\mathcal{P}(K)}. \tag{1}$$

Now condition on the training sample $(\boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)})$. Because

$$\boldsymbol{x}_{0,\mathcal{P}(K)} \sim \mathcal{N}\left(0, \frac{\sigma_x^2}{K} I_{P(K)}\right)$$

and is independent of the training sample,

$$\mathbb{E}\left[\boldsymbol{x}_{0,\mathcal{P}(K)}^\top \Sigma_{K,P}^{post} \boldsymbol{x}_{0,\mathcal{P}(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right] = \frac{\sigma_x^2}{K} \operatorname{tr}(\Sigma_{K,P}^{post}).$$

Taking conditional expectations in (1),

$$\mathbb{E}\left[v(\boldsymbol{d}_{D(K)}) \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right] = \frac{\sigma_x^2}{K} \operatorname{tr}\left(\sigma_\beta^2 I_{P(K)} - \Sigma_{K,P}^{post}\right). \tag{2}$$

By the asymptotic trace characterization of weight information,

$$\frac{1}{P(K)\sigma_\beta^2} \operatorname{tr}\left(\sigma_\beta^2 I_{P(K)} - \Sigma_{K,P}^{post}\right) \xrightarrow{p} \mathcal{I}(n,t).$$

Since $P(K)/K \to p$, (2) implies

$$\mathbb{E}\left[v(\boldsymbol{d}_{D(K)}) \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right] \xrightarrow{p} \sigma_x^2\, p\, \sigma_\beta^2\, \mathcal{I}(n,t). \tag{3}$$

It remains to show that the realized value is asymptotically equal to its conditional expectation. Subtracting (2) from (1),

$$v(\boldsymbol{d}_{D(K)}) - \mathbb{E}\left[v(\boldsymbol{d}_{D(K)}) \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right] = -\left(\boldsymbol{x}_{0,\mathcal{P}(K)}^\top \Sigma_{K,P}^{post} \boldsymbol{x}_{0,\mathcal{P}(K)} - \frac{\sigma_x^2}{K} \operatorname{tr}(\Sigma_{K,P}^{post})\right).$$

Conditional on the training sample, the variance of the quadratic form on the right-hand side is

$$2\left(\frac{\sigma_x^2}{K}\right)^2 \operatorname{tr}\left((\Sigma_{K,P}^{post})^2\right).$$

Moreover,

$$0 \preceq \Sigma_{K,P}^{post} \preceq \sigma_\beta^2 I_{P(K)},$$

so

$$\operatorname{tr}\left((\Sigma_{K,P}^{post})^2\right) \le \sigma_\beta^4 P(K).$$

Therefore

$$\operatorname{Var}\left(v(\boldsymbol{d}_{D(K)}) - \mathbb{E}\left[v(\boldsymbol{d}_{D(K)}) \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right] \,\middle|\, \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\right) \le 2\sigma_x^4 \sigma_\beta^4 \frac{P(K)}{K^2},$$

and the right-hand side converges to 0 because $P(K)/K \to p < \infty$. Hence

$$v(\boldsymbol{d}_{D(K)}) - \mathbb{E}\big[v(\boldsymbol{d}_{D(K)}) \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\big] \xrightarrow{p} 0. \tag{4}$$

Combining (3) and (4),

$$v(\boldsymbol{d}_{D(K)}) \xrightarrow{p} \sigma_x^2\, p\, \sigma_\beta^2\, \mathcal{I}(n,t).$$

This proves the claim.

B Proof of Proposition 2

For each K, recall that

$$\mathcal{S}_K \equiv \frac{1}{T(K)} \operatorname{tr}\big(\boldsymbol{X}_{N(K),\mathcal{T}(K)}^\top \boldsymbol{X}_{N(K),\mathcal{T}(K)}\big),$$

and

$$\mathcal{E}_K \equiv \frac{1}{\sigma_\beta^2} \operatorname{Var}\big(u_0^{(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\big),$$

where

$$u_0^{(K)} \equiv y_0^{(K)} - \boldsymbol{x}_{0,\mathcal{T}(K)}^\top \mathbb{E}\big[\boldsymbol{\beta}_{\mathcal{T}(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\big].$$

We first establish the limit of $\mathcal{S}_K$. Since

$$\mathcal{S}_K = \frac{1}{T(K)} \sum_{k=1}^{T(K)} \sum_{i=1}^{N(K)} (x_{ik}^{(K)})^2,$$

and each $x_{ik}^{(K)} \sim N(0, \sigma_x^2/K)$, we have

$$\mathbb{E}[\mathcal{S}_K] = \frac{N(K)T(K)}{T(K)} \frac{\sigma_x^2}{K} = \frac{N(K)}{K} \sigma_x^2 \to n\sigma_x^2.$$

Moreover, using $\operatorname{Var}((x_{ik}^{(K)})^2) = 2(\sigma_x^2/K)^2$ and independence across (i,k),

$$\operatorname{Var}(\mathcal{S}_K) = \frac{1}{T(K)^2} N(K)T(K)\, 2\frac{\sigma_x^4}{K^2} = 2\sigma_x^4 \frac{N(K)}{T(K)K^2} \to 0.$$

Hence, by Chebyshev's inequality,

$$\mathcal{S}_K \xrightarrow{p} n\sigma_x^2 \equiv \mathcal{S}(n).$$

We next establish the limit of $\mathcal{E}_K$. Write

$$\delta_K \equiv \boldsymbol{\beta}_{\mathcal{T}(K)} - \mathbb{E}\big[\boldsymbol{\beta}_{\mathcal{T}(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\big].$$

Using

$$y_0^{(K)} = \boldsymbol{x}_{0,\mathcal{T}(K)}^\top \boldsymbol{\beta}_{\mathcal{T}(K)} + \varepsilon_0^{(K)},$$

we obtain

$$u_0^{(K)} = \varepsilon_0^{(K)} + \boldsymbol{x}_{0,\mathcal{T}(K)}^\top \delta_K.$$

Conditional on $(\boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)})$, the random vector $\boldsymbol{x}_{0,\mathcal{T}(K)}$ is independent of both $\varepsilon_0^{(K)}$ and δ_K, has mean zero, and $\varepsilon_0^{(K)}$ is independent of the training sample. Therefore the cross term vanishes, and

$$\mathrm{Var}\big(u_0^{(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\big) = \mathrm{Var}(\varepsilon_0^{(K)}) + \mathrm{Var}\big(\boldsymbol{x}_{0,\mathcal{T}(K)}^\top \delta_K \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\big).$$

Since

$$\mathrm{Var}(\varepsilon_0^{(K)}) = \sigma_\beta^2 \left[\left(1 - \frac{T(K)}{K}\right) \sigma_x^2 + \sigma_\eta^2 \right],$$

and, conditional on the training sample,

$$\mathrm{Var}\big(\boldsymbol{x}_{0,\mathcal{T}(K)}^\top \delta_K \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\big) = \frac{\sigma_x^2}{K} \mathrm{tr}\big[\mathrm{Var}\big(\boldsymbol{\beta}_{\mathcal{T}(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\big)\big],$$

it follows that

$$\mathcal{E}_K = \left(1 - \frac{T(K)}{K}\right) \sigma_x^2 + \sigma_\eta^2 + \frac{\sigma_x^2}{K\sigma_\beta^2} \mathrm{tr}\big[\mathrm{Var}\big(\boldsymbol{\beta}_{\mathcal{T}(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\big)\big].$$

The first term converges to

$$(1-t)\sigma_x^2 + \sigma_\eta^2 = \sigma_\varepsilon^2(t).$$

For the second term, by the definition of $\mathcal{I}(n,t)$ as the asymptotic fraction of prior variance learned about the training weights,

$$\frac{1}{T(K)\sigma_\beta^2} \mathrm{tr}\big[\mathrm{Var}\big(\boldsymbol{\beta}_{\mathcal{T}(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\big)\big] \xrightarrow{p} 1 - \mathcal{I}(n,t).$$

Since $T(K)/K \to t$,

$$\frac{\sigma_x^2}{K\sigma_\beta^2} \mathrm{tr}\big[\mathrm{Var}\big(\boldsymbol{\beta}_{\mathcal{T}(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\big)\big] = \frac{T(K)}{K}\sigma_x^2 \cdot \frac{1}{T(K)\sigma_\beta^2} \mathrm{tr}\big[\mathrm{Var}\big(\boldsymbol{\beta}_{\mathcal{T}(K)} \mid \boldsymbol{y}_{N(K)}, \boldsymbol{X}_{N(K),\mathcal{T}(K)}\big)\big] \xrightarrow{p} t\sigma_x^2(1-\mathcal{I}(n,t$$

Therefore,

$$\mathcal{E}_K \xrightarrow{p} \sigma_\varepsilon^2(t) + t\sigma_x^2(1 - \mathcal{I}(n,t)) \equiv \mathcal{E}(\mathcal{I}(n,t); t).$$

Finally, by definition,

$$\Phi(\mathcal{I}; n, t) = \frac{\mathcal{S}(n)}{\mathcal{S}(n) + \mathcal{E}(\mathcal{I}; t)} = \frac{n\sigma_x^2}{n\sigma_x^2 + \sigma_\varepsilon^2(t) + t\sigma_x^2(1-\mathcal{I})}.$$

Hence the fixed-point equation

$$\mathcal{I} = \Phi(\mathcal{I}; n, t)$$

is equivalent to

$$t\sigma_x^2 \mathcal{I}^2 - (n\sigma_x^2 + \sigma_x^2 + \sigma_\eta^2)\mathcal{I} + n\sigma_x^2 = 0,$$

where we used

$$\sigma_\varepsilon^2(t) = (1-t)\sigma_x^2 + \sigma_\eta^2.$$

This is exactly the quadratic characterization of $\mathcal{I}(n, t)$ established earlier. Therefore $\mathcal{I}(n, t)$ is a fixed point of $\Phi(\cdot; n, t)$, and uniqueness in $[0, 1]$ follows from the uniqueness result proved there.

Proposition 4 (Marginal rate of technical substitution)**.** *Fix a feasible target value $v \in (0, \hat{p})$ and a training scope $t \in (v, 1]$. Then*

$$\mu(v, t) = \begin{cases} \dfrac{v(\xi + 1 - v)}{(t - v)^2}, & \text{if } v < t < \hat{p}, \\[2ex] \dfrac{v^2}{\hat{p}(\hat{p} - v)}, & \text{if } \hat{p} < t \leq 1. \end{cases}$$

Therefore:

1. *For fixed v, $\mu(v, t)$ is strictly decreasing in t on $(v, \hat{p})$ and constant on $(\hat{p}, 1]$. Moreover,*

$$\lim_{t \to \hat{p}^-} \mu(v, t) = \frac{v(\xi + 1 - v)}{(\hat{p} - v)^2} > \lim_{t \to \hat{p}^+} \mu(v, t) = \frac{v^2}{\hat{p}(\hat{p} - v)}.$$

2. *For fixed $t > v$, $\mu(v, t)$ is strictly increasing in v.*

Proof. When $t < \hat{p}$, target and training scope coincide, so $\tilde{p}(t) = t$ and

$$\tilde{V}(n, t) = V(n, t, t).$$

Solving $\tilde{V}(n, t) = v$ for n yields

$$\tilde{n}(v, t) = \frac{v(\xi + 1 - v)}{t - v}, \qquad v < t < \hat{p}.$$

When $t > \hat{p}$, the target constraint binds, so $\tilde{p}(t) = \hat{p}$ and

$$\tilde{V}(n, t) = V(n, t, \hat{p}).$$

Solving again for n gives

$$\tilde{n}(v,t) = \frac{v\left(\xi + 1 - \dfrac{tv}{\hat{p}}\right)}{\hat{p} - v}, \qquad \hat{p} < t \leq 1.$$

By definition,

$$MRTS_{tn}(v,t) = -\frac{d\tilde{n}}{dt}(v,t),$$

so

$$MRTS_{tn}(v,t) = \begin{cases} \dfrac{v(\xi + 1 - v)}{(t-v)^2}, & \text{if } v < t < \hat{p}, \\[2ex] \dfrac{v^2}{\hat{p}(\hat{p} - v)}, & \text{if } \hat{p} < t \leq 1. \end{cases}$$

For part (i), direct differentiation gives

$$\partial_t MRTS_{tn}(v,t) = \begin{cases} -\dfrac{2v(\xi + 1 - v)}{(t-v)^3} < 0, & \text{if } v < t < \hat{p}, \\[2ex] 0, & \text{if } \hat{p} < t \leq 1, \end{cases}$$

and

$$\partial_\xi MRTS_{tn}(v,t) = \begin{cases} \dfrac{v}{(t-v)^2} > 0, & \text{if } v < t < \hat{p}, \\[2ex] 0, & \text{if } \hat{p} < t \leq 1. \end{cases}$$

For part (ii), taking one-sided limits at $t = \hat{p}$ gives

$$MRTS_{tn}^{-}(v) = \frac{v(\xi + 1 - v)}{(\hat{p} - v)^2}, \qquad MRTS_{tn}^{+}(v) = \frac{v^2}{\hat{p}(\hat{p} - v)}.$$

Moreover,

$$MRTS_{tn}^{-}(v) - MRTS_{tn}^{+}(v) = \frac{v\left[(\hat{p} - v)^2 + \hat{p}(\xi + 1 - \hat{p})\right]}{\hat{p}(\hat{p} - v)^2} > 0.$$

For part (iii), direct differentiation yields

$$\partial_v MRTS_{tn}(v,t) = \begin{cases} \dfrac{(\xi + 1)(t + v) - 2tv}{(t-v)^3} > 0, & \text{if } v < t < \hat{p}, \\[2ex] \dfrac{v(2\hat{p} - v)}{\hat{p}(\hat{p} - v)^2} > 0, & \text{if } \hat{p} < t \leq 1. \end{cases}$$

Hence $MRTS_{tn}(v,t)$ is strictly increasing in v for every fixed $t > v$. □

Lemma 2. *The function $\gamma(v,t)$ is continuous, strictly decreasing in v and strictly increasing in t.*

Furthermore, for every fixed $v \in (0, \hat{p})$,

$$\lim_{t \to v^+} \gamma(v, t) \in [0, \infty), \qquad \lim_{t \to 1^-} \gamma(v, t) = +\infty,$$

and for every fixed $t \in (0, 1)$,

$$\lim_{v \to 0^+} \gamma(v, t) = +\infty, \qquad \lim_{v \to \min\{t, \hat{p}\}^-} \gamma(v, t) \in [0, \infty).$$

Proof. Let

$$MCR(n, t) \equiv \frac{\Gamma_t(n, t)}{\Gamma_n(n, t)}.$$

By the maintained assumptions, MCR is continuous, strictly decreasing in n, and strictly increasing in t. By Proposition 4, the function $\tilde{n}(v, t)$ is continuous on

$$\mathcal{D} = \{(v, t) \,:\, 0 < v < \hat{p},\ v < t < 1\},$$

strictly increasing in v, and strictly decreasing in t. Since

$$\gamma(v, t) = MCR(\tilde{n}(v, t), t),$$

continuity of γ follows immediately from continuity of MCR and $\tilde{n}$.
To prove monotonicity in v, fix $t \in (0, 1)$ and take

$$0 < v_1 < v_2 < \min\{t, \hat{p}\}.$$

Since $\tilde{n}(v, t)$ is strictly increasing in v,

$$\tilde{n}(v_1, t) < \tilde{n}(v_2, t).$$

Because MCR is strictly decreasing in its first argument,

$$\gamma(v_2, t) = MCR(\tilde{n}(v_2, t), t) < MCR(\tilde{n}(v_1, t), t) = \gamma(v_1, t).$$

Hence $\gamma(v, t)$ is strictly decreasing in v.
To prove monotonicity in t, fix $v \in (0, \hat{p})$ and take

$$v < t_1 < t_2 < 1.$$

Since $\tilde{n}(v, t)$ is strictly decreasing in t,

$$\tilde{n}(v, t_2) < \tilde{n}(v, t_1).$$

Then

$$\begin{aligned}
\gamma(v, t_2) &= MCR(\tilde{n}(v, t_2), t_2) \\
&> MCR(\tilde{n}(v, t_2), t_1) && \text{since } MCR \text{ is strictly increasing in } t \\
&> MCR(\tilde{n}(v, t_1), t_1) && \text{since } MCR \text{ is strictly decreasing in } n \text{ and } \tilde{n}(v, t_2) < \tilde{n}(v, t_1) \\
&= \gamma(v, t_1).
\end{aligned}$$

Thus $\gamma(v, t)$ is strictly increasing in t.
Next fix $v \in (0, \hat{p})$. Since $\gamma(v, t)$ is increasing in t on $(v, 1)$ and strictly positive, the one-sided limit

$$\lim_{t \to v^+} \gamma(v, t)$$

exists in $[0, \infty]$. In fact it is finite: for any $t_0 \in (v, 1)$ and all $t \in (v, t_0]$,

$$0 < \gamma(v, t) \leq \gamma(v, t_0) < \infty.$$

Therefore

$$\lim_{t \to v^+} \gamma(v, t) \in [0, \infty).$$

Now fix $t \in (0, 1)$. Since $\gamma(v, t)$ is decreasing in v on $(0, \min\{t, \hat{p}\})$ and strictly positive, the one-sided limit

$$\lim_{v \to \min\{t, \hat{p}\}^-} \gamma(v, t)$$

exists in $[0, \infty]$. It is again finite: for any $v_0 \in (0, \min\{t, \hat{p}\})$ and all $v \in [v_0, \min\{t, \hat{p}\})$,

$$0 < \gamma(v, t) \leq \gamma(v_0, t) < \infty.$$

Hence

$$\lim_{v \to \min\{t, \hat{p}\}^-} \gamma(v, t) \in [0, \infty).$$

Finally, fix $t \in (0, 1)$. By the formula for $\tilde{n}(v, t)$, one has

$$\tilde{n}(v, t) \to 0 \qquad \text{as } v \to 0^+.$$

Because Γ_t is continuous and strictly positive at $(0, t)$, there exist $\varepsilon > 0$ and $c > 0$ such that

$$\Gamma_t(n, t) \geq c \qquad \text{for all } n \in [0, \varepsilon].$$

Since $\Gamma_n(n, t) \to 0$ as $n \to 0^+$, for v small enough one has $\tilde{n}(v, t) \in [0, \varepsilon]$, and therefore

$$\gamma(v, t) = \frac{\Gamma_t(\tilde{n}(v, t), t)}{\Gamma_n(\tilde{n}(v, t), t)} \geq \frac{c}{\Gamma_n(\tilde{n}(v, t), t)}.$$

The denominator converges to zero as $v \to 0^+$, so

$$\lim_{v \to 0^+} \gamma(v, t) = +\infty.$$

This proves the lemma. □

Proposition 6 (Cost-minimizing design)**.** *The cost-minimization problem admits a solution. The cost-minimizing design is* $p^*(v) = \min\{t^*(v), \hat{p}\}$, $n^*(v) = \tilde{n}(v, t^*(v))$ *and*

$$t^*(v) \in \begin{cases} \left\{ v + \sqrt{\dfrac{v(\xi + 1 - v)}{\gamma}} \right\}, & \text{if } 0 \le v \le v_-(\hat{p}, \gamma, \xi), \\ \left\{ \hat{p} \right\}, & \text{if } v_-(\hat{p}, \gamma, \xi) \le v < v_+(\hat{p}, \gamma), \\ [\hat{p}, 1], & \text{if } v = v_+(\hat{p}, \gamma), \\ \{1\}, & \text{if } v_+(\hat{p}, \gamma) < v < \hat{p}. \end{cases}$$

Proof. The proof has three steps.

Step 1: for each feasible $v \in (0, \hat{p})$*, the reduced cost minimization problem has a unique solution.* Fix $v \in (0, \hat{p})$, and define

$$\tilde{\Gamma}(v, t) \equiv \Gamma(\tilde{n}(v, t), t), \qquad t \in (v, 1).$$

First, $\tilde{\Gamma}(v, t) \to +\infty$ at both ends of the feasible interval.
As $t \to v^+$, we have $\tilde{n}(v, t) \to +\infty$. Since Γ is strictly increasing in t,

$$\tilde{\Gamma}(v, t) = \Gamma(\tilde{n}(v, t), t) \ge \Gamma(\tilde{n}(v, t), v),$$

and the right-hand side tends to $+\infty$ because $\Gamma(n, v) \to +\infty$ as $n \to +\infty$. Hence

$$\lim_{t \to v^+} \tilde{\Gamma}(v, t) = +\infty.$$

As $t \to 1^-$, the scale $\tilde{n}(v, t)$ remains bounded, since on $(\hat{p}, 1]$,

$$\tilde{n}(v, t) = \frac{v\left(\xi + 1 - \frac{tv}{\hat{p}}\right)}{\hat{p} - v}.$$

Moreover, the Inanda conditions imply that collecting training scope close to 1 is prohibitively expensive. Therefore

$$\lim_{t \to 1^-} \tilde{\Gamma}(v, t) = +\infty.$$

Next, $\tilde{\Gamma}(v, t)$ is continuous on $(v, 1)$, including at $t = \hat{p}$, because the two formulas for $\tilde{n}(v, t)$

agree at $t = \hat{p}$:

$$\lim_{t \to \hat{p}^-} \tilde{n}(v, t) = \frac{v(\xi + 1 - v)}{\hat{p} - v} = \lim_{t \to \hat{p}^+} \tilde{n}(v, t).$$

Hence $\tilde{\Gamma}(v, \cdot)$ attains a global minimum on $(v, 1)$.
For $t \neq \hat{p}$, the chain rule gives

$$\partial_t \tilde{\Gamma}(v, t) = \Gamma_n(\tilde{n}(v, t), t)\, \partial_t \tilde{n}(v, t) + \Gamma_t(\tilde{n}(v, t), t).$$

Using $\mu(v, t) = -\partial_t \tilde{n}(v, t)$ and

$$\gamma(v, t) = \frac{\Gamma_t(\tilde{n}(v, t), t)}{\Gamma_n(\tilde{n}(v, t), t)},$$

we obtain

$$\partial_t \tilde{\Gamma}(v, t) = -\Gamma_n(\tilde{n}(v, t), t)\big(\mu(v, t) - \gamma(v, t)\big) = -\Gamma_n(\tilde{n}(v, t), t)\phi(v, t).$$

Since $\Gamma_n > 0$, the sign of $\partial_t \tilde{\Gamma}(v, t)$ is the opposite of the sign of $\phi(v, t)$.
By the previous corollary, for each fixed v, the function $t \mapsto \phi(v, t)$ is strictly decreasing, with

$$\lim_{t \to v^+} \phi(v, t) = +\infty, \qquad \lim_{t \to 1^-} \phi(v, t) = -\infty.$$

Therefore $t \mapsto \tilde{\Gamma}(v, t)$ first decreases and then increases. Hence the minimizer $t^*(v)$ is unique.

Step 2: existence and uniqueness of the thresholds v_- and v_+.
Define the one-sided values of the net scope benefit at the kink:

$$\phi_-(v) \equiv \lim_{t \to \hat{p}^-} \phi(v, t), \qquad \phi_+(v) \equiv \lim_{t \to \hat{p}^+} \phi(v, t).$$

By Proposition 4 and the monotonicity of $\gamma(v, t)$,

$$\phi_-(v) = \frac{v(\xi + 1 - v)}{(\hat{p} - v)^2} - \gamma(v, \hat{p}), \qquad \phi_+(v) = \frac{v^2}{\hat{p}(\hat{p} - v)} - \gamma(v, \hat{p}).$$

Both ϕ_- and ϕ_+ are continuous and strictly increasing in v, because the MRTS terms are strictly increasing in v and $\gamma(v, \hat{p})$ is strictly decreasing in v.
At $v = 0$,

$$\phi_-(0) = \phi_+(0) = -\gamma(0, \hat{p}) < 0.$$

As $v \to \hat{p}^-$, both MRTS terms diverge to $+\infty$, whereas $\gamma(v, \hat{p}) \leq \gamma(0, \hat{p})$. Thus

$$\lim_{v \to \hat{p}^-} \phi_-(v) = +\infty, \qquad \lim_{v \to \hat{p}^-} \phi_+(v) = +\infty.$$

By the intermediate value theorem, there exist unique

$$v_- \in (0, \hat{p}), \qquad v_+ \in (0, \hat{p})$$

such that

$$\phi_-(v_-) = 0, \qquad \phi_+(v_+) = 0.$$

Finally, for every $v \in (0, \hat{p})$,

$$\phi_-(v) - \phi_+(v) = \frac{v(\xi + 1 - v)}{(\hat{p} - v)^2} - \frac{v^2}{\hat{p}(\hat{p} - v)} > 0$$

by Proposition 4. Hence $\phi_-(v) > \phi_+(v)$ for all v. Evaluating at $v = v_-$ gives

$$\phi_+(v_-) < \phi_-(v_-) = 0.$$

Since ϕ_+ is strictly increasing and $\phi_+(v_+) = 0$, it follows that

$$v_- < v_+.$$

Step 3: characterization of the minimizer.
Consider three cases.

Case 1: $0 < v < v_-$. Then $\phi_-(v) < 0$. Since $\phi(v, t) \to +\infty$ as $t \to v^+$ and $t \mapsto \phi(v, t)$ is strictly decreasing, there exists a unique

$$t^*(v) \in (v, \hat{p})$$

such that $\phi(v, t^*(v)) = 0$, equivalently

$$\gamma(v, t^*(v)) = \mu(v, t^*(v)).$$

Because $\tilde{\Gamma}$ decreases where $\phi > 0$ and increases where $\phi < 0$, this is the unique minimizer.

Case 2: $v_- \leq v \leq v_+$. Then

$$\phi_-(v) \geq 0 \geq \phi_+(v).$$

Since $\phi(v, t)$ is strictly decreasing in t, we have

$$\phi(v, t) > 0 \quad \text{for } t \in (v, \hat{p}), \qquad \phi(v, t) < 0 \quad \text{for } t \in (\hat{p}, 1).$$

Therefore

$$\partial_t \tilde{\Gamma}(v, t) < 0 \quad \text{for } t < \hat{p}, \qquad \partial_t \tilde{\Gamma}(v, t) > 0 \quad \text{for } t > \hat{p}.$$

So $\tilde{\Gamma}(v, t)$ decreases up to the kink and increases afterwards, and the unique minimizer is

$$t^*(v) = \hat{p}.$$

Case 3: $v_+ < v < \hat{p}$. Then $\phi_+(v) > 0$. Since $\phi(v, t) \to -\infty$ as $t \to 1^-$ and $t \mapsto \phi(v, t)$ is strictly decreasing, there exists a unique

$$t^*(v) \in (\hat{p}, 1)$$

such that $\phi(v, t^*(v)) = 0$, equivalently

$$\gamma(v, t^*(v)) = \mu(v, t^*(v)).$$

Again, this is the unique minimizer.
Combining the three cases yields the stated characterization of $t^*(v)$. Finally,

$$p^*(v) = \min\{t^*(v), \hat{p}\}, \qquad n^*(v) = \tilde{n}(v, t^*(v))$$

follow directly from the reduction of the problem established above. □

Corollary 9 (Cost of predictive value)**.** *For every target $v \in (0, \hat{p})$,*

$$\tilde{C}(v) = \begin{cases} \gamma v + 2\sqrt{\gamma v(\xi + 1 - v)}, & \text{if } v \in (0, v_-(\hat{p}, \gamma, \xi)), \\ \gamma \hat{p} + \dfrac{v(\xi + 1 - v)}{\hat{p} - v}, & \text{if } v \in [v_-(\hat{p}, \gamma, \xi), v_+(\hat{p}, \gamma)], \\ \gamma + v\left(\dfrac{\xi}{\hat{p} - v} + \dfrac{1}{\hat{p}}\right), & \text{if } v \in (v_+(\hat{p}, \gamma), \hat{p}). \end{cases}$$

Proof. Substitute the optimizer from Proposition 6 into

$$\tilde{C}(v) = \gamma t^*(v) + n^*(v).$$

In the low-value region,

$$t^* = v + \sqrt{\frac{v(\xi + 1 - v)}{\gamma}}, \qquad n^* = \frac{v(\xi + 1 - v)}{t^* - v} = \sqrt{\gamma v(\xi + 1 - v)},$$

so

$$\tilde{C}(v) = \gamma v + 2\sqrt{\gamma v(\xi + 1 - v)}.$$

In the medium-value region,

$$t^* = \hat{p}, \qquad n^* = \frac{v(\xi + 1 - v)}{\hat{p} - v},$$

so

$$\tilde{C}(v) = \gamma \hat{p} + \frac{v(\xi + 1 - v)}{\hat{p} - v}.$$

In the high-value region,

$$t^* = 1, \qquad n^* = \frac{v\left(\xi + 1 - \frac{v}{\hat{p}}\right)}{\hat{p} - v} = v\left(\frac{\xi}{\hat{p} - v} + \frac{1}{\hat{p}}\right),$$

so

$$\tilde{C}(v) = \gamma + v\left(\frac{\xi}{\hat{p} - v} + \frac{1}{\hat{p}}\right).$$

This is the stated piecewise formula. □

Proof. Differentiate the three branches of $\tilde{C}(v)$.
On $(0, v_-)$,

$$\tilde{C}''(v) = -\frac{\sqrt{\gamma}(\xi + 1)^2}{2\left[v(\xi + 1 - v)\right]^{3/2}} < 0,$$

so $MC(v) = \tilde{C}'(v)$ is strictly decreasing.
On (v_-, v_+),

$$\tilde{C}''(v) = \frac{2\hat{p}(\xi + 1 - \hat{p})}{(\hat{p} - v)^3} > 0,$$

so $MC(v)$ is strictly increasing.
On $(v_+, \hat{p})$,

$$\tilde{C}''(v) = \frac{2\hat{p}\,\xi}{(\hat{p} - v)^3} > 0,$$

so $MC(v)$ is again strictly increasing.
Finally, the one-sided marginal costs at v_+ satisfy

$$MC(v_+^-) - MC(v_+^+) = \frac{2\sqrt{\gamma}(1 - \hat{p})\sqrt{\gamma + 4}}{\hat{p}\left(\gamma + 2 - \sqrt{\gamma(\gamma + 4)}\right)} > 0,$$

so marginal cost falls discontinuously at v_+. □

Proposition 6 (Cost-minimizing design)**.** *The cost-minimization problem admits a solution.*

The cost-minimizing design is $p^*(v) = \min\{t^*(v), \hat{p}\}$, $n^*(v) = \tilde{n}(v, t^*(v))$ *and*

$$t^*(v) \in \begin{cases} \left\{ v + \sqrt{\dfrac{v(\xi + 1 - v)}{\gamma}} \right\}, & \text{if } 0 \le v \le v_-(\hat{p}, \gamma, \xi), \\ \left\{ \hat{p} \right\}, & \text{if } v_-(\hat{p}, \gamma, \xi) \le v < v_+(\hat{p}, \gamma), \\ [\hat{p}, 1], & \text{if } v = v_+(\hat{p}, \gamma), \\ \{1\}, & \text{if } v_+(\hat{p}, \gamma) < v < \hat{p}. \end{cases}$$

Proof. Fix $v \in (0, \hat{p})$, and define

$$g_v(t) \equiv \gamma t + \tilde{n}(v, t), \qquad t \in [v, 1].$$

Since g_v is continuous on the compact set $[v, 1]$, an optimum exists.
On each differentiable region,

$$g_v'(t) = \gamma - MRTS_{tn}(v, t).$$

First consider $t \in (v, \hat{p})$. By Proposition 4,

$$g_v'(t) = \gamma - \frac{v(\xi + 1 - v)}{(t - v)^2}.$$

There is at most one interior critical point because $MRTS_{tn}(v, t)$ is strictly decreasing in t. Solving $g_v'(t) = 0$ gives

$$t = v + \sqrt{\frac{v(\xi + 1 - v)}{\gamma}}.$$

This lies in $(v, \hat{p})$ if and only if $v \le v_-$.
At the kink $t = \hat{p}$, the one-sided derivatives are

$$g_{v,-}'(\hat{p}) = \gamma - MRTS_{tn}^-(v), \qquad g_{v,+}'(\hat{p}) = \gamma - MRTS_{tn}^+(v).$$

Hence:

- if $v_- < v < v_+$, then $MRTS_{tn}^-(v) > \gamma > MRTS_{tn}^+(v)$, so $g_{v,-}'(\hat{p}) < 0 < g_{v,+}'(\hat{p})$, and therefore $t^* = \hat{p}$;
- if $v = v_+$, then $g_{v,-}'(\hat{p}) < 0 = g_{v,+}'(\hat{p})$, while $g_v' \equiv 0$ on $(\hat{p}, 1]$, so every $t \in [\hat{p}, 1]$ is optimal;
- if $v > v_+$, then $MRTS_{tn}^+(v) > \gamma$, so $g_v'(t) < 0$ on $(\hat{p}, 1]$, and therefore $t^* = 1$.

Collecting the cases gives

$$t^* \in \begin{cases} \left\{ v + \sqrt{\dfrac{v(\xi + 1 - v)}{\gamma}} \right\}, & \text{if } 0 \le v \le v_-, \\ \{\hat{p}\}, & \text{if } v_- < v < v_+, \\ [\hat{p}, 1], & \text{if } v = v_+, \\ \{1\}, & \text{if } v_+ < v < \hat{p}. \end{cases}$$

The associated choices are $p^* = \min\{t^*, \hat{p}\}$ and $n^* = \tilde{n}(v, t^*)$. □

Proposition 7 (Minimum efficient scale)**.** *A minimum efficient scale exists and cannot lie in the unconstrained-scope region* $(0, v_-)$*. Moreover*

- *If* $\xi \ge 1$ *or* $\gamma \ge (1 - \xi)^2/\xi$*, the minimum efficient scale is in the same-scope region:*

$$v_{MES}(\hat{p}, \gamma, \xi) \in [v_-(\hat{p}, \gamma, \xi), v_+(\hat{p}, \gamma)).$$

- *Otherwise, let*

$$p^\dagger(\gamma, \xi) = \frac{\gamma + 4\sqrt{\xi}\left(\sqrt{\gamma} + \sqrt{\xi}\right)}{\gamma + 4} \in [0, 1).$$

  The minimum efficient scale is

$$v_{MES}(\hat{p}, \gamma, \xi) \in \begin{cases} [v_-(\hat{p}, \gamma, \xi), v_+(\hat{p}, \gamma)), & \text{if } \hat{p} \le p^\dagger(\gamma, \xi), \\ (v_+(\hat{p}, \gamma), \hat{p}), & \text{if } \hat{p} > p^\dagger(\gamma, \xi). \end{cases}$$

Proof. Average cost on the three regions is

$$AC_L(v) = \gamma + 2\sqrt{\frac{\gamma(\xi + 1 - v)}{v}},$$

$$AC_M(v) = \frac{\gamma \hat{p}}{v} + \frac{\xi + 1 - v}{\hat{p} - v},$$

and

$$AC_H(v) = \frac{\gamma}{v} + \frac{\xi}{\hat{p} - v} + \frac{1}{\hat{p}}.$$

On the low-value region,

$$AC_L'(v) = -\frac{\sqrt{\gamma}(\xi + 1)}{v^{3/2}\sqrt{\xi + 1 - v}} < 0,$$

so the MES cannot lie in $(0, v_-)$.
On the medium-value branch,

$$AC'_M(v) = -\frac{\gamma\hat{p}}{v^2} + \frac{\xi + 1 - \hat{p}}{(\hat{p} - v)^2}.$$

Hence there is a unique minimizer,

$$v_M^{MES} = \frac{\hat{p}}{1 + \sqrt{(\xi + 1 - \hat{p})/(\gamma\hat{p})}}.$$

On the high-value branch,

$$AC'_H(v) = -\frac{\gamma}{v^2} + \frac{\xi}{(\hat{p} - v)^2},$$

so there is a unique minimizer,

$$v_H^{MES} = \frac{\hat{p}}{1 + \sqrt{\xi/\gamma}}.$$

The corresponding minimum average costs are

$$\underline{\sigma}_\beta^2 = 1 + \frac{\left(\sqrt{\gamma\hat{p}} + \sqrt{\xi + 1 - \hat{p}}\right)^2}{\hat{p}},$$

and

$$\bar{\sigma}_\beta^2 = \frac{1 + \left(\sqrt{\gamma} + \sqrt{\xi}\right)^2}{\hat{p}}.$$

Comparing them,

$$\underline{\sigma}_\beta^2 \leq \bar{\sigma}_\beta^2 \iff \hat{p} \leq \frac{\gamma + 4\sqrt{\xi}(\sqrt{\gamma} + \sqrt{\xi})}{\gamma + 4} \equiv p^\dagger(\gamma, \xi).$$

Therefore the global MES is the medium-branch minimizer when $\hat{p} \leq p^\dagger(\gamma, \xi)$, and the high-branch minimizer when $\hat{p} > p^\dagger(\gamma, \xi)$.
Finally,

$$p^\dagger(\gamma, \xi) \geq 1 \iff \sqrt{\xi\gamma} + \xi \geq 1.$$

Thus if $\xi \geq 1$, then $p^\dagger(\gamma, \xi) \geq 1$ automatically; and if $\xi < 1$, this is equivalent to

$$\gamma \geq \frac{(1 - \xi)^2}{\xi}.$$

□

Proposition 8 (Profit-maximizing relative value)**.** *Fix* $(\gamma, \hat{p}, \xi)$.

If $\hat{p} > p^{\dagger}(\gamma, \xi)$, then the unique global maximizer is

$$v^*(\sigma_\beta^2) = \begin{cases} 0, & \text{if } \sigma_\beta^2 < \bar{\sigma}_\beta^2, \\ \hat{p} - \sqrt{\dfrac{\xi \hat{p}}{\sigma_\beta^2 - 1/\hat{p}}}, & \text{if } \sigma_\beta^2 \geq \bar{\sigma}_\beta^2. \end{cases}$$

If $\hat{p} \leq p^{\dagger}(\gamma, \xi)$, then the unique global maximizer is

$$v^*(\sigma_\beta^2) = \begin{cases} 0, & \text{if } \sigma_\beta^2 < \underline{\sigma}_\beta^2, \\ \hat{p} - \sqrt{\dfrac{\hat{p}(\xi + 1 - \hat{p})}{\sigma_\beta^2 - 1}}, & \text{if } \underline{\sigma}_\beta^2 \leq \sigma_\beta^2 < \tilde{\sigma}_\beta^2, \\ \hat{p} - \sqrt{\dfrac{\xi \hat{p}}{\sigma_\beta^2 - 1/\hat{p}}}, & \text{if } \sigma_\beta^2 > \tilde{\sigma}_\beta^2. \end{cases}$$

At $\sigma_\beta^2 = \tilde{\sigma}_\beta^2$, both branch maximizers are optimal.

Proof. By Proposition 7, the low-value region is never globally optimal, so it is enough to compare the same-scope and expanded-scope branches.
On the same-scope branch,

$$\pi_M(v; \sigma_\beta^2) = \sigma_\beta^2 v - \gamma \hat{p} - \frac{v(\xi + 1 - v)}{\hat{p} - v}, \qquad v \in [v_-, v_+).$$

On the expanded-scope branch,

$$\pi_H(v; \sigma_\beta^2) = \sigma_\beta^2 v - \gamma - v \left(\frac{\xi}{\hat{p} - v} + \frac{1}{\hat{p}} \right), \qquad v \in (v_+, \hat{p}).$$

Each branch objective is strictly concave in v, because

$$\frac{\partial^2 \pi_M}{\partial v^2} = -\frac{2\hat{p}(\xi + 1 - \hat{p})}{(\hat{p} - v)^3} < 0, \qquad \frac{\partial^2 \pi_H}{\partial v^2} = -\frac{2\hat{p}\xi}{(\hat{p} - v)^3} < 0.$$

Hence each branch has a unique branchwise maximizer, characterized by the first-order condition

$$\sigma_\beta^2 = MC(v).$$

Solving this condition on the same-scope branch yields

$$v_M(\sigma_\beta^2) = \hat{p} - \sqrt{\frac{\hat{p}(\xi + 1 - \hat{p})}{\sigma_\beta^2 - 1}},$$

while solving it on the expanded-scope branch yields

$$v_H(\sigma_\beta^2) = \hat{p} - \sqrt{\frac{\xi \hat{p}}{\sigma_\beta^2 - 1/\hat{p}}}.$$

Substituting these back into profits gives the indirect profit functions

$$\Pi_M(\sigma_\beta^2) = \left(\sqrt{\hat{p}(\sigma_\beta^2 - 1)} - \sqrt{\xi + 1 - \hat{p}}\right)^2 - \gamma \hat{p}$$

and

$$\Pi_H(\sigma_\beta^2) = \left(\sqrt{\hat{p}\sigma_\beta^2 - 1} - \sqrt{\xi}\right)^2 - \gamma.$$

The same-scope branch becomes profitable at

$$\underline{\sigma}_\beta^2 = \min_{v \in [v_-, v_+)} AC(v),$$

while the expanded-scope branch becomes profitable at

$$\bar{\sigma}_\beta^2 = \min_{v \in [v_+, \hat{p})} AC(v).$$

If $\hat{p} > p^\dagger(\gamma, \xi)$, then $\bar{\sigma}_\beta^2 < \underline{\sigma}_\beta^2$, so prediction first becomes profitable on the expanded-scope branch. Since

$$\Pi_H'(\sigma_\beta^2) = v_H(\sigma_\beta^2), \qquad \Pi_M'(\sigma_\beta^2) = v_M(\sigma_\beta^2),$$

and $v_H(\sigma_\beta^2) > v_M(\sigma_\beta^2)$ whenever both are defined, the difference $\Pi_H - \Pi_M$ is strictly increasing. Therefore, once the expanded-scope branch is profitable, it dominates globally, which proves the first part.

If $\hat{p} \leq p^\dagger(\gamma, \xi)$, then $\underline{\sigma}_\beta^2 \leq \bar{\sigma}_\beta^2$, so prediction first becomes profitable on the same-scope branch. Moreover, $\Pi_H - \Pi_M$ is continuous and strictly increasing whenever both branches are active, while for sufficiently large σ_β^2 the expanded-scope branch dominates because it allows values arbitrarily close to $\hat{p}$, whereas the same-scope branch is bounded above by v_+. It follows that there exists a unique threshold $\tilde{\sigma}_\beta^2$ such that

$$\Pi_H(\tilde{\sigma}_\beta^2) = \Pi_M(\tilde{\sigma}_\beta^2).$$

Hence the global maximizer is 0, then $v_M(\sigma_\beta^2)$, and then $v_H(\sigma_\beta^2)$, with both branch maximizers optimal at $\sigma_\beta^2 = \tilde{\sigma}_\beta^2$. □

Lemma 4 (Monotonicity of the low-value scale)**.** *Let*

$$n_L(v) \equiv \sqrt{\frac{v(\xi + 1 - v)}{\gamma}}, \qquad \gamma > 0,\ \xi > 0,$$

and define

$$v_-(\gamma, \hat{p}, \xi) = \frac{2\hat{p} + \gamma(\xi + 1) - \sqrt{\gamma^2(\xi + 1)^2 + 4\gamma\hat{p}(\xi + 1 - \hat{p})}}{2(1 + \gamma)}.$$

Then

$$n_L'(v) = \frac{\xi + 1 - 2v}{2\sqrt{\gamma\, v(\xi + 1 - v)}}.$$

Hence n_L is strictly increasing on $\big(0, (\xi + 1)/2\big)$ and strictly decreasing on $\big((\xi + 1)/2, \xi + 1\big)$. In particular, n_L is weakly increasing on $[0, v_-(\gamma, \hat{p}, \xi)]$ whenever

$$v_-(\gamma, \hat{p}, \xi) \leq \frac{\xi + 1}{2},$$

or equivalently,

$$\hat{p} \leq \frac{\xi + 1}{2}(1 + \sqrt{\gamma}).$$

Proof. Differentiating gives

$$n_L'(v) = \frac{1}{2}\left(\frac{v(\xi + 1 - v)}{\gamma}\right)^{-1/2} \frac{\xi + 1 - 2v}{\gamma} = \frac{\xi + 1 - 2v}{2\sqrt{\gamma\, v(\xi + 1 - v)}}.$$

Since the denominator is positive for $v \in (0, \xi + 1)$, the sign of $n_L'(v)$ is the sign of $\xi + 1 - 2v$. This proves the first claim.

Therefore n_L is increasing on $[0, v_-]$ exactly when $v_- \leq (\xi + 1)/2$. Substituting the expression for v_- and rearranging yields

$$\hat{p} \leq \frac{\xi + 1}{2}(1 + \sqrt{\gamma}),$$

which proves the last claim. □

Proposition 14 (Price subgame). *Fix $(v_1, v_2) \in [0, \hat{p})^2$.*

1. *On the captive segment, the unique equilibrium prices are*

$$P_i^{C*} = \sigma_\beta^2 v_i, \qquad i \in \{1, 2\},$$

   so captive-segment profits are

$$R_i^C(v_i) = \frac{1 - \lambda}{2}\sigma_\beta^2 v_i.$$

2. *If $v_i > v_j$, the unique equilibrium on the shopper segment that survives trembling-hand refinement is*

$$P_i^{S*} = \sigma_\beta^2(v_i - v_j), \qquad P_j^{S*} = 0,$$

   all shoppers buy from firm i, and shopper-segment profits are

$$R_i^S(v_i, v_j) = \lambda\sigma_\beta^2(v_i - v_j), \qquad R_j^S(v_j, v_i) = 0.$$

3. *If $v_1 = v_2$, the unique equilibrium payoff vector on the shopper segment is*

$$(R_1^S, R_2^S) = (0, 0).$$

Proof. The captive segment is independent across firms. Firm i solves

$$\max_{P_i^C \geq 0} \frac{1-\lambda}{2} P_i^C$$

subject to full coverage on its captive users, which requires

$$P_i^C \leq \sigma_\beta^2 v_i.$$

Hence the unique optimal captive price is

$$P_i^{C*} = \sigma_\beta^2 v_i,$$

and captive-segment profit is

$$R_i^C(v_i) = \frac{1-\lambda}{2} \sigma_\beta^2 v_i.$$

Now consider the shopper segment. Fix $v_i > v_j$, and define

$$d \equiv \sigma_\beta^2 (v_i - v_j) > 0.$$

If shopper prices are (P_i^S, P_j^S), then firm i wins all shoppers if and only if

$$\sigma_\beta^2 v_i - P_i^S \geq \sigma_\beta^2 v_j - P_j^S,$$

that is, if and only if

$$P_i^S \leq P_j^S + d.$$

Since ties are broken in favor of the higher-value firm i, shopper-segment profits are

$$\pi_i^S(P_i^S, P_j^S) = \lambda P_i^S \mathbf{1}\{P_i^S \leq P_j^S + d\},$$

$$\pi_j^S(P_j^S, P_i^S) = \lambda P_j^S \mathbf{1}\{P_i^S > P_j^S + d\}.$$

We first characterize Nash equilibrium in pure strategies.
For firm i, given any $P_j^S \geq 0$, any price $P_i^S < P_j^S + d$ still wins all shoppers but yields strictly less revenue than $P_i^S = P_j^S + d$, while any price $P_i^S > P_j^S + d$ loses all shoppers. Hence firm i has the unique best reply

$$BR_i(P_j^S) = \{P_j^S + d\}.$$

For firm j, given $P_i^S \geq 0$, three cases are possible.

First, if $P_i^S < d$, then for every $P_j^S \geq 0$,

$$P_i^S > P_j^S + d \quad \Longleftrightarrow \quad P_j^S < P_i^S - d < 0,$$

which is impossible. So firm j cannot win at any nonnegative price, and every $P_j^S \geq 0$ yields zero profit.
Second, if $P_i^S = d$, then firm j would need $P_j^S < 0$ to attract shoppers, again impossible. So every $P_j^S \geq 0$ yields zero profit.
Third, if $P_i^S > d$, then firm j wins all shoppers whenever

$$P_j^S < P_i^S - d.$$

Over the feasible set $[0, \infty)$, its profit is then

$$\lambda P_j^S \quad \text{for } P_j^S \in [0, P_i^S - d),$$

and zero otherwise. The supremum is $\lambda(P_i^S - d)$, approached as

$$P_j^S \uparrow P_i^S - d,$$

but not attained because equality gives the market to firm i. Thus for $P_i^S > d$, firm j has no pure best reply.
It follows that in any pure-strategy Nash equilibrium one must have

$$P_i^S \leq d.$$

Combining this with firm i's best-reply condition

$$P_i^S = P_j^S + d$$

yields

$$P_j^S = 0 \qquad \text{and} \qquad P_i^S = d.$$

Therefore the shopper subgame has the unique pure-strategy Nash equilibrium

$$(P_i^{S*}, P_j^{S*}) = (d, 0) = \left(\sigma_\beta^2(v_i - v_j), 0\right).$$

The associated shopper profits are

$$R_i^S(v_i, v_j) = \lambda d = \lambda \sigma_\beta^2(v_i - v_j), \qquad R_j^S(v_j, v_i) = 0.$$

We now verify that this equilibrium is the unique equilibrium surviving trembling-hand re-

finement. Since every trembling-hand perfect equilibrium is a Nash equilibrium, and the unperturbed shopper game has a unique Nash equilibrium, $(d, 0)$ is the only possible candidate. It therefore suffices to show that $(d, 0)$ is trembling-hand perfect.
Because prices are continuous, we verify perfection by a standard finite-action perturbation around the candidate equilibrium. For each

$$\epsilon \in \big(0, \min\{1/2, d\}\big),$$

consider the finite price set

$$A^\epsilon \equiv \{0, d - \epsilon, d\}.$$

Let the perturbed mixed strategies be

$$\mu_i^\epsilon(d) = 1 - 2\epsilon, \qquad \mu_i^\epsilon(d - \epsilon) = \epsilon, \qquad \mu_i^\epsilon(0) = \epsilon,$$

and

$$\mu_j^\epsilon(0) = 1 - 2\epsilon, \qquad \mu_j^\epsilon(d - \epsilon) = \epsilon, \qquad \mu_j^\epsilon(d) = \epsilon.$$

Thus each pure price in A^ϵ is chosen with strictly positive probability ϵ, and

$$\mu_i^\epsilon \to \delta_d, \qquad \mu_j^\epsilon \to \delta_0 \qquad \text{as } \epsilon \downarrow 0.$$

Against μ_j^ϵ, price d wins shoppers against every price in the support of μ_j^ϵ, so its expected shopper profit is

$$\mathbb{E}_{\mu_j^\epsilon}\left[\pi_i^S(d, P_j^S)\right] = \lambda d.$$

Price $d - \epsilon$ also wins shoppers against every price in the support, but yields only

$$\mathbb{E}_{\mu_j^\epsilon}\left[\pi_i^S(d - \epsilon, P_j^S)\right] = \lambda(d - \epsilon) < \lambda d,$$

and price 0 yields zero. Hence d is the unique best reply to μ_j^ϵ.
Against μ_i^ϵ, no nonnegative price of firm j can win shoppers against any price in the support of μ_i^ϵ, because for every

$$P_i^S \in \{0, d - \epsilon, d\} \quad \text{and} \quad P_j^S \in \{0, d - \epsilon, d\},$$

the strict inequality required for firm j to win,

$$P_i^S > P_j^S + d,$$

never holds. Therefore every action in A^ϵ gives firm j zero expected profit, and in particular 0 is a best reply to μ_i^ϵ.

Hence, for every $\epsilon > 0$,

$$d \in BR_i(\mu_j^\epsilon), \qquad 0 \in BR_j(\mu_i^\epsilon),$$

with

$$\mu_i^\epsilon \to \delta_d, \qquad \mu_j^\epsilon \to \delta_0 \qquad \text{as } \epsilon \downarrow 0.$$

This proves that $(d, 0)$ is trembling-hand perfect. Since it is also the unique Nash equilibrium of the unperturbed shopper game, it is the unique equilibrium that survives trembling-hand refinement.

Finally, if $v_1 = v_2$, then the shopper segment is a homogeneous-good Bertrand game with zero marginal cost. Therefore every Nash equilibrium yields zero shopper profits, so the equilibrium payoff vector is uniquely

$$(R_1^S, R_2^S) = (0, 0).$$

□

Proposition 9 (No symmetric pure equilibrium)**.** *Suppose $\mu > 0$. Let $\tilde{C}$ be continuous on $[0, \hat{p})$, differentiable on $(0, \hat{p}) \setminus \{v_+\}$, and admit one-sided derivatives at v_+. If*

$$\tilde{C}'_-(v_+) > \tilde{C}'_+(v_+),$$

then there is no symmetric pure-strategy equilibrium with

$$v_1 = v_2 \in (0, \hat{p}).$$

Proof. Write

$$\rho_F \equiv \frac{1-\lambda}{2}\sigma_\beta^2, \qquad \rho_L \equiv \frac{1+\lambda}{2}\sigma_\beta^2,$$

so that $\rho_L > \rho_F$ whenever $\lambda > 0$.

If firm 2 chooses v, firm 1's payoff from choosing w is

$$\Pi_1(w, v) = \begin{cases} \rho_F w - \tilde{C}(w), & \text{if } w \le v, \\ \rho_L w - \lambda\sigma_\beta^2 v - \tilde{C}(w), & \text{if } w > v. \end{cases}$$

In particular,

$$\Pi_1(v, v) = \rho_F v - \tilde{C}(v).$$

Suppose, toward a contradiction, that (v, v) is a symmetric pure equilibrium.

First consider the case $v \neq v_+$. Since $\tilde{C}$ is differentiable at v, for every sufficiently small $\varepsilon > 0$,

$$\Pi_1(v+\varepsilon, v) - \Pi_1(v, v) = \rho_L \varepsilon - \big(\tilde{C}(v+\varepsilon) - \tilde{C}(v)\big),$$

and

$$\Pi_1(v-\varepsilon, v) - \Pi_1(v, v) = -\rho_F \varepsilon + \big(\tilde{C}(v) - \tilde{C}(v-\varepsilon)\big).$$

Therefore

$$\lim_{\varepsilon \downarrow 0} \frac{\Pi_1(v+\varepsilon, v) - \Pi_1(v, v)}{\varepsilon} = \rho_L - \tilde{C}'(v),$$

and

$$\lim_{\varepsilon \downarrow 0} \frac{\Pi_1(v-\varepsilon, v) - \Pi_1(v, v)}{\varepsilon} = \tilde{C}'(v) - \rho_F.$$

Since (v, v) is an equilibrium, neither a small upward nor a small downward deviation can be profitable. Hence one must have

$$\rho_L - \tilde{C}'(v) \leq 0 \qquad \text{and} \qquad \tilde{C}'(v) - \rho_F \leq 0,$$

that is,

$$\tilde{C}'(v) \geq \rho_L \qquad \text{and} \qquad \tilde{C}'(v) \leq \rho_F.$$

This is impossible because $\rho_L > \rho_F$.
Now consider the kink $v = v_+$. For every sufficiently small $\varepsilon > 0$,

$$\Pi_1(v_+ + \varepsilon, v_+) - \Pi_1(v_+, v_+) = \rho_L \varepsilon - \big(\tilde{C}(v_+ + \varepsilon) - \tilde{C}(v_+)\big),$$

and

$$\Pi_1(v_+ - \varepsilon, v_+) - \Pi_1(v_+, v_+) = -\rho_F \varepsilon + \big(\tilde{C}(v_+) - \tilde{C}(v_+ - \varepsilon)\big).$$

Dividing by ε and taking limits gives

$$\lim_{\varepsilon \downarrow 0} \frac{\Pi_1(v_+ + \varepsilon, v_+) - \Pi_1(v_+, v_+)}{\varepsilon} = \rho_L - \tilde{C}'_+(v_+),$$

and

$$\lim_{\varepsilon \downarrow 0} \frac{\Pi_1(v_+ - \varepsilon, v_+) - \Pi_1(v_+, v_+)}{\varepsilon} = \tilde{C}'_-(v_+) - \rho_F.$$

Again, if (v_+, v_+) were an equilibrium, neither deviation could be profitable, so one would need

$$\tilde{C}'_+(v_+) \geq \rho_L \qquad \text{and} \qquad \tilde{C}'_-(v_+) \leq \rho_F.$$

But these inequalities imply

$$\tilde{C}'_-(v_+) \leq \rho_F < \rho_L \leq \tilde{C}'_+(v_+),$$

hence

$$\tilde{C}'_-(v_+) < \tilde{C}'_+(v_+),$$

contradicting the assumed downward kink

$$\tilde{C}'_-(v_+) > \tilde{C}'_+(v_+).$$

Therefore no symmetric pure-strategy equilibrium exists in $(0, \hat{p})$. □

Proposition 10 (Asymmetric pure equilibria)**.** *Define*

$$v_F \in \arg\max_{v \in [0,\hat{p})} \left\{ \rho_F v - \tilde{C}(v) \right\},$$

and

$$v_L \in \arg\max_{v \in [0,\hat{p})} \left\{ \rho_L v - \tilde{C}(v) \right\},$$

which are unique by Proposition 8 and such that

$$v_L > v_F.$$

Then the period-1 game has exactly two pure-strategy Nash equilibria:

$$(v_1^*, v_2^*) = (v_L, v_F) \qquad \textit{and} \qquad (v_1^*, v_2^*) = (v_F, v_L).$$

Proof. Consider a candidate equilibrium with $v_1 > v_2$. By (5), firm 1 solves

$$\max_{v_1 \in [0,\hat{p})} \left\{ \rho_L v_1 - \tilde{C}(v_1) \right\},$$

up to the constant term $-\lambda \sigma_\beta^2 v_2$, so its unique best response is v_L. Firm 2 solves

$$\max_{v_2 \in [0,\hat{p})} \left\{ \rho_F v_2 - \tilde{C}(v_2) \right\},$$

so its unique best response is v_F. Since $v_L > v_F$, the ranking is consistent, hence (v_L, v_F) is an equilibrium. By symmetry, so is (v_F, v_L).
Conversely, any pure equilibrium with $v_1 \neq v_2$ must assign the leader the unique maximizer of the leader problem and the follower the unique maximizer of the follower problem. Hence the only such equilibria are the two stated pairs. Proposition 9 rules out symmetric interior pure equilibria. □

Corollary 12 (Equilibrium technology regimes under shoppers and price discrimination)**.** *Fix* $(\gamma, \hat{p}, \xi, \lambda)$ *with* $\lambda > 0$*, and define*

$$\rho_F \equiv \frac{1-\lambda}{2}\sigma_\beta^2, \qquad \rho_L \equiv \frac{1+\lambda}{2}\sigma_\beta^2.$$

When $\hat{p} \leq p^{\dagger}(\gamma, \xi)$, suppose additionally that

$$\rho_F \neq \tilde{\sigma}_\beta^2 \qquad \textit{and} \qquad \rho_L \neq \tilde{\sigma}_\beta^2,$$

so that Proposition 8 yields unique maximizers for both the follower and the leader. Then the pure-strategy equilibria are exactly the following.

1. *If $\hat{p} > p^{\dagger}(\gamma, \xi)$, there are three possible regimes:*

   (a) ***Both firms use expanded-scope training*** *if*

   $$\bar{\sigma}_\beta^2 \leq \rho_F < \rho_L.$$

   (b) ***Only the leader uses expanded-scope training*** *if*

   $$\rho_F < \bar{\sigma}_\beta^2 \leq \rho_L.$$

   (c) ***No firm enters*** *if*

   $$\rho_L < \bar{\sigma}_\beta^2.$$

2. *If $\hat{p} \leq p^{\dagger}(\gamma, \xi)$, there are six possible regimes:*

   (a) ***Both firms use expanded-scope training*** *if*

   $$\tilde{\sigma}_\beta^2 < \rho_F < \rho_L.$$

   (b) ***The leader uses expanded-scope training and the follower uses same-scope training*** *if*

   $$\underline{\sigma}_\beta^2 \leq \rho_F < \tilde{\sigma}_\beta^2 < \rho_L.$$

   (c) ***Only the leader uses expanded-scope training*** *if*

   $$\rho_F < \underline{\sigma}_\beta^2 < \tilde{\sigma}_\beta^2 < \rho_L.$$

   (d) ***Both firms use same-scope training*** *if*

   $$\underline{\sigma}_\beta^2 \leq \rho_F < \rho_L < \tilde{\sigma}_\beta^2.$$

   (e) ***Only the leader uses same-scope training*** *if*

   $$\rho_F < \underline{\sigma}_\beta^2 \leq \rho_L < \tilde{\sigma}_\beta^2.$$

(f) ***No firm enters*** *if*

$$\rho_L < \underline{\sigma}_\beta^2.$$

Proof. By Proposition 8, the follower and leader solve the monopoly problem at effective prices ρ_F and ρ_L, respectively. Since

$$\rho_L > \rho_F,$$

the leader chooses weakly more value than the follower, and strictly more whenever both maximizers are unique and not both equal to zero.
If ρ_L lies below the relevant entry threshold in Proposition 8, then even a firm that captures all shoppers prefers $v = 0$, so $(0, 0)$ is the unique equilibrium.
If instead ρ_L is above the relevant entry threshold, then $v_L > 0$. Proposition 9 rules out symmetric pure equilibria, and Proposition 10 implies that the only pure-strategy equilibria are

$$(v_L, v_F) \qquad \text{and} \qquad (v_F, v_L).$$

The stated cases follow directly from Proposition 8, according to whether ρ_F and ρ_L lie below the entry threshold, on the same-scope branch, or on the expanded-scope branch. □